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(54) **ENGINEERING B CELL-BASED PROTEIN FACTORIES TO TREAT SERIOUS DISEASES**

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A61K 38/51	(2006.01)
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(52) **U.S. Cl.**

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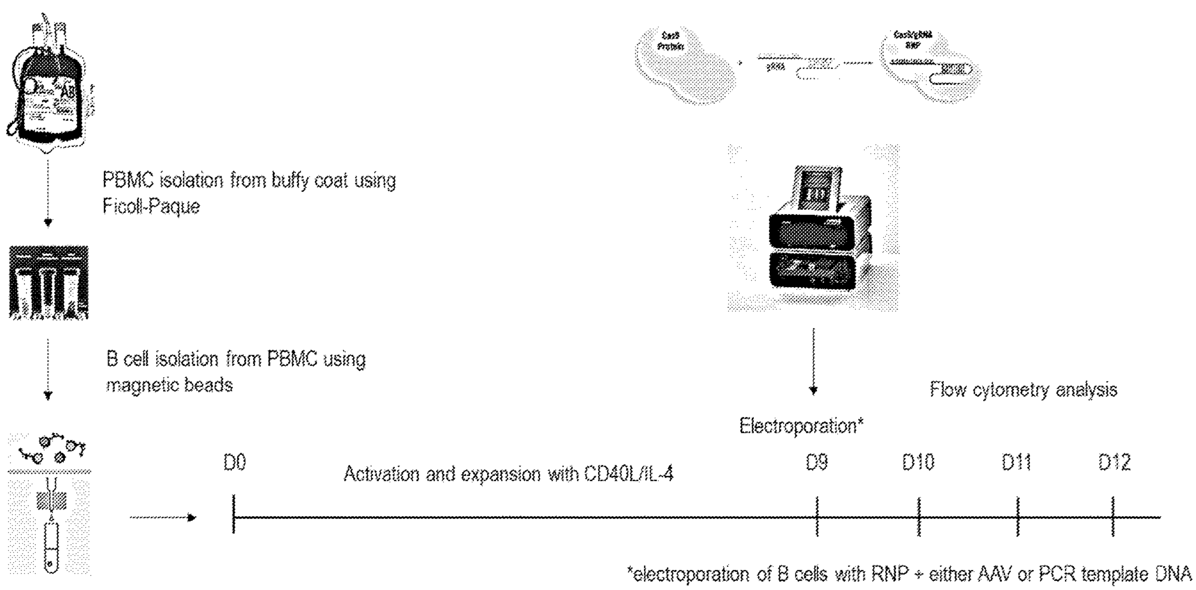
(57) **ABSTRACT**

The invention(s) disclosed herein relate to improved methods for expanding cell populations, particularly B cell populations. The invention further relates comprising improved cell media, compositions thereof, and methods of using such expanded B cells. Wherein a population of cells comprises engineered human B cells, wherein the engineered human B cells comprise a therapeutic protein, whose gene has been inserted into the beta-2M locus.

Specification includes a Sequence Listing.



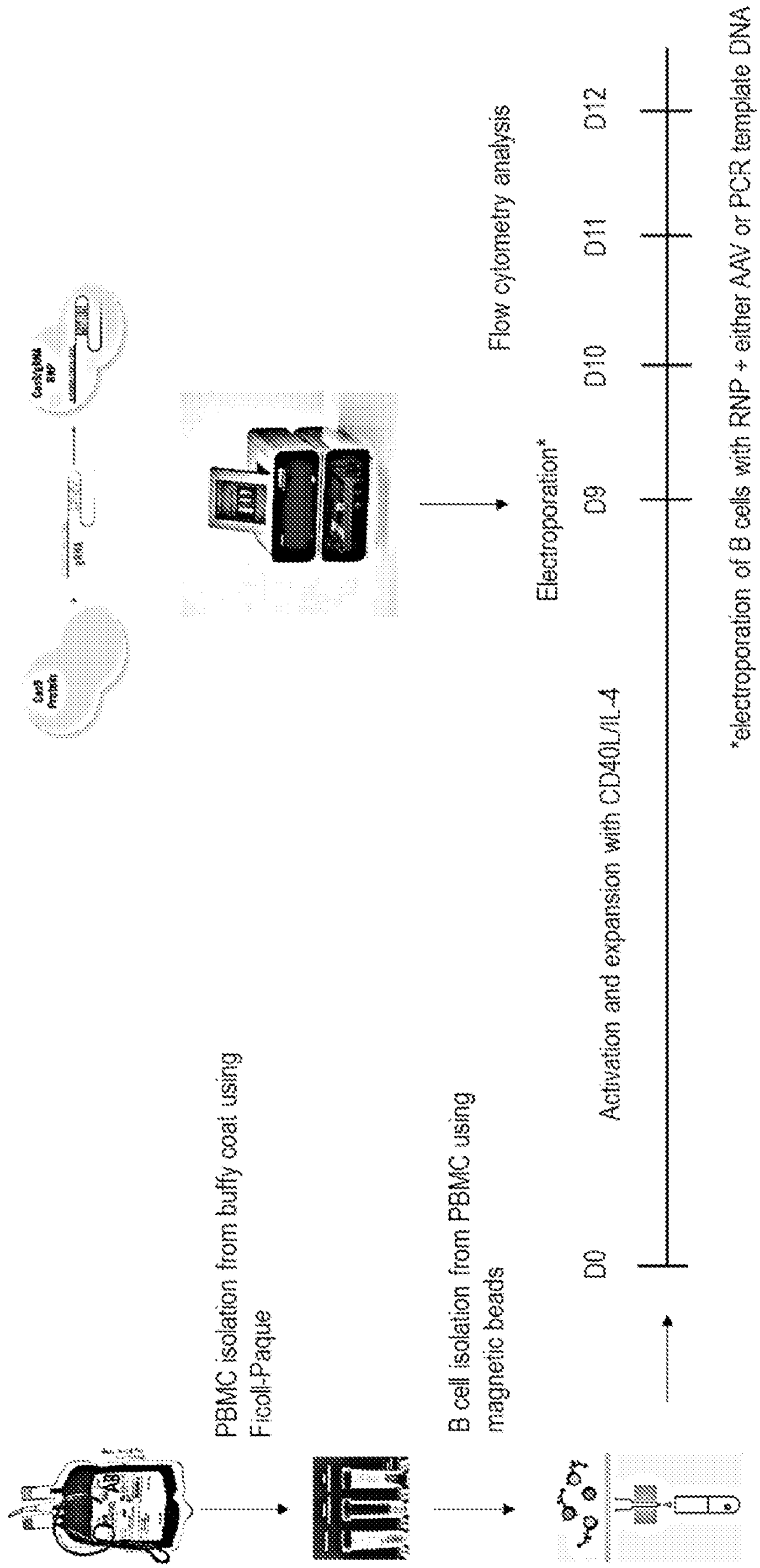

PBMC isolation from buffy coat using Ficoll-Paque
B cell isolation from PBMC using magnetic beads
D0
Activation and expansion with CD40L/IL-4
Electroporation*
D9
D10
D11
D12
Flow cytometry analysis
*electroporation of B cells with RNP + either AAV or PCR template DNA

FIG. 1

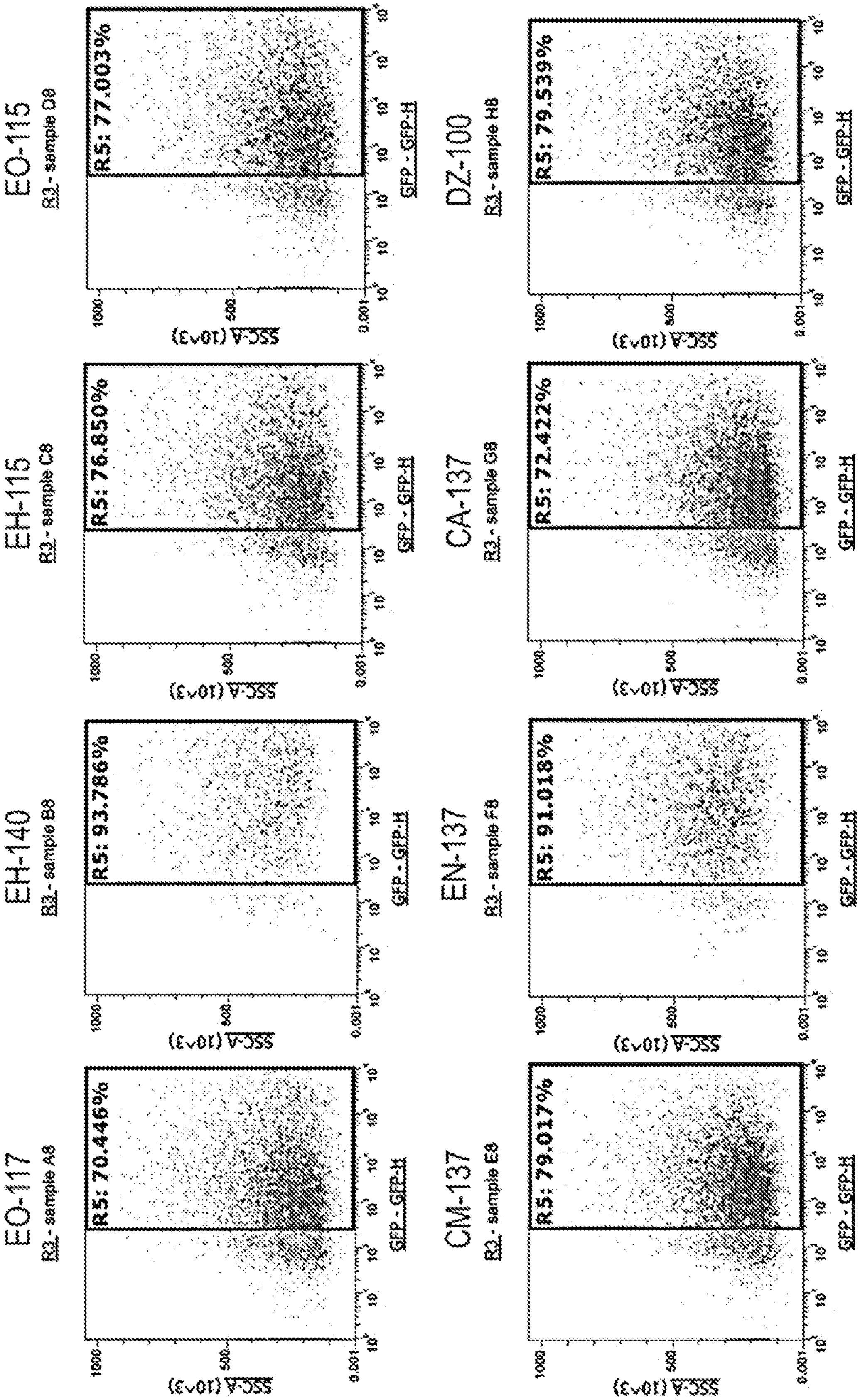

EO-117
R3 - sample A8
R5: 70.446%
EH-140
R3 - sample B8
R5: 93.786%
EH-115
R3 - sample C8
R5: 76.850%
EO-115
R3 - sample D8
R5: 77.003%
CM-137
R3 - sample E8
R5: 79.017%
EN-137
R3 - sample F8
R5: 91.018%
CA-137
R3 - sample G8
R5: 72.422%
DZ-100
R3 - sample H8
R5: 79.539%
SSC-A (10^3)
GFP - GFP-H

FIG. 2A

	Donor 466	
Program	GFP+%	Viability%
EO-117	70.446	98.037
EH-140	93.786	92.978
EH-115	76.850	98.119
EO-115	77.003	98.350
CM-137	**79.017**	**98.484**
EN-137	91.018	96.632
CA-137	72.422	98.208
DZ-100	79.539	98.252
Control	1.444	96.394

FIG. 2B

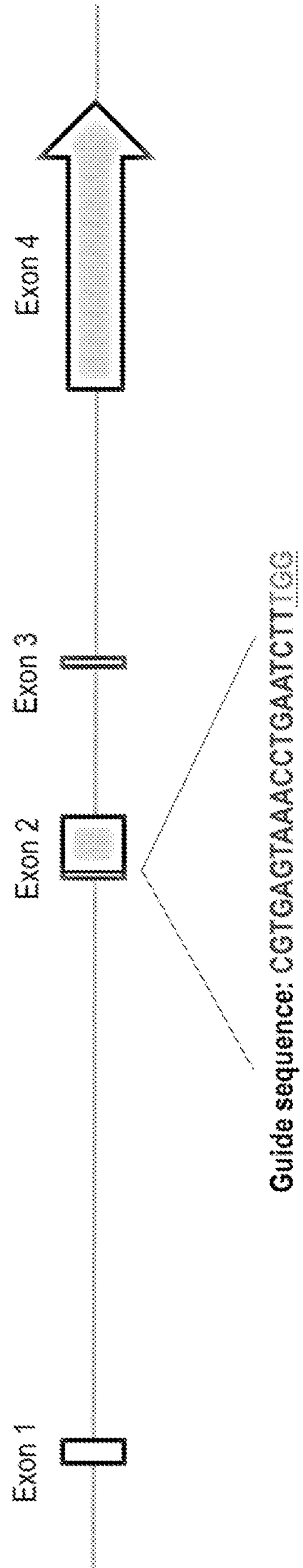

Exon 1
Exon 2
Exon 3
Exon 4
Guide sequence: CGTGAGTAACCTGAATCTTTGG

FIG. 3

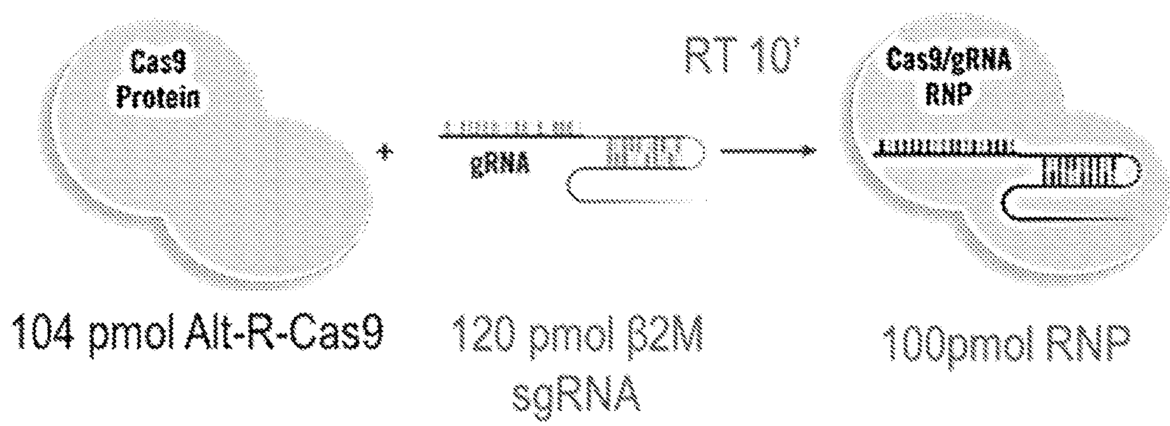

Cas9
Protein
+
gRNA
RT 10'
Cas9/gRNA
RNP
104 pmol Alt-R-Cas9
120 pmol β2M
sgRNA
100pmol RNP

FIG. 4A

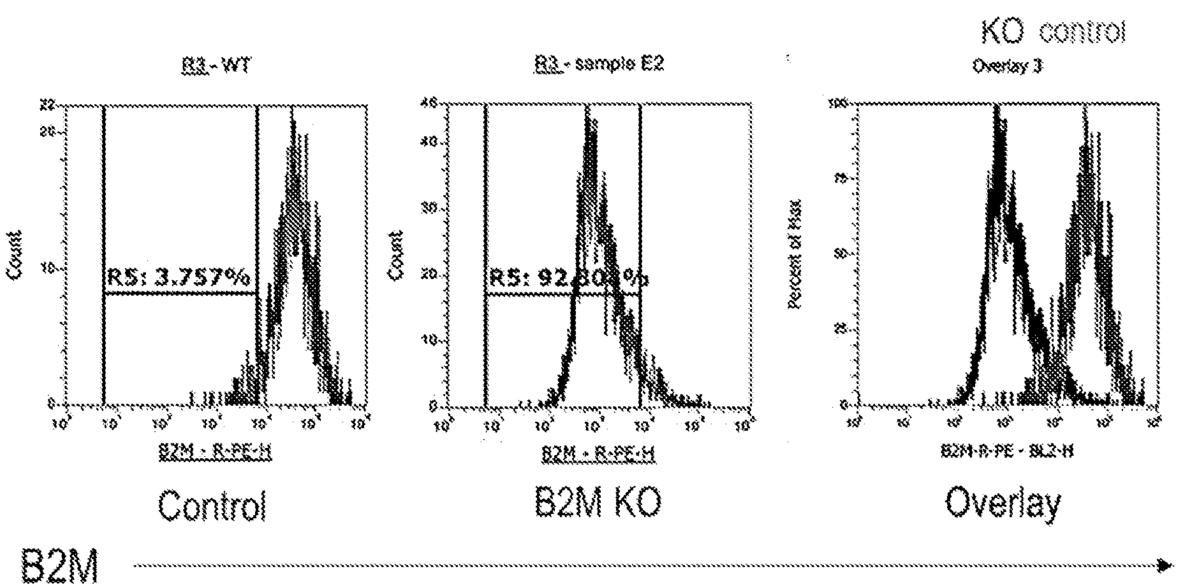

KO control
R3 - WT
R3 - sample E2
Overlay 3
Count
Percent of Max
R5: 3.757%
B2M - R-PE-H
B2M-R-PE - BL2-H
Control
B2M KO
Overlay
B2M

FIG. 4B

	Donor 47	Donor 47	Donor 48	Donor 48
Program	KO%*	Viability	KO%	Viability
EO-117	33.6	95	44	93.1
EH-140	79.7	92.3	87.6	93.6
EH-115	37.9	89.6	47.5	93.7
EO-115	37.2	93.1	44.3	94
CM-137	**80.2**	**91.8**	**85.6**	**92.8**
EN-137	63.4	93.9	69.2	92.5
CA-137	76.8	92	81.5	91.6
DZ-100	75.9	92.9	82.6	93.4
Control	3.8	91.3	1.2	94.5
*% b2M KnockOut by flow cytometry				

FIG. 4C

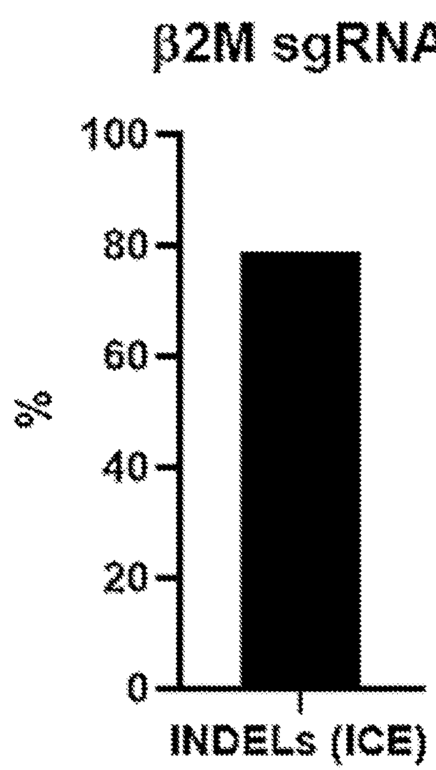


FIG. 5A

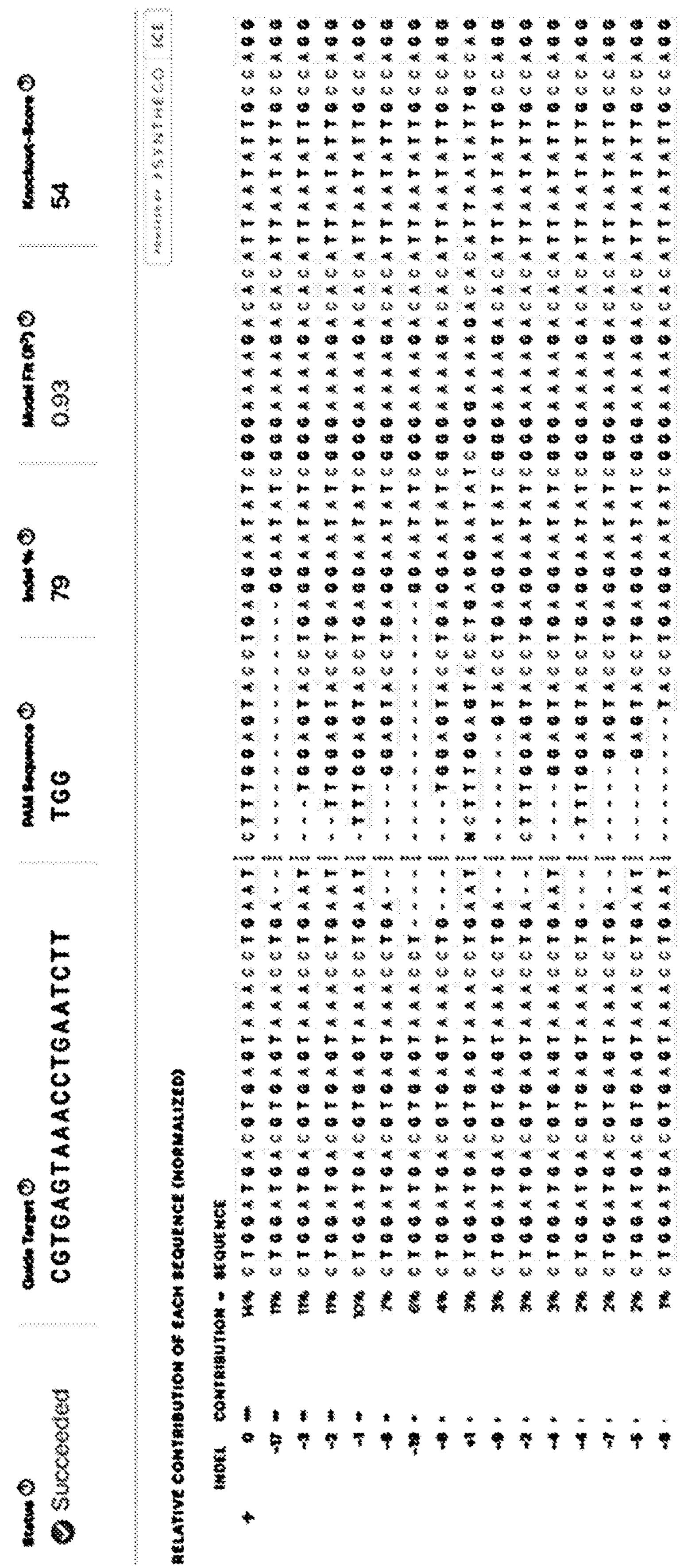

Status
Succeeded
Guide Target
CGTGAGTAAACCTGAATCTT
PAM Sequence
TGG
Indel %
79
Model Fit (R²)
0.93
Knockout-Score
54
RELATIVE CONTRIBUTION OF EACH SEQUENCE (NORMALIZED)
INDEL
CONTRIBUTION
SEQUENCE

FIG. 5B

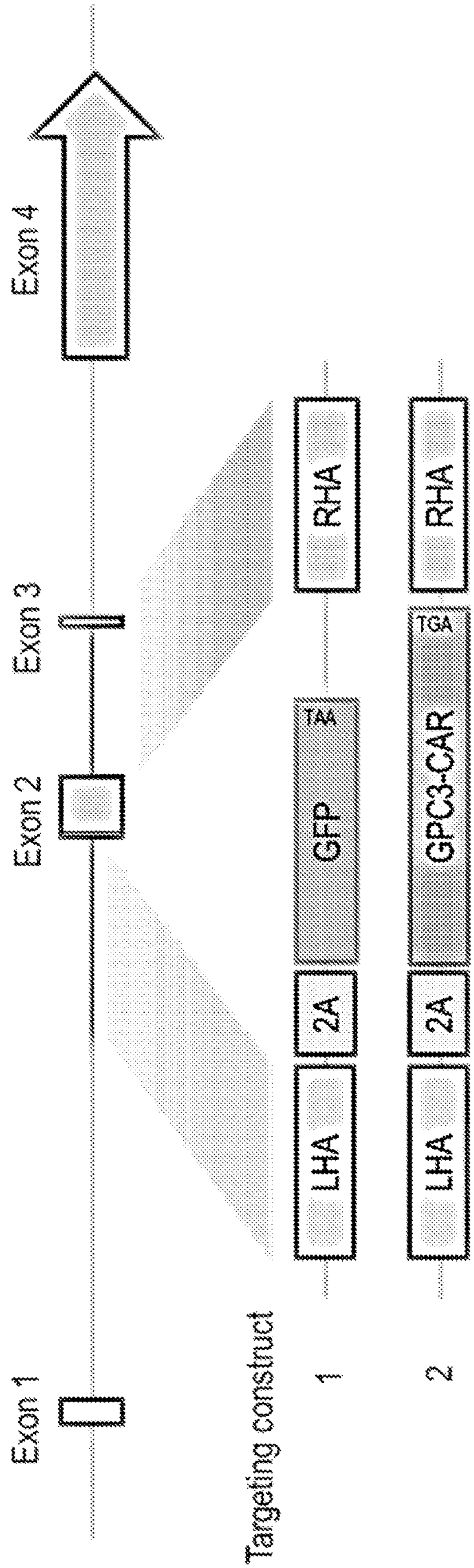

Exon 1
Exon 2
Exon 3
Exon 4
Targeting construct
1
LHA
2A
GFP
TAA
RHA
2
LHA
2A
GPC3-CAR
TGA
RHA

FIG. 6

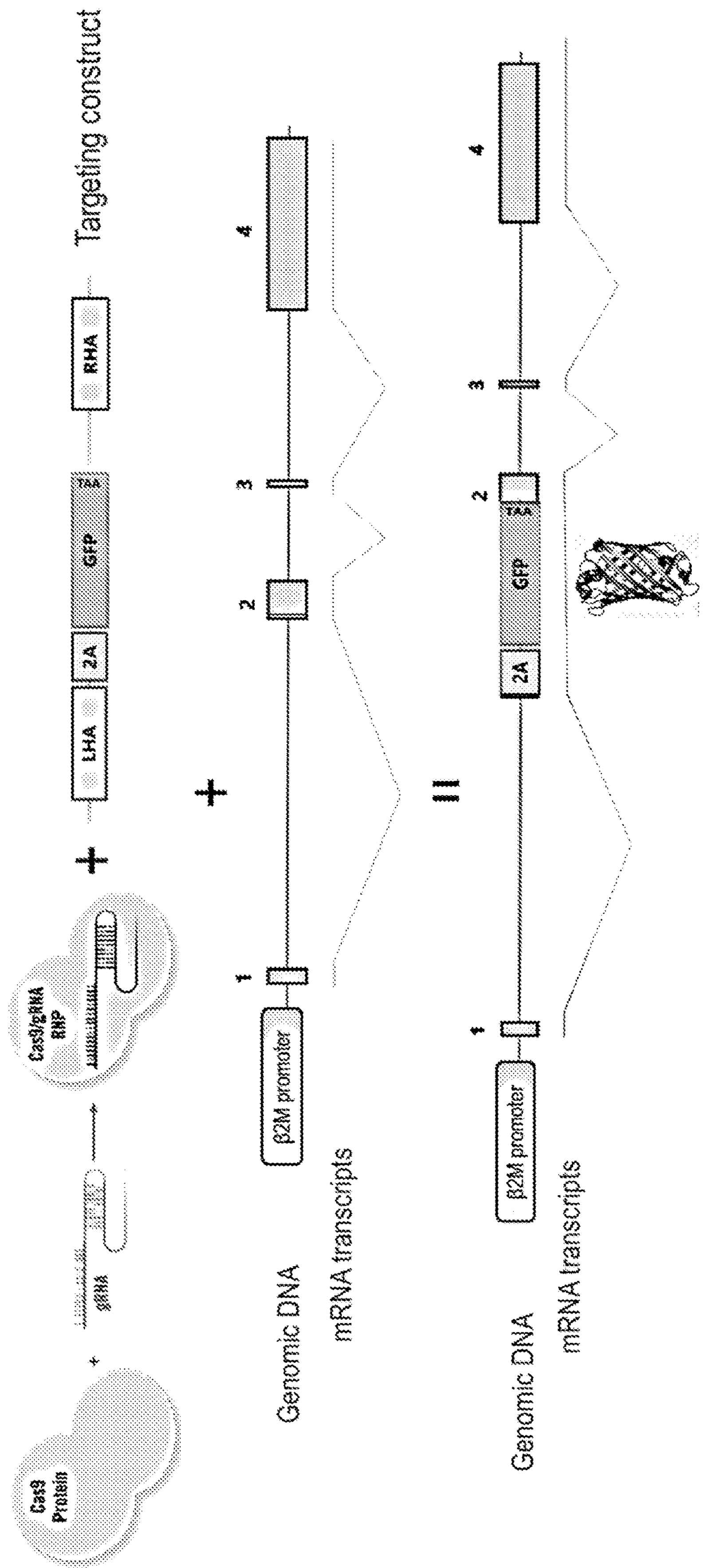

Cas9 Protein
gRNA
Cas9/gRNA RNP
LHA
2A
GFP
TAA
RHA
Targeting construct
Genomic DNA
mRNA transcripts
β2M promoter
1
2
3
4
Genomic DNA
mRNA transcripts
β2M promoter
1
2A
GFP
TAA
2
3
4

FIG. 7

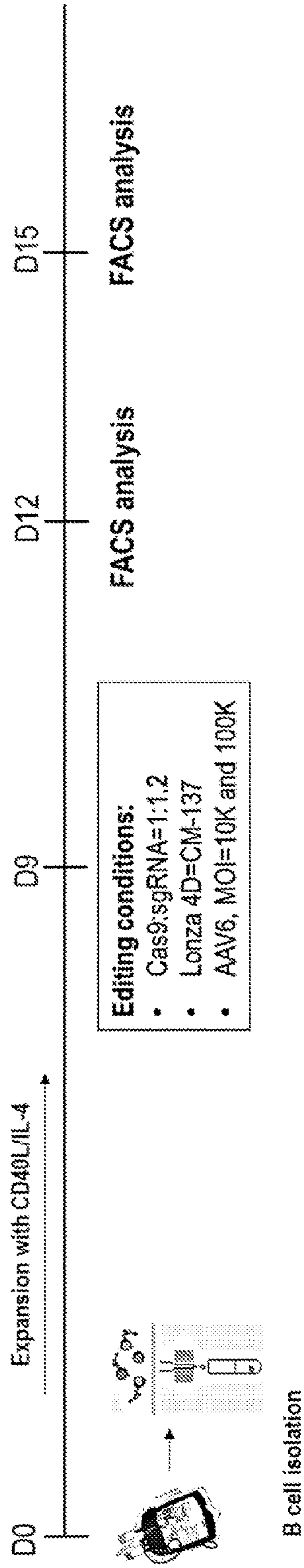

D0
D9
D12
D15
Expansion with CD40L/IL-4
B cell isolation
Editing conditions:
• Cas9:sgRNA=1:1.2
• Lonza 4D=CM-137
• AAV6, MOI=10K and 100K
FACS analysis
FACS analysis

FIG. 8A

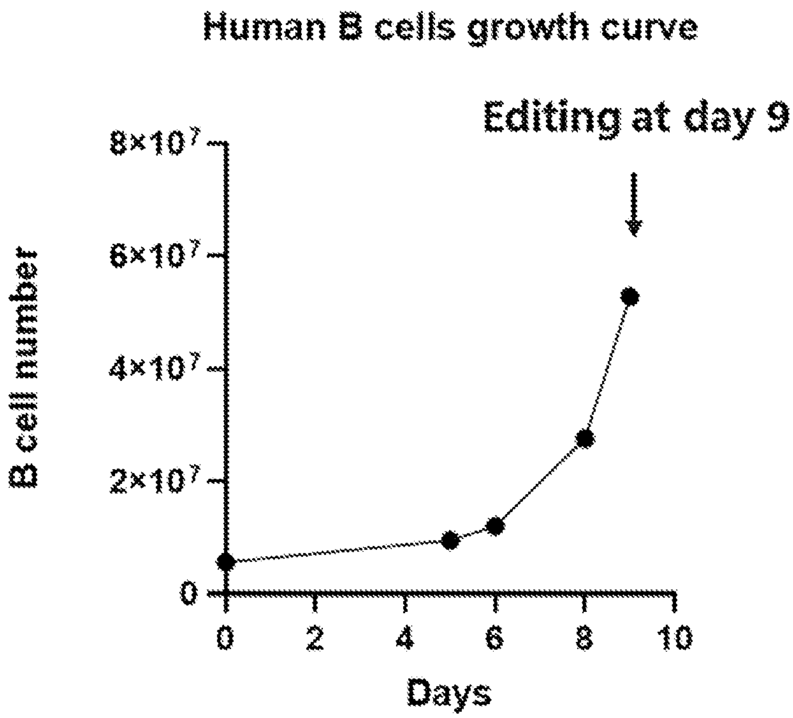

Human B cells growth curve
Editing at day 9
B cell number
8×10^7
6×10^7
4×10^7
2×10^7
0
0
2
4
6
8
10
Days

FIG. 8B

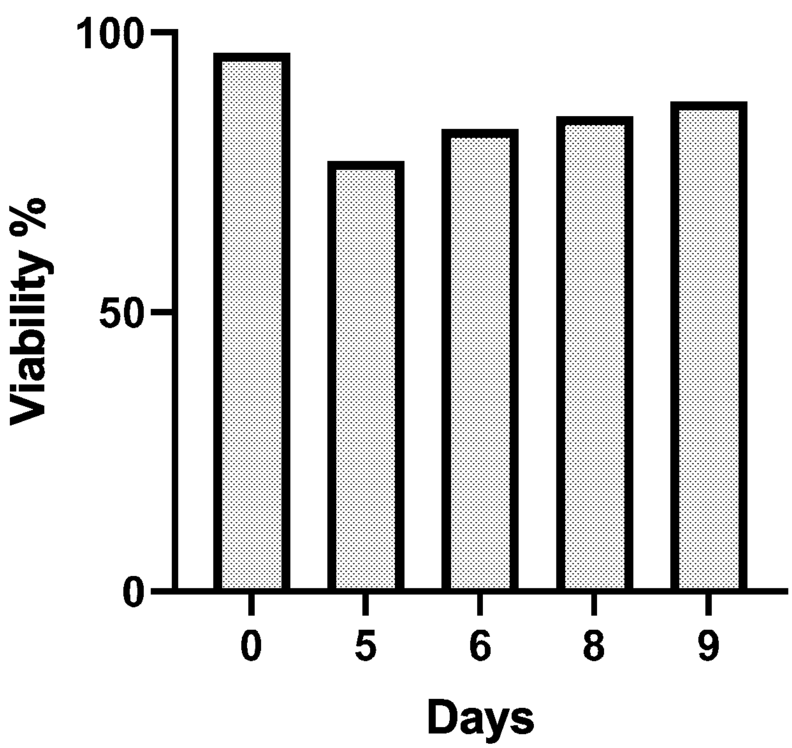

Viability %
100
50
0
0
5
6
8
9
Days

FIG. 8C

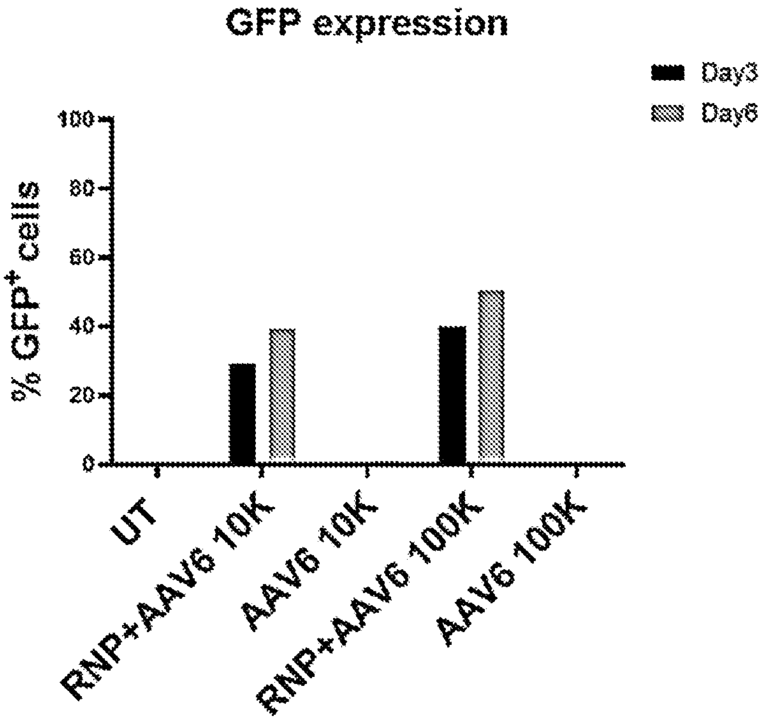

GFP expression
Day3
Day6
% GFP+ cells
100
80
60
40
20
0
UT
RNP+AAV6 10K
AAV6 10K
RNP+AAV6 100K
AAV6 100K

FIG. 9A

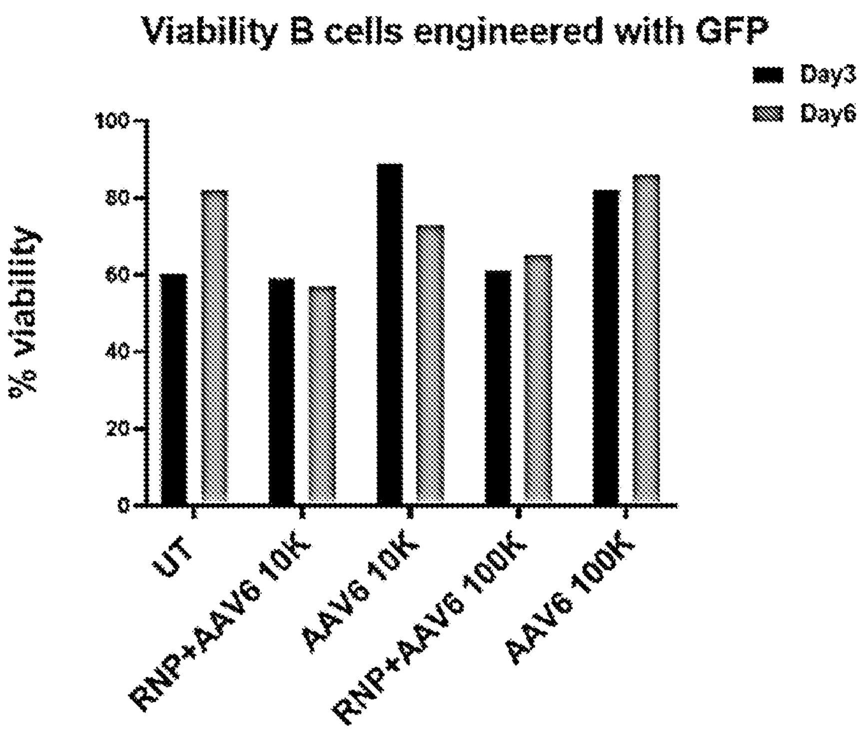

Viability B cells engineered with GFP
Day3
Day6
% viability
100
80
60
40
20
0
UT
RNP+AAV6 10K
AAV6 10K
RNP+AAV6 100K
AAV6 100K

FIG. 9B

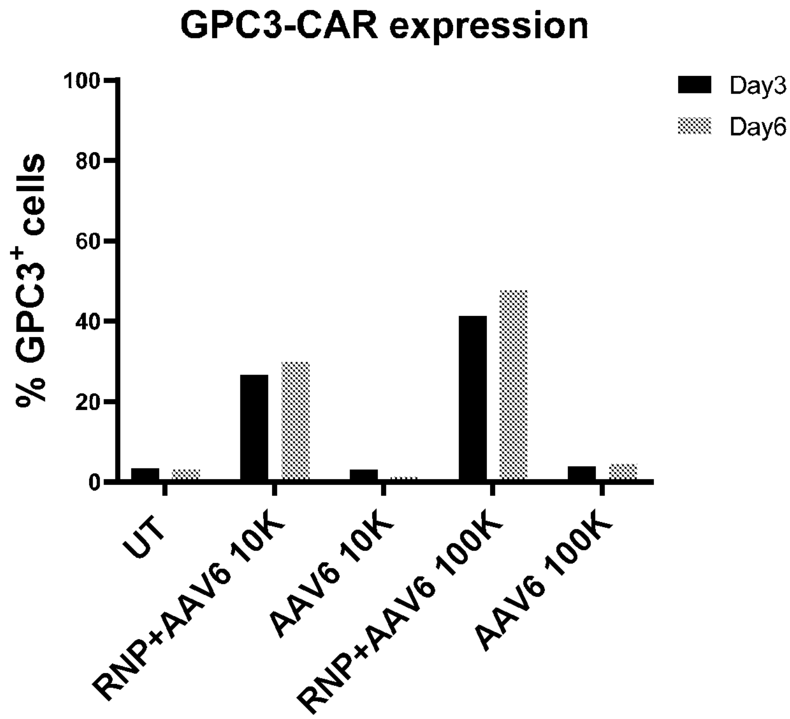

GPC3-CAR expression
% GPC3$^+$ cells
100
80
60
40
20
0
Day3
Day6
UT
RNP+AAV6 10K
AAV6 10K
RNP+AAV6 100K
AAV6 100K

FIG. 9C

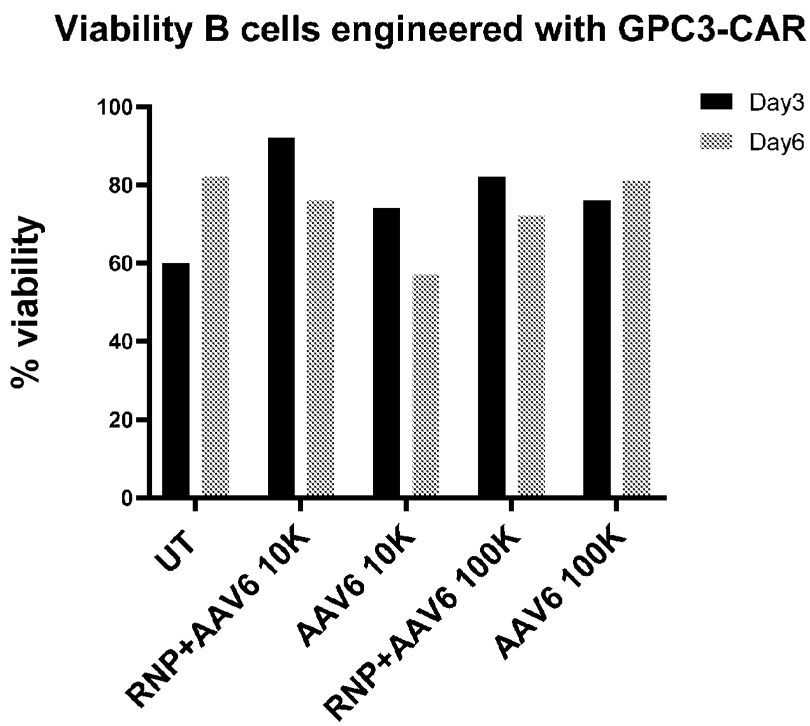

Viability B cells engineered with GPC3-CAR
% viability
100
80
60
40
20
0
Day3
Day6
UT
RNP+AAV6 10K
AAV6 10K
RNP+AAV6 100K
AAV6 100K

FIG. 9D

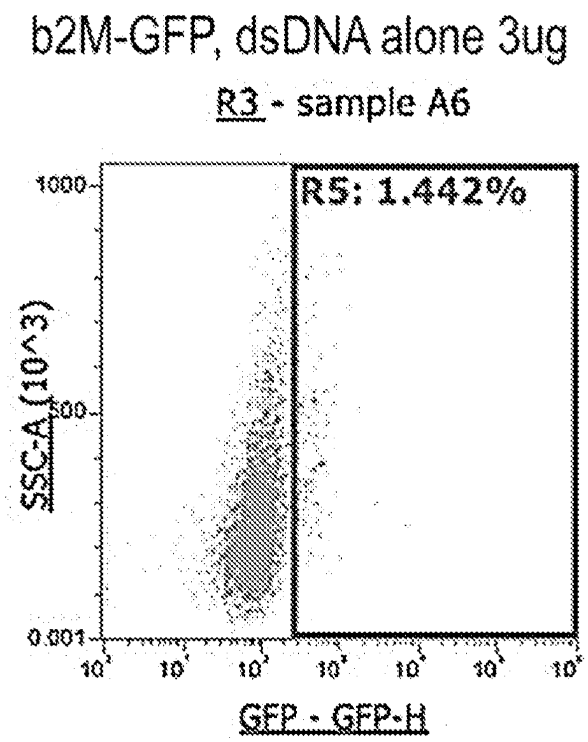


FIG. 10A

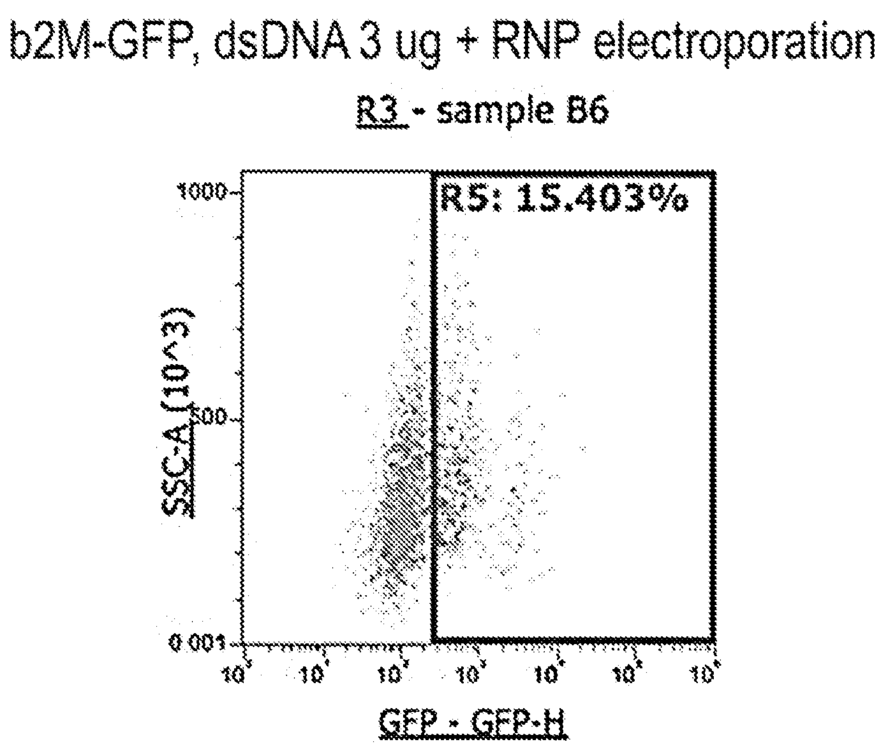


FIG. 10B

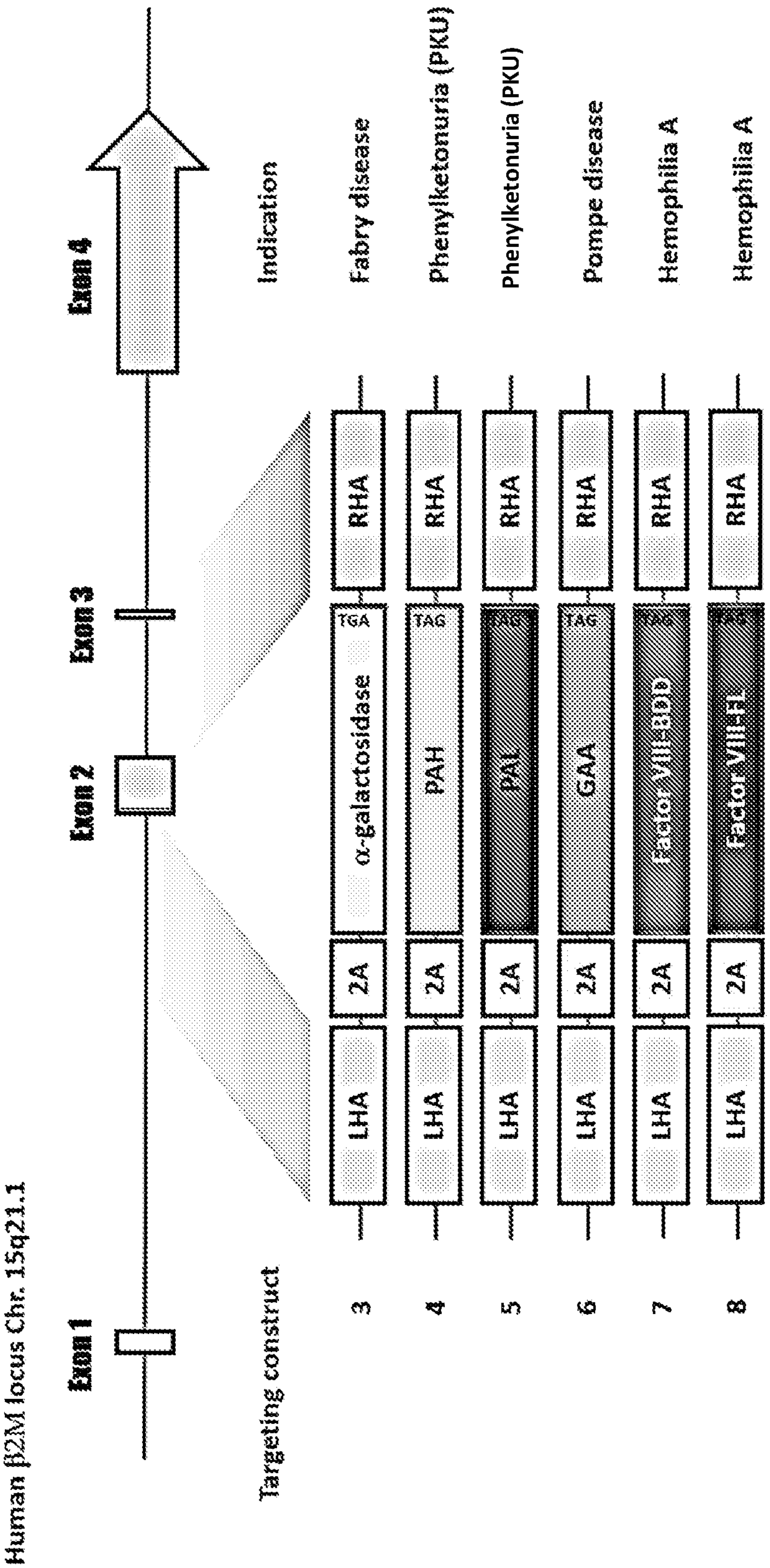

Human β2M locus Chr. 15q21.1
Exon 1
Exon 2
Exon 3
Exon 4
Targeting construct
Indication
3
LHA
2A
α-galactosidase
TGA
RHA
Fabry disease
4
LHA
2A
PAH
TAG
RHA
Phenylketonuria (PKU)
5
LHA
2A
PAL
TAG
RHA
Phenylketonuria (PKU)
6
LHA
2A
GAA
TAG
RHA
Pompe disease
7
LHA
2A
Factor VIII-BDD
TAG
RHA
Hemophilia A
8
LHA
2A
Factor VIII-FL
TAG
RHA
Hemophilia A

FIG. 11

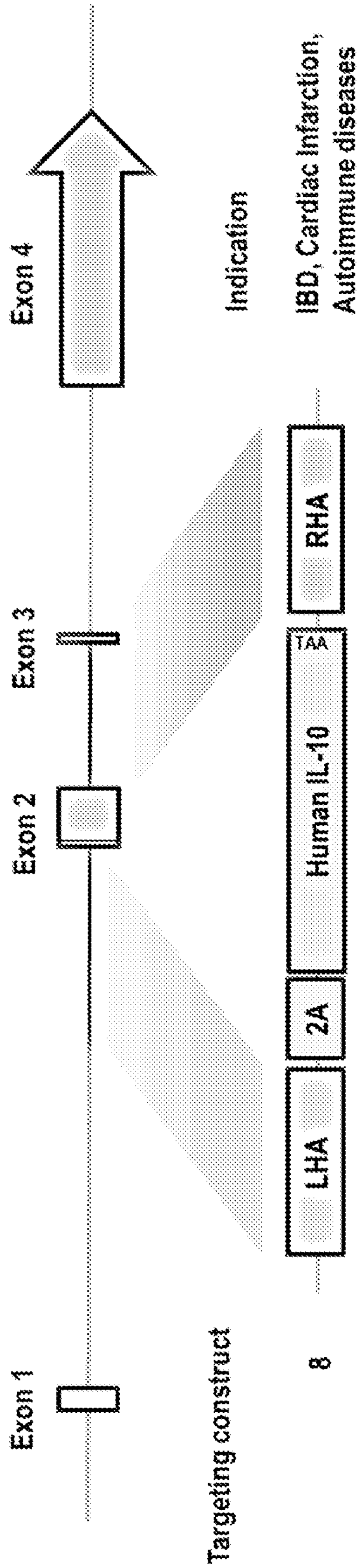

Exon 1
Exon 2
Exon 3
Exon 4
Targeting construct
Indication
8
LHA
2A
Human IL-10
TAA
RHA
IBD, Cardiac Infarction, Autoimmune diseases

FIG. 12

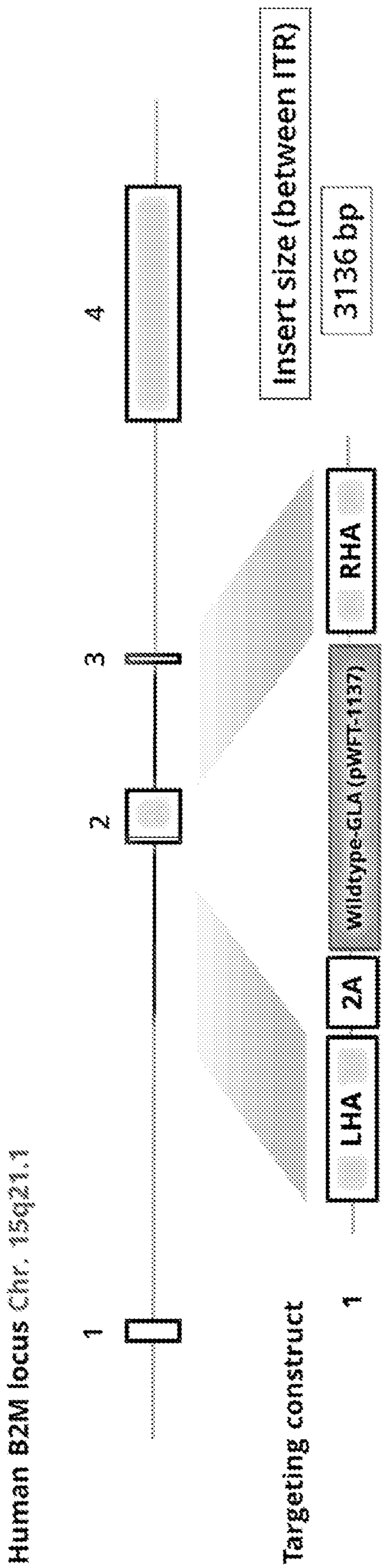

Human B2M locus Chr. 15q21.1
1
2
3
4
Targeting construct
1
LHA
2A
Wildtype-GLA (pWFT-1137)
RHA
Insert size (between ITR)
3136 bp

FIG. 13

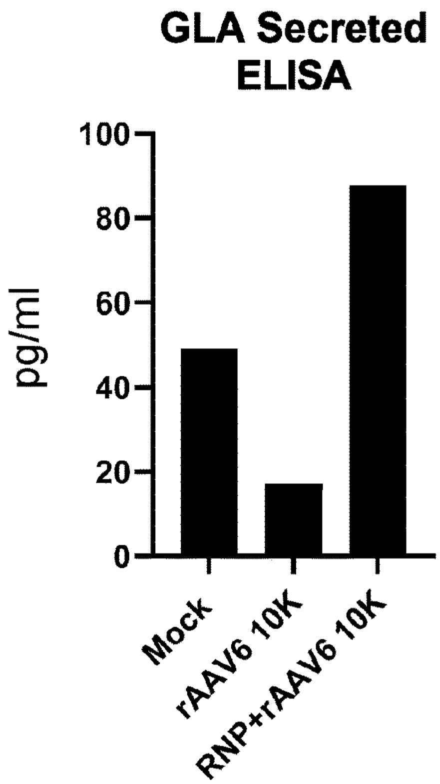


FIG. 14A

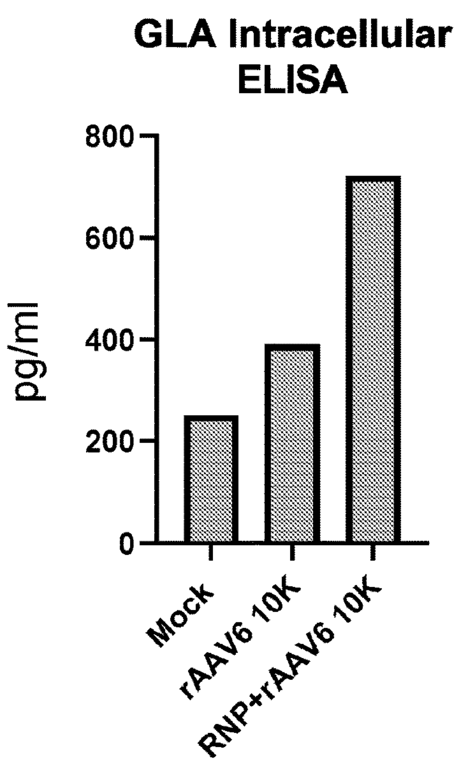


FIG. 14B

ENGINEERING B CELL-BASED PROTEIN FACTORIES TO TREAT SERIOUS DISEASES

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This patent application is a national stage application of International Application No. PCT/US2022/025471 entitled "ENGINEERING B CELL-BASED PROTEIN FACTORIES TO TREAT SERIOUS DISEASES," filed on Apr. 20, 2022, which claims the benefit of U.S. Provisional Patent Application Ser. No. 63/176,944, filed on Apr. 20, 2021, each of which are incorporated by reference herein in their entirety for all purposes.

SEQUENCE LISTING

[0002] The instant application contains a Sequence Listing which has been submitted electronically in ASCII format and is hereby incorporated by reference in its entirety. Said ASCII copy, created on Jan. 31, 2024, is named 2219235_00121US2_SL.txt and is 144,951 bytes in size.

BACKGROUND OF THE INVENTION

[0003] B cells are naturally hardwired to present antigens and secrete immunoglobulins. Theoretically, B cells should have great potential as a cellular therapy for targeting certain diseased cell types and expressing therapeutic proteins. There thus exists a need for alternative treatments, such as genetically engineered B cells, for the treatment of a variety of diseases and disorders, including cancer, heart disease, inflammatory disease, muscle wasting disease, neurological disease, and the like. Modifying B cells for the treatment of various diseases, however, is a technique that has not been extensively studied, despite the critical role of B cells in immune responses. Human B cells are easily isolated and can be expanded, making them viable candidates for engineering. Importantly, B cells can be matured to long-lived cells that are ideal for provision of therapeutic proteins that are required for extended periods. Indeed, injected B cells can traffic to their normal tissue niches including spleen, lymph nodes and the bone marrow where they can persist. Ex vivo modification is desirable to avoid in vivo use of recombinant viruses to which immune responses can inactivate. Collectively, B cells are ideal as therapeutic delivery vehicles.

SUMMARY OF THE INVENTION

[0004] In various embodiments, the invention disclosed herein relates to genetically engineering B cells to express a therapeutic protein.

[0005] In various embodiments, the invention relates to a population of cells comprising engineered human B cells, wherein the engineered human B cells comprise a therapeutic protein, whose gene has been inserted into the β2M locus.

[0006] In various embodiments, the engineered human B cells further comprise a disrupted β2M gene. In various embodiments, the nucleic acid sequence capable of expressing the therapeutic payload has been inserted into exon 2 of the β2M locus. In various embodiments, the nucleic acid sequence capable of expressing the therapeutic payload has been inserted into an intron of the β2M locus, such that β2M expression is maintained at a percentage of greater than 50%. In various embodiments, the therapeutic protein is alpha-galactosidase A (GLA), acid alpha-glucosidase (GAA), phenylalanine hydroxylase (PAH), phenylalanine ammonia-lyase (PAL) or full length or B domain deleted (BDD) FVIII. In various embodiments, the therapeutic protein is selected from the amino acid sequences consisting of SEQ ID NOs. 2-6. In various embodiments, the therapeutic protein is a GPC3 chimeric receptor. In various embodiments, the GPC3 chimeric receptor comprises an amino acid sequence of SEQ ID NO. 16. In various embodiments, the therapeutic protein is a cytokine or a chemokine. In various embodiments, the cytokine is IL-10. In various embodiments, the cytokine comprises an amino acid sequence of SEQ ID NO. 7. In various embodiments, the expression of the endogenous β2M has been reduced by at least 40%. In various embodiments, the endogenous β2M has been reduced by at least 80%. In various embodiments, the population of cells, express said therapeutic proteins.

[0007] In various embodiments, the invention relates to a population of cells comprising engineered human B cells, wherein the engineered human B cells comprise, a disrupted β2M gene; and a therapeutic protein, whose gene has been inserted into the β2M locus, wherein the therapeutic protein selected from the amino acid sequences consisting of SEQ ID NOs. 2-7, wherein a nucleic acid sequence capable of expressing the therapeutic payload has been inserted into exon 2 of the β2M locus, wherein expression of the endogenous β2M has been reduced by at least 40%; and wherein at least 20% of the population of cells, express said therapeutic proteins.

[0008] In various embodiments, the invention relates to a method of producing an engineered B cell expressing a therapeutic protein, the method comprising delivering to a human B cell, an RNA-guided nuclease, a gRNA targeting the β2M gene, a construct comprising a nucleic acid sequence encoding a therapeutic protein.

[0009] In various embodiments, the RNA-guided nuclease and gRNA targeting the β2M gene are delivered to the B cell as an RNP. In various embodiments, the RNA-guided nuclease and gRNA targeting the β2M gene are delivered to the B cell as a nanoparticle. In various embodiments, the RNA-guided nuclease and gRNA targeting the β2M gene are delivered to the B cell via electroporation. In various embodiments, the construct delivered to the B cell using a viral vector. In various embodiments, the construct delivered to the B cell as DNA. In various embodiments, the RNA-guided nuclease comprises the nucleotide sequence of SEQ ID NO. 18. In various embodiments, the gRNA comprises the nucleic acid sequence of SEQ ID NO. 19. In various embodiments, the gRNA specifically targets exon 2 of the B2M locus. In various embodiments, the gRNA specifically targets an intron of the B2M locus. In various embodiments, the β2M expression is maintained at a percentage of greater than 50%. In various embodiments, the targeting construct comprises a codon-optimized nucleic acid sequence selected from the group consisting of SEQ ID NOs. 10-17 and 31. In various embodiments, the construct comprises a left homology arm of SEQ ID NO. 20 and a right homology arm of SEQ ID NO. 21. In various embodiments, the expression of the endogenous β2M has been reduced by at least 40%. In various embodiments, the expression of the endogenous β2M has been reduced by at least 80%. In various embodiments, the at least 20% of the engineered B cells, express said therapeutic protein.

[0010] In various embodiments, the invention relates to a method of producing an engineered B cell expressing a therapeutic protein, the method comprising delivering to a human B cell, comprising a RNA-guided nuclease, wherein the RNA-guided nuclease comprises the amino acid sequence of SEQ ID NO. 18; a gRNA targeting the B2M gene, wherein the gRNA comprises the nucleic acid sequence of SEQ ID NO. 19; a construct comprising a nucleic acid sequence encoding a therapeutic protein; wherein the construct comprises a nucleic acid sequence selected from the group consisting of SEQ ID NOs. 10-17 and 31; wherein the construct further comprises a left homology arm of SEQ ID NO. 22 and a right homology arm of SEQ ID NO. 21, wherein expression of the endogenous B2M has been reduced by at least 40%; and wherein at least 20% of the engineered B cells, express said therapeutic protein.

[0011] In various embodiments, the invention relates to a method of treating a patient in need thereof, by administering to said patient a population of cells comprising engineered human B cells, wherein the engineered human B cells comprise a therapeutic payload, whose gene has been inserted into the β2M locus.

[0012] In various embodiments, the engineered human B cells further comprise a disrupted β2M gene. In various embodiments, the nucleic acid sequence capable of expressing the therapeutic payload has been inserted into exon 2 of the β2M locus. In various embodiments, the nucleic acid sequence capable of expressing the therapeutic payload has been inserted into an intron of the β2M locus, such that β2M expression is maintained at a percentage of greater than 50%. In various embodiments, the nucleic acid sequence capable of expressing the therapeutic payload has been inserted into exon 2 of the β2M locus. In various embodiments, the RNA-guided nuclease and gRNA targeting the β2M gene are delivered to the B cell as an RNP. In various embodiments, the RNA-guided nuclease and gRNA targeting the β2M gene are delivered to the B cell as a nanoparticle. In various embodiments, the RNA-guided nuclease and gRNA targeting the β2M gene are delivered to the B cell via electroporation. In various embodiments, the construct delivered to the B cell using a viral vector. In various embodiments, the construct delivered to the B cell as DNA. In various embodiments, the therapeutic protein is alpha-galactosidase A (GLA), acid alpha-glucosidase (GAA), phenylalanine hydroxylase (PAH), phenylalanine ammonia-lyase (PAL) or B domain deleted (BDD) FVIII. In various embodiments, the therapeutic protein selected from the amino acid sequences consisting of SEQ ID NOs. 2-7. In various embodiments, the therapeutic protein is a GPC3 chimeric receptor. In various embodiments, the GPC3 chimeric receptor comprises an amino acid sequence of SEQ ID NO. 16. In various embodiments, the therapeutic protein is a cytokine or a chemokine. In various embodiments, the cytokine is IL-10. In various embodiments, the cytokine is SEQ ID NO. 9. In various embodiments, the expression of the endogenous β2M has been reduced by at least 40%. In various embodiments, the expression of the endogenous β2M has been reduced by at least 80%. In various embodiments, at least 20% of the population of cells, express said therapeutic protein. In various embodiments, the disease or disorder is Fabry disease, Pompe disease, Phenylketonuria (PKU) or Hemophilia A.

[0013] In various embodiments, the invention relates to method of treating a patient in need thereof comprising administering to said patient a population of cells comprising engineered human B cells, wherein the engineered human B cells comprise, a disrupted β2M gene; and a therapeutic payload, whose gene has been inserted into the β2M locus, wherein the therapeutic protein selected from the amino acid sequences consisting of SEQ ID NOs. 2-7; wherein a nucleic acid sequence capable of expressing the therapeutic payload has been inserted into exon 2 of the β2M locus; wherein expression of the endogenous β2M has been reduced by at least 40%; and wherein at least 20% of the population of cells, express said therapeutic proteins.

[0014] In various embodiments, the invention relates to a genome editing system, comprising an RNA-guided nuclease; a gRNA targeting the β2M gene; and a construct comprising a nucleic acid sequence encoding a therapeutic protein.

[0015] In various embodiments, the RNA-guided nuclease comprises the amino acid sequence of SEQ ID NO. 18. In various embodiments, the gRNA comprises the nucleic acid sequence of SEQ ID NO. 19. In various embodiments, the construct comprises a nucleic acid sequence selected from the group consisting of SEQ ID NOs. 10-17 and 31. In various embodiments, the construct comprises a left homology arm of SEQ ID NO. 21 and a right homology arm of SEQ ID NO. 22. In various embodiments, the expression of the endogenous β2M has been reduced by at least 40%. In various embodiments, the expression of the endogenous β2M has been reduced by at least 80%. In various embodiments, at least 20% of the engineered B cells, express said therapeutic protein.

[0016] In various embodiments, the invention relates to a genome editing system, comprising, a RNA-guided nuclease, wherein the RNA-guided nuclease comprises the amino acid sequence of SEQ ID NO. 8; a gRNA targeting the β2M gene, wherein the gRNA comprises the nucleic acid sequence of SEQ ID NO. 9; and a construct comprising a nucleic acid sequence encoding a therapeutic protein; wherein the construct comprises a nucleic acid sequence selected from the group consisting of SEQ ID NOs. 10-16; wherein the construct further comprises a left homology arm of SEQ ID NO. 17 and a right homology arm of SEQ ID NO. 18; wherein expression of the endogenous β2M has been reduced by at least 40%; and wherein at least 20% of the engineered B cells, express said therapeutic protein.

[0017] In various embodiments, the invention relates to an engineered B cell comprising a nucleic acid sequence capable of expressing an amino acid sequence comprising SEQ ID NO. 2. In various embodiments, the invention relates to an engineered B cell comprising a nucleic acid sequence capable of expressing an amino acid sequence comprising SEQ ID NO. 3. In various embodiments, the invention relates to an engineered B cell comprising a nucleic acid sequence capable of expressing an amino acid sequence comprising SEQ ID NO. 4. In various embodiments, the invention relates to an engineered B cell comprising a nucleic acid sequence capable of expressing an amino acid sequence comprising SEQ ID NO. 5. In various embodiments, the invention relates to an engineered B cell comprising a nucleic acid sequence capable of expressing an amino acid sequence comprising SEQ ID NO. 6. In various embodiments, the invention relates to an engineered B cell comprising a nucleic acid sequence capable of expressing an amino acid sequence comprising SEQ ID NO. 7.

[0018] In various embodiments, the invention relates to a nucleic acid construct capable of insertion into the β2M locus of a B cell, comprising the nucleic acid sequence of SEQ ID NO. 10. In various embodiments, the invention relates to a nucleic acid construct capable of insertion into the β2M locus of a B cell, comprising the nucleic acid sequence of SEQ ID NO. 11. In various embodiments, the invention relates to a nucleic acid construct capable of insertion into the β2M locus of a B cell, comprising the nucleic acid sequence of SEQ ID NO. 12. In various embodiments, the invention relates to a nucleic acid construct capable of insertion into the β2M locus of a B cell, comprising the nucleic acid sequence of SEQ ID NO. 13. In various embodiments, the invention relates to a nucleic acid construct capable of insertion into the β2M locus of a B cell, comprising the nucleic acid sequence of SEQ ID NO. 14. In various embodiments, the invention relates to a nucleic acid construct capable of insertion into the β2M locus of a B cell, comprising the nucleic acid sequence of SEQ ID NO. 15. In various embodiments, the invention relates to a nucleic acid construct capable of insertion into the β2M locus of a B cell, comprising the nucleic acid sequence of SEQ ID NO. 16. In various embodiments, the invention relates to a nucleic acid construct capable of insertion into the β2M locus of a B cell, comprising the nucleic acid sequence of SEQ ID NO. 17. In various embodiments, the invention relates to a nucleic acid construct capable of insertion into the β2M locus of a B cell, comprising the nucleic acid sequence of SEQ ID NO. 31.

BRIEF DESCRIPTION OF THE FIGURES

[0019] FIG. **1** shows a schematic of human B cell preparation for CRISPR engineering: isolation, activation and expansion. PBMCs were isolated from buffy coats using Ficoll-Paque. Primary human B cells were isolated using the EASYSEP™ Human B Cell Isolation Kit. Isolated B cells were activated and expanded using the human B Cell Expansion Kit for over 9 days. Harvested B cells were engineered with nucleofection using AMAXA™ 4D-NUCLEOFACTOR™. Engineered B cells were cultured and analyzed by PCR and flow cytometry.

[0020] FIGS. **2**A-**2**B show optimal human B cell nucleofection protocol development. FIG. **2**A shows the screen of indicated electroporation programs for optimal human B cell nucleofection using Amaxa 4D. 1 μg pMAX-GFP was used for each condition with 1 million activated human B cells in buffer P3. Nuclefection efficiency was determined by GFP expression using flow cytometry. FIG. **2**B shows a table summary of nucleofection efficiency and cell viability for each electroporation program. Program CM-137 was selected as the optimal electroporation program.

[0021] FIG. **3** shows a schematic of the design of the CRISPR guide RNA for engineering of human β2M locus. CRISPR guide sequence CGTGAGTAAACCTGAATCTT (SEQ ID NO: 34) was selected to target the beginning of exon 2 of the human β2M gene to efficiently knock-out express of functional β2M protein. Phosphorothioated 2' O-Methyl modified single guide RNA (sgRNA) oligomer was synthesized by IDT to improve CRISPR efficiency. Figure discloses SEQ ID No: 24.

[0022] FIGS. **4**A-**4**C show optimal human B cell CRISPR editing protocol development. FIG. **4**A shows a schematic of RNP formation for CRISPR gene editing. Cas9 enzyme was incubated with sgRNA at a 1:1.2 ratio for 10 minutes at room temperature to form RNP. 100 pmol RNA was used for 1 million B cells in a 20 μL reaction. FIG. **4**B shows flow profiles of control and β2M CRISPR edited B cells. Successful β2M CRISPR editing resulted in loss of β2M expression on the cell surface, which was determined by flow cytometry. Overlay of the profiles illustrated efficient knock-out of β2M. FIG. **4**C shows a table summary of β2M knock-out efficiency and cell viability for each electroporation program. Program CM-137 was selected as the optimal β2M knock-out program.

[0023] FIGS. **5**A-**5**B show validation of β2M KO frequency at the genomic level. FIG. **5**A shows human B cells were isolated from PBMCs (1 healthy donor) and electroporated with WT-Cas9 complex with β2M sgRNA. Two days after targeting, genomic DNA was extracted Sanger sequencing was used to quantify INDELs. FIG. **5**B shows an overview of the insertions and deletions generated at the cut site of the β2M sgRNA. Analysis was performed using ICE synthego (ice.synthego.com) web-based software. (SEQ ID NOs: 34-50, respectively, in order of appearance.)

[0024] FIG. **6** shows a schematic of the design of promoter-less β2M targeting constructs, based on the following principles: 1) only correctly targeted alleles express the transgene (GFP or GPC3-CAR), 2) high level constitutive expression of transgene driven from endogenous β2M promoter, and 3) loss of β2M expression on engineered B cells allows for easy detection of successful editing.

[0025] FIG. **7** shows a schematic of β2M gene pre- and post-editing where GFP expression is driven from the endogenous β2M promoter.

[0026] FIGS. **8**A-**8**C show human B-cell expansion and β2M editing protocol. FIG. **8**A shows a schematic of the B cells editing procedure. Human B cells were isolated from healthy donor PBMCs and expanded in presence of CD40L and IL-4 for 9 days. Expanded cells were electroporated with RNP (WT-Cas9 and β2M sgRNA) and transduced with AAV6 to deliver GFP and GPC3-CAR HDR-donor cassettes. FIG. **8**B shows a growth curve of cultured human B cells. FIG. **8**C shows the viability of cultured human B cells over the course of expansion.

[0027] FIGS. **9**A-**9**D show the efficient AAV6-mediated integration at the β2M locus in activated human B cells. To determine the rates of targeted integration of the GFP and the GPC3-CAR promoter-less HDR-donors, expression of GFP (FIG. **9**A) and GPC3-CAR (FIG. **9**B) in engineered B cells was evaluated using flow cytometry at 3 or 6 days after editing. Using an MOI of 100K, AAV6 mediated targeting efficiency was >40% for GFP or aGPC3-CAR at β2M locus. Viability of GFP (FIG. **9**C) and GPC3-CAR (FIG. **9**D) engineered B cells was measured using Trypan Blue at 3 or 6 days after editing.

[0028] FIGS. **10**A-**10**B show the results of flow cytometry analysis after using PCR dsDNA as an HDR template for CRISPR-mediated targeting of human B cells at β2M locus. GFP dsDNA PCR product was used as an HDR template for CRISPR engineering of human B cells in combination with Cas9 RNP. Three μg of donor DNA resulted in 15% targeting efficiency as determined by flow cytometry analysis of GFP expression. Flow cytometry analysis is shown without (FIG. **10**A) and with RNP electroporation (FIG. **10**B).

[0029] FIG. **11** shows schematic of the design of CRISPR-mediated β2M locus targeting for the stable expression of enzymes for replacement therapy.

[0030] FIG. **12** shows a schematic of the design of CRISPR-mediated β2M locus targeting for the stable expression of cytokines as payload.

[0031] FIG. 13 shows a schematic of the design of CRISPR-mediated β2M locus targeting for the stable expression of wild type GLA as payload.

[0032] FIG. 14 shows secreted (FIG. 14A) and intracellular (FIG. 14B) GLA expression in B cells engineered using Cas9-rAAV to express wild type GLA. The targeting loci was exon 2 of the β2M locus and GLA expression was driven by the endogenous β2M promoter.

DETAILED DESCRIPTION

[0033] The present disclosure provides an efficient gene method for transgene integration into the β2M locus for cell therapy. The present disclosure is based, at least in part, on the discovery that insertion of a therapeutic protein into the β2M locus in B cells using gene editing technologies enhances several characteristics important for cell-based immunotherapy. For example, targeted expression of a therapeutic protein from the β2M locus takes advantage of the high basal level of β2M expression in B cells to achieve a high level and ubiquitous expression of the transgene/therapeutic protein across different B cell types independent of developmental stage and activation status.

[0034] Disclosed herein are a number of constructs for insertion into the β2M locus. The invention disclosed herein would be suitable for any number of therapies that require delivery or replacement of a therapeutic protein such as a therapeutic enzyme, an antibody, a cytokine a selection marker, a suicide gene, etc.

[0035] In certain embodiments, the optimized gene editing methods deliver the transgene, which is inserted into an exon of the β2M gene. Such methods are capable of achieving a greater than 50% knockout of the endogenous β2M gene in human B cells. In other embodiments of the present disclosure, the transgene is inserted using gene editing methods into an intron of the β2M gene, such that β2M gene expression is not disrupted or is only minimally disrupted.

[0036] The present disclose is capable of achieving over 50% targeted integration efficiency.

I. Definitions

[0037] The section headings used herein are for organizational purposes only and are not to be construed as limiting the subject matter described. All documents, or portions of documents, cited in this application, including but not limited to patents, patent applications, articles, books, and treatises, are hereby expressly incorporated by reference in their entirety for any purpose. As utilized in accordance with the present disclosure, the following terms, unless otherwise indicated, shall be understood to have the following meanings:

[0038] In this application, the use of "or" means "and/or" unless stated otherwise. Furthermore, the use of the term "including", as well as other forms, such as "includes" and "included", is not limiting. Also, terms such as "element" or "component" encompass both elements and components comprising one unit and elements and components that comprise more than one subunit unless specifically stated otherwise.

[0039] The term "polynucleotide", "nucleotide", or "nucleic acid" includes both single-stranded and double-stranded nucleotide polymers. The nucleotides comprising the polynucleotide can be ribonucleotides or deoxyribonucleotides or a modified form of either type of nucleotide. Said modifications include base modifications such as bromouridine and inosine derivatives, ribose modifications such as 2', 3'-dideoxyribose, and internucleotide linkage modifications such as phosphorothioate, phosphorodithioate, phosphoroselenoate, phosphoro-diselenoate, phosphoro-anilothioate, phoshoraniladate and phosphoroamidate.

[0040] The term "oligonucleotide" refers to a polynucleotide comprising 200 or fewer nucleotides. Oligonucleotides can be single stranded or double stranded, e.g., for use in the construction of a mutant gene. Oligonucleotides can be sense or antisense oligonucleotides. An oligonucleotide can include a label, including a radiolabel, a fluorescent label, a hapten or an antigenic label, for detection assays. Oligonucleotides can be used, for example, as PCR primers, cloning primers or hybridization probes.

[0041] The term "control sequence" refers to a polynucleotide sequence that can affect the expression and processing of coding sequences to which it is ligated. The nature of such control sequences can depend upon the host organism. In particular embodiments, control sequences for prokaryotes can include a promoter, a ribosomal binding site, and a transcription termination sequence. For example, control sequences for eukaryotes can include promoters comprising one or a plurality of recognition sites for transcription factors, transcription enhancer sequences, and transcription termination sequence. Control sequences can include leader sequences (signal peptides) and/or fusion partner sequences.

[0042] As used herein, "operably linked" means that the components to which the term is applied are in a relationship that allows them to carry out their inherent functions under suitable conditions.

[0043] The term "vector" means any molecule or entity (e.g., nucleic acid, plasmid, bacteriophage or virus) used to transfer protein coding information into a host cell. The term "expression vector" or "expression construct" refers to a vector that is suitable for transformation of a host cell and contains nucleic acid sequences that direct and/or control (in conjunction with the host cell) expression of one or more heterologous coding regions operatively linked thereto. An expression construct can include, but is not limited to, sequences that affect or control transcription, translation, and, if introns are present, affect RNA splicing of a coding region operably linked thereto.

[0044] The term "host cell" refers to a cell that has been transformed, or is capable of being transformed, with a nucleic acid sequence and thereby expresses a gene of interest. The term includes the progeny of the parent cell, whether or not the progeny is identical in morphology or in genetic make-up to the original parent cell, so long as the gene of interest is present.

[0045] The term "transformation" refers to a change in a cell's genetic characteristics, and a cell has been transformed when it has been modified to contain new DNA or RNA. For example, a cell is transformed where it is genetically modified from its native state by introducing new genetic material via transfection, transduction, or other techniques. Following transfection or transduction, the transforming DNA can recombine with that of the cell by physically integrating into a chromosome of the cell, or can be maintained transiently as an episomal element without being replicated, or can replicate independently as a plasmid. A cell is considered to have been "stably transformed" when the transforming DNA is replicated with the division of the cell.

[0046] The term "transfection" refers to the uptake of foreign or exogenous DNA by a cell. A number of transfection techniques are well known in the art and are disclosed herein. See, e.g., Graham et al., 1973, *Virology,* 1973, 52:456; Sambrook et al., Molecular Cloning: A Laboratory Manual, 2001, supra; Davis et al., Basic Methods in Molecular Biology, 1986, Elsevier; Chu et al., 1981, Gene, 13:197.

[0047] The term "transduction" refers to the process whereby foreign DNA is introduced into a cell via viral vector. See, e.g., Jones et al., Genetics: Principles and Analysis, 1998, Boston: Jones & Bartlett Publ.

[0048] The terms "polypeptide" or "protein" refer to a macromolecule having the amino acid sequence of a protein, including deletions from, additions to, and/or substitutions of one or more amino acids of the native sequence. The terms "polypeptide" and "protein" specifically encompass antigen-binding molecules, antibodies, or sequences that have deletions from, additions to, and/or substitutions of one or more amino acid of antigen-binding protein. The term "polypeptide fragment" refers to a polypeptide that has an amino-terminal deletion, a carboxyl-terminal deletion, and/or an internal deletion as compared with the full-length native protein. Such fragments can also contain modified amino acids as compared with the native protein. Useful polypeptide fragments include immunologically functional fragments of antigen-binding molecules.

[0049] The term "isolated" means (i) free of at least some other proteins with which it would normally be found, (ii) is essentially free of other proteins from the same source, e.g., from the same species, (iii) separated from at least about 50 percent of polynucleotides, lipids, carbohydrates, or other materials with which it is associated in nature, (iv) operably associated (by covalent or noncovalent interaction) with a polypeptide with which it is not associated in nature, or (v) does not occur in nature.

[0050] A "variant" of a polypeptide (e.g., an antigen-binding molecule) comprises an amino acid sequence wherein one or more amino acid residues are inserted into, deleted from and/or substituted into the amino acid sequence relative to another polypeptide sequence. Variants include fusion proteins.

[0051] The term "identity" refers to a relationship between the sequences of two or more polypeptide molecules or two or more nucleic acid molecules, as determined by aligning and comparing the sequences. "Percent identity" means the percent of identical residues between the amino acids or nucleotides in the compared molecules and is calculated based on the size of the smallest of the molecules being compared. For these calculations, gaps in alignments (if any) are preferably addressed by a particular mathematical model or computer program (i.e., an "algorithm").

[0052] To calculate percent identity, the sequences being compared are typically aligned in a way that gives the largest match between the sequences. One example of a computer program that can be used to determine percent identity is the GCG program package, which includes GAP (Devereux et al., *Nucl. Acid Res.,* 1984, 12, 387; Genetics Computer Group, University of Wisconsin, Madison, Wis.). The computer algorithm GAP is used to align the two polypeptides or polynucleotides for which the percent sequence identity is to be determined. The sequences are aligned for optimal matching of their respective amino acid or nucleotide (the "matched span", as determined by the algorithm). In certain embodiments, a standard comparison matrix (see, e.g., Dayhoff et al., 1978, Atlas of Protein Sequence and Structure, 5:345-352 for the PAM 250 comparison matrix; Henikoff et al., 1992, *Proc. Natl. Acad. Sci. U.S.A.,* 89, 10915-10919 for the BLO-SUM 62 comparison matrix) is also used by the algorithm.

[0053] As used herein, the twenty conventional (e.g., naturally occurring) amino acids and their abbreviations follow conventional usage. See, e.g., Immunology A Synthesis (2nd Edition, Golub and Green, Eds., Sinauer Assoc., Sunderland, Mass. (1991)), which is incorporated herein by reference for any purpose. Stereoisomers (e.g., D-amino acids) of the twenty conventional amino acids, unnatural amino acids such as alpha-, alpha-disubstituted amino acids, N-alkyl amino acids, lactic acid, and other unconventional amino acids can also be suitable components for polypeptides of the present invention. Examples of unconventional amino acids include: 4-hydroxyproline, gamma.-carboxyglutamate, epsilon-N,N,N-trimethyllysine, e-N-acetyllysine, 0-phosphoserine, N-acetylserine, N-formylmethionine, 3-methylhistidine, 5-hydroxylysine, sigma.-N-methylarginine, and other similar amino acids and imino acids (e.g., 4-hydroxyproline). In the polypeptide notation used herein, the left-hand direction is the amino terminal direction and the right-hand direction is the carboxy-terminal direction, in accordance with standard usage and convention.

[0054] Conservative amino acid substitutions can encompass non-naturally occurring amino acid residues, which are typically incorporated by chemical peptide synthesis rather than by synthesis in biological systems. These include peptidomimetics and other reversed or inverted forms of amino acid moieties. Naturally occurring residues can be divided into classes based on common side chain properties:

[0055] a) hydrophobic: norleucine, Met, Ala, Val, Leu, Ile;

[0056] b) neutral hydrophilic: Cys, Ser, Thr, Asn, Gln;

[0057] c) acidic: Asp, Glu;

[0058] d) basic: His, Lys, Arg;

[0059] e) residues that influence chain orientation: Gly, Pro; and

[0060] f) aromatic: Trp, Tyr, Phe.

[0061] For example, non-conservative substitutions can involve the exchange of a member of one of these classes for a member from another class.

[0062] In making changes to the antigen-binding molecule, the costimulatory or activating domains of the engineered T cell, according to certain embodiments, the hydropathic index of amino acids can be considered. Each amino acid has been assigned a hydropathic index on the basis of its hydrophobicity and charge characteristics. They are: isoleucine (+4.5); valine (+4.2); leucine (+3.8); phenylalanine (+2.8); cysteine/cystine (+2.5); methionine (+1.9); alanine (+1.8); glycine (−0.4); threonine (−0.7); serine (−0.8); tryptophan (−0.9); tyrosine (−1.3); proline (−1.6); histidine (−3.2); glutamate (−3.5); glutamine (−3.5); aspartate (−3.5); asparagine (−3.5); lysine (−3.9); and arginine (−4.5). See, e.g., Kyte et al., 1982, *J. Mol. Biol.,* 157, 105-131. It is known that certain amino acids can be substituted for other amino acids having a similar hydropathic index or score and still retain a similar biological activity. It is also understood in the art that the substitution of like amino acids can be made effectively on the basis of hydrophilicity, particularly where the biologically functional protein or peptide thereby created is intended for use in immunological embodiments, as in the present case. Exemplary amino acid substitutions are set forth in Table 1.

TABLE 1

Original Residues	Exemplary Substitutions	Preferred Substitutions
Ala	Val, Leu, Ile	Val
Arg	Lys, Gin, Asn	Lys
Asn	Gln	Gln
Asp	Glu	Glu
Cys	Ser, Ala	Ser
Gln	Asn	Asn
Glu	Asp	Asp
Gly	Pro, Ala	Ala
His	Asn, Gln, Lys, Arg	Arg
Ile	Leu, Val, Met, Ala, Phe, Norleucine	Leu
Leu	Norleucine, Ile, Va, Met, Ala, Phe	Ile
Lys	Arg, 1,4 Diamino-butyric Acid, Gin, Asn	Arg
Met	Leu, Phe, Ile	Leu
Phe	Leu, Val, Ile, Ala, Tyr	Leu
Pro	Ala	Gly
Ser	Thr, Ala, Cys	Thr
Thr	Ser	Ser
Trp	Tyr, Phe	Tyr
Tyr	Trp, Phe, Thr, Ser	Phe
Val	Ile, Met, Leu, Phe, Ala, Norleucine	Leu

[0063] The term "derivative" refers to a molecule that includes a chemical modification other than an insertion, deletion, or substitution of amino acids (or nucleic acids). In certain embodiments, derivatives comprise covalent modifications, including, but not limited to, chemical bonding with polymers, lipids, or other organic or inorganic moieties. In certain embodiments, a chemically modified antigen-binding molecule can have a greater circulating half-life than an antigen-binding molecule that is not chemically modified. In some embodiments, a derivative antigen-binding molecule is covalently modified to include one or more water-soluble polymer attachments, including, but not limited to, polyethylene glycol, polyoxyethylene glycol, or polypropylene glycol.

[0064] Peptide analogs are commonly used in the pharmaceutical industry as non-peptide drugs with properties analogous to those of the template peptide. These types of non-peptide compound are termed "peptide mimetics" or "peptidomimetics." Fauchere, J. L., 1986, Adv. Drug Res., 1986, 15, 29; Veber, D. F. & Freidinger, R. M., 1985, Trends in Neuroscience, 8, 392-396; and Evans, B. E., et al., 1987, J Med. Chem., 30, 1229-1239, which are incorporated herein by reference for any purpose.

[0065] The term "therapeutically effective amount" refers to the amount of immune cells or other therapeutic agent determined to produce a therapeutic response in a mammal. Such therapeutically effective amounts are readily ascertained by one of ordinary skill in the art.

[0066] The terms "patient" and "subject" are used interchangeably and include human and non-human animal subjects as well as those with formally diagnosed disorders, those without formally recognized disorders, those receiving medical attention, those at risk of developing the disorders, etc.

[0067] The term "treat" and "treatment" includes therapeutic treatments, prophylactic treatments, and applications in which one reduces the risk that a subject will develop a disorder or other risk factor. Treatment does not require the complete curing of a disorder and encompasses embodiments in which one reduces symptoms or underlying risk factors. The term "prevent" does not require the 100% elimination of the possibility of an event. Rather, it denotes that the likelihood of the occurrence of the event has been reduced in the presence of the compound or method.

[0068] Standard techniques can be used for recombinant DNA, oligonucleotide synthesis, and tissue culture and transformation (e.g., electroporation, lipofection). Enzymatic reactions and purification techniques can be performed according to manufacturer's specifications or as commonly accomplished in the art or as described herein. The foregoing techniques and procedures can be generally performed according to conventional methods well known in the art and as described in various general and more specific references that are cited and discussed throughout the present specification. See, e.g., Sambrook et al., Molecular Cloning: A Laboratory Manual (2d ed., Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N.Y. (1989)), which is incorporated herein by reference for any purpose.

[0069] As used herein, the term "substantially" or "essentially" refers to a quantity, level, value, number, frequency, percentage, dimension, size, amount, weight or length that is about 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% higher compared to a reference quantity, level, value, number, frequency, percentage, dimension, size, amount, weight or length. In one embodiment, the terms "essentially the same" or "substantially the same" refer to a range of quantity, level, value, number, frequency, percentage, dimension, size, amount, weight or length that is about the same as a reference quantity, level, value, number, frequency, percentage, dimension, size, amount, weight or length.

[0070] As used herein, the terms "substantially free of" and "essentially free of" are used interchangeably, and when used to describe a composition, such as a cell population or culture media, refer to a composition that is free of a specified substance, such as, 95% free, 96% free, 97% free, 98% free, 99% free of the specified substance, or is undetectable as measured by conventional means. Similar meaning can be applied to the term "absence of," where referring to the absence of a particular substance or component of a composition.

[0071] As used herein, the term "appreciable" refers to a range of quantity, level, value, number, frequency, percentage, dimension, size, amount, weight or length or an event that is readily detectable by one or more standard methods. The terms "not-appreciable" and "not appreciable" and equivalents refer to a range of quantity, level, value, number, frequency, percentage, dimension, size, amount, weight or length or an event that is not readily detectable or undetectable by standard methods. In one embodiment, an event is not appreciable if it occurs less than 5%, 4%, 3%, 2%, 1%, 0.1%, 0.001%, or less of the time.

[0072] Throughout this specification, unless the context requires otherwise, the words "comprise," "comprises" and "comprising" will be understood to imply the inclusion of stated step or element or group of steps or elements but not the exclusion of any other step or element or group of steps or elements. In particular embodiments, the terms "include," "has," "contains," and "comprise" are used synonymously.

[0073] As used herein, "consisting of" is meant including, and limited to, whatever follows the phrase "consisting of".

Thus, the phrase "consisting of" indicates that the listed elements are required or mandatory, and that no other elements may be present.

[0074] By "consisting essentially of" is meant including any elements listed after the phrase, and limited to other elements that do not interfere with or contribute to the activity or action specified in the disclosure for the listed elements. Thus, the phrase "consisting essentially of" indicates that the listed elements are required or mandatory, but that no other elements are optional and may or may not be present depending upon whether or not they affect the activity or action of the listed elements.

[0075] Reference throughout this specification to "one embodiment," "an embodiment," "a particular embodiment," "a related embodiment," "a certain embodiment," "an additional embodiment," or "a further embodiment" or combinations thereof means that a particular feature, structure or characteristic described in connection with the embodiment is included in at least one embodiment of the present invention. Thus, the appearances of the foregoing phrases in various places throughout this specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, or characteristics may be combined in any suitable manner in one or more embodiments.

[0076] As used herein, the term "about" or "approximately" refers to a quantity, level, value, number, frequency, percentage, dimension, size, amount, weight or length that varies by as much as 30, 25, 20, 15, 10, 9, 8, 7, 6, 5, 4, 3, 2 or 1% to a reference quantity, level, value, number, frequency, percentage, dimension, size, amount, weight or length. In particular embodiments, the terms "about" or "approximately" when preceding a numerical value indicates the value plus or minus a range of 15%, 10%, 5% or 1%, or any intervening ranges thereof.

[0077] As used herein, the term "introducing" refers to a process that comprises contacting a cell with a polynucleotide, polypeptide, or small molecule. An introducing step may also comprise microinjection of polynucleotides or polypeptides into the cell, use of liposomes to deliver polynucleotides or polypeptides into the cell, or fusion of polynucleotides or polypeptides to cell permeable moieties to introduce them into a cell.

II. Gene Editing Methods

[0078] Gene editing (including genomic editing) is a type of genetic engineering in which nucleotide(s)/nucleic acid(s) is/are inserted, deleted, and/or substituted in a DNA sequence, such as in the genome of a targeted cell. Targeted gene editing enables insertion, deletion and/or substitution at pre-selected sites in the genome of a targeted cell (e.g., in a targeted gene or targeted DNA sequence). When a sequence of an endogenous gene is edited, for example by deletion, insertion or substitution of nucleotide(s)/nucleic acid(s), the endogenous gene comprising the affected sequence may be knocked-out or knocked-down due to the sequence alteration. Therefore, targeted editing may be used to disrupt endogenous gene expression. "Targeted integration" refers to a process involving insertion of one or more exogenous sequences, with or without deletion of an endogenous sequence at the insertion site. Targeted integration can result from targeted gene editing when a donor template containing an exogenous sequence is present. As used herein, a "disrupted gene" refers to a gene comprising an insertion, deletion, or substitution relative to an endogenous gene such that expression of a functional protein from the endogenous gene is reduced or inhibited. As used herein, "disrupting a gene" refers to a method of inserting, deleting or substituting at least one nucleotide/nucleic acid in an endogenous gene such that expression of a functional protein from the endogenous gene is reduced or inhibited. Methods of disrupting a gene are known to those of skill in the art and described herein.

[0079] Targeted editing can be achieved either through a nuclease-independent approach, or through a nuclease-dependent approach. In the nuclease-independent targeted editing approach, homologous recombination is guided by homologous sequences flanking an exogenous polynucleotide to be introduced into an endogenous sequence through the enzymatic machinery of the host cell. The exogenous polynucleotide may introduce deletions, insertions or replacement of nucleotides in the endogenous sequence.

[0080] Alternatively, the nuclease-dependent approach can achieve targeted editing with higher frequency through the specific introduction of double strand breaks (DSBs) by specific rare-cutting nucleases (e.g., endonucleases). Such nuclease-dependent targeted editing also utilizes DNA repair mechanisms, for example, non-homologous end joining (NHEJ), which occurs in response to DSBs. DNA repair by NHEJ often leads to random insertions or deletions (indels) of a small number of endogenous nucleotides. In contrast to NHEJ mediated repair, repair can also occur by a homology directed repair (HDR). When a donor template containing exogenous genetic material flanked by a pair of homology arms is present, the exogenous genetic material can be introduced into the genome by HDR, which results in targeted integration of the exogenous genetic material.

[0081] Available endonucleases capable of introducing specific and targeted DSBs include, but are not limited to, zinc-finger nucleases (ZFN), meganucleases, transcription activator-like effector nucleases (TALEN), and RNA-guided CRISPR Cas9 nuclease (CRISPR/Cas9; Clustered Regular Interspaced Short Palindromic Repeats Associated 9). Additionally, DICE (dual integrase cassette exchange) system utilizing phiC31 and Bxb1 integrases may also be used for targeted integration.

[0082] ZFNs are targeted nucleases comprising a nuclease fused to a zinc finger DNA binding domain (ZFBD), which is a polypeptide domain that binds DNA in a sequence specific manner through on more zinc fingers. A zinc finger is a domain of about 30 amino acids within the zinc finger-binding domain whose structure is stabilized through coordination of a zinc ion. Examples of zinc fingers include, but not limited to, C2H2 zinc fingers, C3H zinc fingers, and C4 zinc fingers. A designed zinc finger domain is a domain not occurring in nature whose design/composition results principally from rational criteria, e.g., application of substitution rules and computerized algorithms for processing information in a database storing information of existing ZFP designs and binding data. See, for example, U.S. Pat. Nos. 6,140,081; 6,453,242; and 6,534,261; see also WO98/53058; WO 98/53059; WO 98/53060; WO 02/016536 and WO 03/016496. A selected zinc finger domain is a domain not found in nature whose production results primarily from an empirical process such as phage display, interaction trap or hybrid selection. ZFNs are described in greater detail in U.S. Pat. Nos. 7,888,121 and 7,972,854. The most recognized example of a ZFN is a fusion of the Fok1 nuclease with a zinc finger DNA binding domain.

[0083] A TALEN is a targeted nuclease comprising a nuclease fused to a TAL effector DNA binding domain. A "transcription activator-like effector DNA binding domain", "TAL effector DNA binding domain", or "TALE DNA binding domain" is a polypeptide domain of TAL effector proteins that is responsible for binding of the TAL effector protein to DNA. TAL effector proteins are secreted by plant pathogens of the genus *Xanthomonas* during infection. These proteins enter the nucleus of the plant cell, bind effector-specific DNA sequences via their DNA binding domain, and activate gene transcription at these sequences via their transactivation domains. TAL effector DNA binding domain specificity depends on an effector-variable number of imperfect 34 amino acid repeats, which comprise polymorphisms at select repeat positions called repeat variable diresidues (RVD). TALENs are described in greater detail in US Patent Application No. 2011/0145940. The most recognized example of a TALEN in the art is a fusion polypeptide of the Fok1 nuclease to a TAL effector DNA binding domain.

[0084] Additional examples of targeted nucleases suitable for use as provided herein include, but are not limited to, Bxb1, phiC31, R4, PhiBT1, and WB/SPBc/TP901-1, whether used individually or in combination.

[0085] Other non-limiting examples of targeted nucleases include naturally-occurring and recombinant nucleases, e.g., CRISPR/Cas9, restriction endonucleases, meganucleases homing endonucleases, and the like.

1. CRISPR-Cas9 Gene Editing

[0086] The CRISPR-Cas9 system is a naturally-occurring defense mechanism in prokaryotes that has been repurposed as an RNA-guided DNA-targeting platform used for gene editing. It relies on the DNA nuclease Cas9, and two noncoding RNAs, crisprRNA (CrRNA) and trans-activating RNA (tracrRNA), to target the cleavage of DNA. CRISPR is an abbreviation for Clustered Regularly Interspaced Short Palindromic Repeats, a family of DNA sequences found in the genomes of bacteria and archaea that contain fragments of DNA (spacer DNA) with similarity to foreign DNA previously exposed to the cell, for example, by viruses that have infected or attacked the prokaryote. These fragments of DNA are used by the prokaryote to detect and destroy similar foreign DNA upon reintroduction, for example, from similar viruses during subsequent attacks. Transcription of the CRISPR locus results in the formation of an RNA molecule comprising the spacer sequence, which associates with and targets Cas (CRISPR-associated) proteins able to recognize and cut the foreign, exogenous DNA. Numerous types and classes of CRISPR/Cas systems have been described (see, e.g., Koonin et al., (2017) Curr Opin Microbiol 37:67-78).

[0087] crRNA drives sequence recognition and specificity of the CRISPR-Cas9 complex through Watson-Crick base pairing typically with a 20 nucleotide (nt) sequence in the target DNA. Changing the sequence of the 5' 20 nt in the crRNA allows targeting of the CRISPR-Cas9 complex to specific loci. The CRISPR-Cas9 complex only binds DNA sequences that contain a sequence match to the first 20 nt of the crRNA, single-guide RNA (sgRNA), if the target sequence is followed by a specific short DNA motif (with the sequence NGG) referred to as a protospacer adjacent motif (PAM).

[0088] TracrRNA hybridizes with the 3' end of crRNA to form an RNA-duplex structure that is bound by the Cas9 endonuclease to form the catalytically active CRISPR-Cas9 complex, which can then cleave the target DNA.

[0089] Once the CRISPR-Cas9 complex is bound to DNA at a target site, two independent nuclease domains within the Cas9 enzyme each cleave one of the DNA strands upstream of the PAM site, leaving a double-strand break (DSB) where both strands of the DNA terminate in a base pair (a blunt end).

[0090] After binding of CRISPR-Cas9 complex to DNA at a specific target site and formation of the site-specific DSB, the next key step is repair of the DSB. Cells use two main DNA repair pathways to repair the DSB: non-homologous end-joining (NHEJ) and homology-directed repair (HDR).

[0091] NHEJ is a robust repair mechanism that appears highly active in the majority of cell types, including nondividing cells. NHEJ is error-prone and can often result in the removal or addition of between one and several hundred nucleotides at the site of the DSB, though such modifications are typically <20 nt. The resulting insertions and deletions (indels) can disrupt coding or noncoding regions of genes. Alternatively, HDR uses a long stretch of homologous donor DNA, provided endogenously or exogenously, to repair the DSB with high fidelity. HDR is active only in dividing cells, and occurs at a relatively low frequency in most cell types. In many embodiments of the present disclosure, NHEJ is utilized as the repair operant.

[0092] In some embodiments, the Cas9 (CRISPR associated protein 9) endonuclease is from *Streptococcus pyogenes*, although other Cas9 homologs may be used. It should be understood, that wild-type Cas9 may be used or modified versions of Cas9 may be used (e.g., evolved versions of Cas9, or Cas9 orthologues or variants), as provided herein. In some embodiments, Cas9 may be substituted with another RNA-guided endonuclease, such as Cpf1 (of a class II CRISPR/Cas system).

[0093] In some embodiments, the CRISPR/Cas system comprise components derived from a Type-1, Type-II, or Type-III system. Updated classification schemes for CRISPR/Cas loci define Class 1 and Class 2 CRISPR/Cas systems, having Types I to V or VI (Makarova et al., (2015) Nat Rev Microbiol, 13(11):722-36; Shmakov et al., (2015)) Mol Cell, 60:385-397). Class 2 CRISPR/Cas systems have single protein effectors. Cas proteins of Types II, V, and VI are single-protein, RNA-guided endonucleases, herein called "Class 2 Cas nucleases." Class 2 Cas nucleases include, for example, Cas9, Cpf1, C2c1, C2c2, and C2c3 proteins. The Cpf1 nuclease (Zetsche et al., (2015) Cell 163: 1-13) is homologous to Cas9, and contains a RuvC-like nuclease domain.

[0094] In some embodiments, the Cas nuclease is from a Type-II CRISPR/Cas system (e.g., a Cas9 protein from a CRISPR/Cas9 system). In some embodiments, the Cas nuclease is from a Class 2 CRISPR Cas system (a single protein Cas nuclease such as a Cas9 protein or a Cpf1 protein). The Cas9 and Cpf1 family of proteins are enzymes with DNA endonuclease activity, and they can be directed to cleave a desired nucleic acid target by designing an appropriate guide RNA, as described further herein.

[0095] In some embodiments, a Cas nuclease may comprise more than one nuclease domain. For example, a Cas9 nuclease may comprise at least one RuvC-like nuclease domain (e.g., Cpf1) and at least one HNH-like nuclease domain (e.g., Cas9). In some embodiments, the Cas9 nuclease introduces a DSB in the target sequence. In some embodiments, the Cas9 nuclease is modified to contain only one functional nuclease domain. For example, the Cas9 nuclease is modified such that one of the nuclease domains is mutated or fully or partially deleted to reduce its nucleic acid cleavage activity. In some embodiments, the Cas9 nuclease is modified to contain no functional RuvC-like nuclease domain. In other embodiments, the Cas9 nuclease is modified to contain no functional HNH-like nuclease domain. In some embodiments in which only one of the nuclease domains is functional, the Cas9 nuclease is a nickase that is capable of introducing a single-stranded break (a "nick") into the target sequence. In some embodiments, a conserved amino acid within a Cas9 nuclease domain is substituted to reduce or alter a nuclease activity. In some embodiments, the Cas nuclease nickase comprises an amino acid substitution in the RuvC-like nuclease domain. Exemplary amino acid substitutions in the RuvC-like nuclease domain include D10A (based on the *S. pyogenes* Cas9 nuclease). In some embodiments, the nickase comprises an amino acid substitution in the HNH-like nuclease domain. Exemplary amino acid substitutions in the HNH-like nuclease domain include E762A, H840A, N863A, H983A, and D986A (based on the *S. pyogenes* Cas9 nuclease).

[0096] In some embodiments, the Cas nuclease is from a Type-I CRISPR/Cas system. In some embodiments, the Cas nuclease is a component of the Cascade complex of a Type-I CRISPR/Cas system. For example, the Cas nuclease is a Cas3 nuclease. In some embodiments, the Cas nuclease is derived from a Type-III CRISPR/Cas system. In some embodiments, the Cas nuclease is derived from Type-IV CRISPR/Cas system. In some embodiments, the Cas nuclease is derived from a Type-V CRISPR/Cas system. In some embodiments, the Cas nuclease is derived from a Type-VI CRISPR/Cas system.

2. Guide RNAs

[0097] The present disclosure provides a genome-targeting nucleic acid that can direct the activities of an associated polypeptide (e.g., a site-directed polypeptide) to a specific target sequence within a target nucleic acid. The genome-targeting nucleic acid can be an RNA. A genome-targeting RNA is referred to as a "guide RNA" or "gRNA" herein. A guide RNA comprises at least a spacer sequence that hybridizes to a target nucleic acid sequence of interest, and a CRISPR repeat sequence. In Type II systems, the gRNA also comprises a second RNA called the tracrRNA sequence. In the Type II gRNA, the CRISPR repeat sequence and tracrRNA sequence hybridize to each other to form a duplex. In the Type V gRNA, the crRNA forms a duplex. In both systems, the duplex binds a site-directed polypeptide, such that the guide RNA and site-direct polypeptide form a complex. In some embodiments, the genome-targeting nucleic acid provides target specificity to the complex by virtue of its association with the site-directed polypeptide. The genome-targeting nucleic acid thus directs the activity of the site-directed polypeptide.

[0098] As is understood by the person of ordinary skill in the art, each guide RNA is designed to include a spacer sequence complementary to its genomic target sequence. See Jinek et al., Science, 337, 816-821 (2012) and Deltcheva et al., Nature, 471, 602-607 (2011).

[0099] In some embodiments, the genome-targeting nucleic acid (e.g., gRNA) is a double-molecule guide RNA. In some embodiments, the genome-targeting nucleic acid (e.g., gRNA) is a single-molecule guide RNA.

[0100] A double-molecule guide RNA comprises two strands of RNA. The first strand comprises in the 5' to 3' direction, an optional spacer extension sequence, a spacer sequence and a minimum CRISPR repeat sequence. The second strand comprises a minimum tracrRNA sequence (complementary to the minimum CRISPR repeat sequence), a 3' tracrRNA sequence and an optional tracrRNA extension sequence.

[0101] A single-molecule guide RNA (referred to as a "sgRNA") in a Type II system comprises, in the 5' to 3' direction, an optional spacer extension sequence, a spacer sequence, a minimum CRISPR repeat sequence, a single-molecule guide linker, a minimum tracrRNA sequence, a 3' tracrRNA sequence and an optional tracrRNA extension sequence. The optional tracrRNA extension may comprise elements that contribute additional functionality (e.g., stability) to the guide RNA. The single-molecule guide linker links the minimum CRISPR repeat and the minimum tracrRNA sequence to form a hairpin structure. The optional tracrRNA extension comprises one or more hairpins.

[0102] A single-molecule guide RNA in a Type V system comprises, in the 5' to 3' direction, a minimum CRISPR repeat sequence and a spacer sequence.

[0103] In some embodiments, the sgRNA comprises a 20 nucleotide spacer sequence at the 5' end of the sgRNA sequence. In some embodiments, the sgRNA comprises a less than 20 nucleotide spacer sequence at the 5' end of the sgRNA sequence. In some embodiments, the sgRNA comprises a more than 20 nucleotide spacer sequence at the 5' end of the sgRNA sequence.

[0104] In some embodiments, the sgRNA comprises comprise no uracil at the 3' end of the sgRNA sequence. In some embodiments, the sgRNA comprises comprise one or more uracil at the 3' end of the sgRNA sequence. For example, the sgRNA can comprise 1 uracil (U) at the 3' end of the sgRNA sequence. The sgRNA can comprise 2 uracil (UU) at the 3' end of the sgRNA sequence. The sgRNA can comprise 3 uracil (UUU) at the 3' end of the sgRNA sequence. The sgRNA can comprise 4 uracil (UUUU) at the 3' end of the sgRNA sequence. The sgRNA can comprise 5 uracil (UUUUU) at the 3' end of the sgRNA sequence. The sgRNA can comprise 6 uracil (UUUUUU) at the 3' end of the sgRNA sequence. The sgRNA can comprise 7 uracil (UUUUUUU) at the 3' end of the sgRNA sequence. The sgRNA can comprise 8 uracil (UUUUUUUU) at the 3' end of the sgRNA sequence.

[0105] The sgRNA can be unmodified or modified. For example, modified sgRNAs can comprise one or more 2'-O-methyl phosphorothioate nucleotides.

[0106] By way of illustration, guide RNAs used in the CRISPR/Cas/Cpf1 system, or other smaller RNAs can be readily synthesized by chemical means, as illustrated below and described in the art. While chemical synthetic procedures are continually expanding, purifications of such RNAs by procedures such as high performance liquid chromatography (HPLC, which avoids the use of gels such as PAGE) tends to become more challenging as polynucleotide lengths increase significantly beyond a hundred or so nucleotides. One approach used for generating RNAs of greater length is to produce two or more molecules that are ligated together. Much longer RNAs, such as those encoding a Cas9 or Cpf1 endonuclease, are more readily generated enzymatically. Various types of RNA modifications can be introduced during or after chemical synthesis and/or enzymatic generation of RNAs, e.g., modifications that enhance stability, reduce the likelihood or degree of innate immune response, and/or enhance other attributes, as described in the art.

[0107] In some embodiments, indel frequency (editing frequency) may be determined using a TIDE analysis, which can be used to identify highly efficient gRNA molecules. In some embodiments, a highly efficient gRNA yields a gene editing frequency of higher than 80%. For example, a gRNA is considered to be highly efficient if it yields a gene editing frequency of at least 80%, at least 85%, at least 90%, at least 95%, or 100%.

[0108] In some embodiments, gene disruption may occur by deletion of a genomic sequence using two guide RNAs. Methods of using CRISPR-Cas gene editing technology to create a genomic deletion in a cell (e.g., to knock out a gene in a cell) are known (Bauer D E et al. *Vis. Exp.* 2015; 95; e52118).

3. Spacer Sequence

[0109] In some embodiments, a gRNA comprises a spacer sequence. A spacer sequence is a sequence (e.g., a 20 nucleotide sequence) that defines the target sequence (e.g., a DNA target sequences, such as a genomic target sequence) of a target nucleic acid of interest. In some embodiments, the spacer sequence is 15 to 30 nucleotides. In some embodiments, the spacer sequence is 15, 16, 17, 18, 19, 29, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30 nucleotides. In some embodiments, a spacer sequence is 20 nucleotides.

[0110] The "target sequence" is adjacent to a PAM sequence and is the sequence modified by an RNA-guided nuclease (e.g., Cas9). The "target nucleic acid" is a double-stranded molecule: one strand comprises the target sequence and is referred to as the "PAM strand," and the other complementary strand is referred to as the "non-PAM strand." One of skill in the art recognizes that the gRNA spacer sequence hybridizes to the reverse complement of the target sequence, which is located in the non-PAM strand of the target nucleic acid of interest. Thus, the gRNA spacer sequence is the RNA equivalent of the target sequence. For example, if the target sequence is 5'-AGAGCAACAGTGCTGTGGCC-3' (SEQ ID NO: 32), then the gRNA spacer sequence is 5'-AGAGCAACAGUGCUGUGGCC-3' (SEQ ID NO: 33). The spacer of a gRNA interacts with a target nucleic acid of interest in a sequence-specific manner via hybridization (i.e., base pairing). The nucleotide sequence of the spacer thus varies depending on the target sequence of the target nucleic acid of interest.

[0111] In a CRISPR/Cas system herein, the spacer sequence is designed to hybridize to a region of the target nucleic acid that is located 5' of a PAM of the Cas9 enzyme used in the system. The spacer may perfectly match the target sequence or may have mismatches. Each Cas9 enzyme has a particular PAM sequence that it recognizes in a target DNA. For example, *S. pyogenes* recognizes in a target nucleic acid a PAM that comprises the sequence 5'-NRG-3', where R comprises either A or G, where N is any nucleotide and N is immediately 3' of the target nucleic acid sequence targeted by the spacer sequence.

[0112] In some embodiments, the target nucleic acid sequence comprises 20 nucleotides. In some embodiments, the target nucleic acid comprises less than 20 nucleotides. In some embodiments, the target nucleic acid comprises more than 20 nucleotides. In some embodiments, the target nucleic acid comprises at least: 5, 10, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 30 or more nucleotides. In some embodiments, the target nucleic acid comprises at most: 5, 10, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 30 or more nucleotides. In some embodiments, the target nucleic acid sequence comprises 20 bases immediately 5' of the first nucleotide of the PAM. For example, in a sequence comprising 5'-NNNNNNNNNNNNNNNNNNNNNRG-3', the target nucleic acid comprises the sequence that corresponds to the Ns, wherein N is any nucleotide, and the underlined NRG sequence is the *S. pyogenes* PAM.

4. Methods of Making gRNAs

[0113] The gRNAs of the present disclosure are produced by a suitable means available in the art, including but not limited to in vitro transcription (IVT), synthetic and/or chemical synthesis methods, or a combination thereof. Enzymatic (IVT), solid-phase, liquid-phase, combined synthetic methods, small region synthesis, and ligation methods are utilized. In one embodiment, the gRNAs are made using IVT enzymatic synthesis methods. Methods of making polynucleotides by IVT are known in the art and are described in International Application PCT/US2013/30062. Accordingly, the present disclosure also includes polynucleotides, e.g., DNA, constructs and vectors are used to in vitro transcribe a gRNA described herein.

[0114] In some embodiments, non-natural modified nucleobases are introduced into polynucleotides, e.g., gRNA, during synthesis or post-synthesis. In certain embodiments, modifications are on internucleoside linkages, purine or pyrimidine bases, or sugar. In some embodiments, a modification is introduced at the terminal of a polynucleotide; with chemical synthesis or with a polymerase enzyme. Examples of modified nucleic acids and their synthesis are disclosed in PCT application No. PCT/US2012/058519. Synthesis of modified polynucleotides is also described in Verma and Eckstein, Annual Review of Biochemistry, vol. 76, 99-134 (1998).

[0115] In some embodiments, enzymatic or chemical ligation methods are used to conjugate polynucleotides or their regions with different functional moieties, such as targeting or delivery agents, fluorescent labels, liquids, nanoparticles, etc. Conjugates of polynucleotides and modified polynucleotides are reviewed in Goodchild, Bioconjugate Chemistry, vol. 1(3), 165-187 (1990).

[0116] Certain embodiments of the invention also provide nucleic acids, e.g., vectors, encoding gRNAs described herein. In some embodiments, the nucleic acid is a DNA molecule. In other embodiments, the nucleic acid is an RNA molecule. In some embodiments, the nucleic acid comprises a nucleotide sequence encoding a crRNA. In some embodiments, the nucleotide sequence encoding the crRNA comprises a spacer flanked by all or a portion of a repeat sequence from a naturally-occurring CRISPR/Cas system. In some embodiments, the nucleic acid comprises a nucleotide sequence encoding a tracrRNA. In some embodiments, the crRNA and the tracrRNA is encoded by two separate nucleic acids. In other embodiments, the crRNA and the tracrRNA is encoded by a single nucleic acid. In some embodiments, the crRNA and the tracrRNA is encoded by opposite strands of a single nucleic acid. In other embodiments, the crRNA and the tracrRNA is encoded by the same strand of a single nucleic acid.

[0117] In some embodiments, the gRNAs provided by the disclosure are chemically synthesized by any means described in the art (see e.g., WO/2005/01248). While chemical synthetic procedures are continually expanding, purifications of such RNAs by procedures such as high performance liquid chromatography (HPLC, which avoids the use of gels such as PAGE) tends to become more challenging as polynucleotide lengths increase significantly beyond a hundred or so nucleotides. One approach used for generating RNAs of greater length is to produce two or more molecules that are ligated together.

[0118] In some embodiments, the gRNAs provided by the disclosure are synthesized by enzymatic methods (e.g., in vitro transcription, IVT).

[0119] Various types of RNA modifications can be introduced during or after chemical synthesis and/or enzymatic generation of RNAs, e.g., modifications that enhance stability, reduce the likelihood or degree of innate immune response, and/or enhance other attributes, as described in the art.

[0120] In certain embodiments, more than one guide RNA can be used with a CRISPR/Cas nuclease system. Each guide RNA may contain a different targeting sequence, such that the CRISPR/Cas system cleaves more than one target nucleic acid. In some embodiments, one or more guide RNAs may have the same or differing properties such as activity or stability within the Cas9 RNP complex. Where more than one guide RNA is used, each guide RNA can be encoded on the same or on different vectors. The promoters used to drive expression of the more than one guide RNA is the same or different.

[0121] The guide RNA may target any sequence of interest via the targeting sequence (e.g., spacer sequence) of the crRNA. In some embodiments, the degree of complementarity between the targeting sequence of the guide RNA and the target sequence on the target nucleic acid molecule is about 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 97%, 98%, 99%, or 100%. In some embodiments, the targeting sequence of the guide RNA and the target sequence on the target nucleic acid molecule is 100% complementary. In other embodiments, the targeting sequence of the guide RNA and the target sequence on the target nucleic acid molecule may contain at least one mismatch. For example, the targeting sequence of the guide RNA and the target sequence on the target nucleic acid molecule may contain 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 mismatches. In some embodiments, the targeting sequence of the guide RNA and the target sequence on the target nucleic acid molecule may contain 1-6 mismatches. In some embodiments, the targeting sequence of the guide RNA and the target sequence on the target nucleic acid molecule may contain 5 or 6 mismatches.

[0122] The length of the targeting sequence may depend on the CRISPR/Cas9 system and components used. For example, different Cas9 proteins from different bacterial species have varying optimal targeting sequence lengths. Accordingly, the targeting sequence may comprise 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 35, 40, 45, 50, or more than 50 nucleotides in length. In some embodiments, the targeting sequence may comprise 18-24 nucleotides in length. In some embodiments, the targeting sequence may comprise 19-21 nucleotides in length. In some embodiments, the targeting sequence may comprise 20 nucleotides in length.

[0123] In some embodiments of the present disclosure, a CRISPR/Cas nuclease system includes at least one guide RNA. In some embodiments, the guide RNA and the Cas protein may form a ribonucleoprotein (RNP), e.g., a CRISPR/Cas complex. The guide RNA may guide the Cas protein to a target sequence on a target nucleic acid molecule (e.g., a genomic DNA molecule), where the Cas protein cleaves the target nucleic acid. In some embodiments, the CRISPR/Cas complex is a Cpf1/guide RNA complex. In some embodiments, the CRISPR complex is a Type-II CRISPR/Cas9 complex. In some embodiments, the Cas protein is a Cas9 protein. In some embodiments, the CRISPR/Cas9 complex is a Cas9/guide RNA complex.

5. Delivery of guide RNA and Nuclease

[0124] In some embodiments, a gRNA and an RNA-guided nuclease are delivered to a cell separately, either simultaneously or sequentially. In some embodiments, a gRNA and an RNA-guided nuclease are delivered to a cell together. In some embodiments, a gRNA and an RNA-guided nuclease are pre-complexed together to form a ribonucleoprotein (RNP).

[0125] RNPs are useful for gene editing, at least because they minimize the risk of promiscuous interactions in a nucleic acid-rich cellular environment and protect the RNA from degradation. Methods for forming RNPs are known in the art. In some embodiments, an RNP containing an RNA-guided nuclease (e.g., a Cas nuclease, such as a Cas9 nuclease) and a gRNA targeting a gene of interest is delivered a cell (e.g.: a T cell). In some embodiments, an RNP is delivered to a T cell by electroporation.

[0126] As used herein, a "β2M targeting RNP" refers to a gRNA that targets the β2M gene pre-complexed with an RNA-guided nuclease. In some embodiments, a β2M targeting RNP is delivered to a cell. In some embodiments, more than one RNP is delivered to a cell. In some embodiments, more than one RNA is delivered to a cell separately. In some embodiments, more than one RNP is delivered to the cell simultaneously.

[0127] In some embodiments, an RNA-guided nuclease is delivered to a cell in a DNA vector that expresses the RNA-guided nuclease, an RNA that encodes the RNA-guided nuclease, or a protein. In some embodiments, a gRNA targeting a gene is delivered to a cell as an RNA, or a DNA vector that expresses the gRNA.

[0128] Delivery of an RNA-guided nuclease, gRNA, and/or an RNP may be through direct injection or cell transfection using known methods, for example, electroporation or chemical transfection. Other cell transfection methods may be used.

6. Multi-Modal or Differential Delivery of Components

[0129] Skilled artisans will appreciate that different components of genome editing systems can be delivered together or separately and simultaneously or nonsimultaneously. Separate and/or asynchronous delivery of genome editing system components may be particularly desirable to provide temporal or spatial control over the function of genome editing systems and to limit certain effects caused by their activity.

[0130] Different or differential modes as used herein refer to modes of delivery that confer different pharmacodynamic or pharmacokinetic properties on the subject component molecule, e.g., a RNA-guided nuclease molecule, gRNA, template nucleic acid, or payload. For example, the modes of delivery can result in different tissue distribution, different half-life, or different temporal distribution, e.g., in a selected compartment, tissue, or organ.

[0131] Some modes of delivery, e.g., delivery by a nucleic acid vector that persists in a cell, or in progeny of a cell, e.g., by autonomous replication or insertion into cellular nucleic acid, result in more persistent expression of and presence of a component. Examples include viral, e.g., AAV or lentivirus, delivery.

[0132] By way of example, the components of a genome editing system, e.g., a RNA-guided nuclease and a gRNA, can be delivered by modes that differ in terms of resulting half-life or persistent of the delivered component the body, or in a particular compartment, tissue or organ. In an embodiment, a gRNA can be delivered by such modes. The RNA-guided nuclease molecule component can be delivered by a mode which results in less persistence or less exposure to the body or a particular compartment or tissue or organ.

[0133] More generally, in an embodiment, a first mode of delivery is used to deliver a first component and a second mode of delivery is used to deliver a second component. The first mode of delivery confers a first pharmacodynamic or pharmacokinetic property. The first pharmacodynamic property can be, e.g., distribution, persistence, or exposure, of the component, or of a nucleic acid that encodes the component, in the body, a compartment, tissue or organ. The second mode of delivery confers a second pharmacodynamic or pharmacokinetic property. The second pharmacodynamic property can be, e.g., distribution, persistence, or exposure, of the component, or of a nucleic acid that encodes the component, in the body, a compartment, tissue or organ.

[0134] In certain embodiments, the first pharmacodynamic or pharmacokinetic property, e.g., distribution, persistence or exposure, is more limited than the second pharmacodynamic or pharmacokinetic property.

[0135] In certain embodiments, the first mode of delivery is selected to optimize, e.g., minimize, a pharmacodynamic or pharmacokinetic property, e.g., distribution, persistence or exposure.

[0136] In certain embodiments, the second mode of delivery is selected to optimize, e.g., maximize, a pharmacodynamic or pharmacokinetic property, e.g., distribution, persistence or exposure.

[0137] In certain embodiments, the first mode of delivery comprises the use of a relatively persistent element, e.g., a nucleic acid, e.g., a plasmid or viral vector, e.g., an AAV, adenovirus or lentivirus. As such vectors are relatively persistent product transcribed from them would be relatively persistent.

[0138] In certain embodiments, the second mode of delivery comprises a relatively transient element, e.g., an RNA or protein.

[0139] In certain embodiments, the first component comprises gRNA, and the delivery mode is relatively persistent, e.g., the gRNA is transcribed from a plasmid or viral vector, e.g., an AAV, adenovirus or lentivirus. Transcription of these genes would be of little physiological consequence because the genes do not encode for a protein product, and the gRNAs are incapable of acting in isolation. The second component, a RNA-guided nuclease molecule, is delivered in a transient manner, for example as mRNA encoding the protein or as protein, ensuring that the full RNA-guided nuclease molecule/gRNA complex is only present and active for a short period of time.

[0140] Furthermore, the components can be delivered in different molecular form or with different delivery vectors that complement one another to enhance safety and tissue specificity.

[0141] Use of differential delivery modes can enhance performance, safety, and/or efficacy, e.g., the likelihood of an eventual off-target modification can be reduced. Delivery of immunogenic components, e.g., Cas9 molecules, by less persistent modes can reduce immunogenicity, as peptides from the bacterially-derived Cas enzyme are displayed on the surface of the cell by WIC molecules. A two-part delivery system can alleviate these drawbacks.

[0142] Differential delivery modes can be used to deliver components to different, but overlapping target regions. The formation active complex is minimized outside the overlap of the target regions. Thus, in an embodiment, a first component, e.g., a gRNA is delivered by a first delivery mode that results in a first spatial, e.g., tissue, distribution. A second component, e.g., a RNA-guided nuclease molecule is delivered by a second delivery mode that results in a second spatial, e.g., tissue, distribution. In an embodiment the first mode comprises a first element selected from a liposome, nanoparticle, e.g., polymeric nanoparticle, and a nucleic acid, e.g., viral vector. The second mode comprises a second element selected from the group. In an embodiment, the first mode of delivery comprises a first targeting element, e.g., a cell specific receptor or an antibody, and the second mode of delivery does not include that element. In certain embodiments, the second mode of delivery comprises a second targeting element, e.g., a second cell specific receptor or second antibody.

[0143] When the RNA-guided nuclease molecule is delivered in a virus delivery vector, a liposome, or polymeric nanoparticle, there is the potential for delivery to and therapeutic activity in multiple tissues, when it may be desirable to only target a single tissue. A two-part delivery system can resolve this challenge and enhance tissue specificity. If the gRNA and the RNA-guided nuclease molecule are packaged in separated delivery vehicles with distinct but overlapping tissue tropism, the fully functional complex is only be formed in the tissue that is targeted by both vectors.

Iii. Knock-Down and/or Insertion into the P2M Loci of Human B Cells

[0144] In various embodiments, the invention relates to a population of cells comprising engineered human B cells, wherein the engineered human B cells comprise a disrupted β2M gene. In various embodiments, the β2M gene to be disrupted comprises SEQ ID NO. 1. In various embodiments, the β2M gene to be disrupted is at least 75%, 80%, 85%, 90%, 95% or 100% identical to the nucleic acid sequence of SEQ ID NO. 1.

[0145] In various embodiments, the disruption in the B2M gene results in an eliminated or decreased expression of the B2M gene. In various embodiments, expression of the β2M gene is reduced by 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or 100%. In preferred embodiments, β2M expression is reduced by at least 85%.

[0146] In various embodiments, the β2M gene is disrupted by deletion of all or part of the β2M gene. In various embodiments the β2M gene is disrupted by insertion of the gene encoding a therapeutic protein into a coding exon of the β2M gene. In various embodiments, the gene encoding a therapeutic protein (which is described in more detail below) is inserted into exon 2 of the β2M gene. In various embodiments, the gene encoding a therapeutic protein is inserted into exon 1 of the β2M gene. In various embodiments, the gene encoding a therapeutic protein is inserted into exon 3 of the β2M gene. In various embodiments, the gene encoding a therapeutic protein is inserted into exon 4 of the β2M gene.

[0147] In various embodiments, the gene encoding a therapeutic protein is inserted into an intron of the β2M gene. In various embodiments, the insertion of the gene encoding a therapeutic protein does not disrupt the expression of the B2M gene in a B cell.

Iv. Therapeutic Proteins

[0148] In various embodiments, the engineered B cell comprises a therapeutic protein to be delivered to a patient in need thereof. See for example FIG. **11**, for a non-exclusive list of targeting constructs capable of expressing a therapeutic protein. As used herein, the term "therapeutic protein" means any protein that may contribute to the treatment, reduction of symptoms, prevention or cure of a disease or disorder in a patient. In certain embodiments, the therapeutic protein may be suitable for treatment of a rare disease or an orphan disease, where said therapy can be achieved by the replacement of a particular protein and/or enzyme. A therapeutic protein may include but is not limited to an enzyme, a ligand, a naturally occurring, engineered and/or chimeric receptor, a cytokine or a chemokine. Such disease include for example, but are not limited to Fabry disease, Pompe disease, Phenylketonuria (PKU) or Hemophilia A.

[0149] In various embodiments said therapeutic protein is a protein for the treatment of Fabry disease. In various embodiments, the therapeutic protein is a-galactosidase. In various embodiments, the therapeutic protein comprises the amino acid sequence of SEQ ID NO. 2. In various embodiments, the therapeutic protein is at least 75%, 80%, 85%, 90% or 95% identical to the amino acid sequence of SEQ ID NO. 2. In various embodiments, the targeting construct which comprises a left homology arm, a 2A cleavable peptide, a codon-optimized therapeutic protein and a right homology arm. In various embodiments, the targeting construct comprises the nucleic acid sequence of SEQ ID NO. 10. In various embodiments, the targeting construct is at least 75%, 80%, 85%, 90% or 95% identical to the codon-optimized nucleic acid sequence of SEQ ID NO. 10. In various embodiments, the targeting construct comprises the nucleic acid sequence of SEQ ID NO. 31. In various embodiments, the targeting construct is at least 75%, 80%, 85%, 90% or 95% identical to the nucleic acid sequence of SEQ ID NO. 31.

[0150] In various embodiments said therapeutic protein is a protein for the treatment of Phenylketonuria (PKU). In various embodiments, the therapeutic protein is phenylalanine hydroxylase (PAH). In various embodiments, the therapeutic protein comprises the amino acid sequence of SEQ ID NO. 3. In various embodiments, the therapeutic protein is at least 75%, 80%, 85%, 90% or 95% identical to the amino acid sequence of SEQ ID NO. 3. In various embodiments, the targeting construct which comprises a left homology arm, a 2A cleavable peptide, a codon-optimized therapeutic protein and a right homology arm comprises the nucleic acid sequence of SEQ ID NO. 11. In various embodiments, the targeting construct is at least 75%, 80%, 85%, 90% or 95% identical to the nucleic acid sequence of SEQ ID NO. 11. In various embodiments, the therapeutic protein is phenylalanine ammonia-lyase (PAL). In various embodiments, the therapeutic protein comprises the codon-optimized amino acid sequence of SEQ ID NO. 4. In various embodiments, the therapeutic protein is at least 75%, 80%, 85%, 90% or 95% identical to the amino acid sequence of SEQ ID NO. 4. In various embodiments, the targeting construct which comprises a left homology arm, a 2A cleavable peptide, a codon-optimized therapeutic protein and a right homology arm comprises the nucleic acid sequence of SEQ ID NO. 12. In various embodiments, the targeting construct is at least 75%, 80%, 85%, 90% or 95% identical to the nucleic acid sequence of SEQ ID NO. 12.

[0151] In various embodiments said therapeutic protein is a protein for the treatment of Pompe disease. In various embodiments, the therapeutic protein is acid alpha-glucosidase (GAA). In various embodiments, the therapeutic protein comprises the amino acid sequence of SEQ ID NO. 5. In various embodiments, the therapeutic protein is at least 75%, 80%, 85%, 90% or 95% identical to the amino acid sequence of SEQ ID NO. 5. In various embodiments, the targeting construct which comprises a left homology arm, a 2A cleavable peptide, a codon-optimized therapeutic protein and a right homology arm comprises the nucleic acid sequence of SEQ ID NO. 13. In various embodiments, the targeting construct is at least 75%, 80%, 85%, 90% or 95% identical to the nucleic acid sequence of SEQ ID NO. 13.

[0152] In various embodiments said therapeutic protein is a protein for the treatment of Hemophilia A. In various embodiments, the therapeutic protein is B domain deleted (BDD) of Factor VIII. In various embodiments, the therapeutic protein comprises the amino acid sequence of SEQ ID NO. 6. In various embodiments, the therapeutic protein is at least 75%, 80%, 85%, 90% or 95% identical to the amino acid sequence of SEQ ID NO. 6. In various embodiments, In various embodiments, the targeting construct which comprises a left homology arm, a 2A cleavable peptide, a codon-optimized therapeutic protein and a right homology arm comprises the nucleic acid sequence of SEQ ID NO. 14. In various embodiments, the targeting construct is at least 75%, 80%, 85%, 90% or 95% identical to the codon-optimized nucleic acid sequence of SEQ ID NO. 14. In various embodiments, the therapeutic protein is the full length domain of Factor VIII. In various embodiments, the therapeutic protein comprises the amino acid sequence of SEQ ID NO. 7. In various embodiments, the therapeutic protein is at least 75%, 80%, 85%, 90% or 95% identical to the amino acid sequence of SEQ ID NO. 7. In various embodiments, the targeting construct which comprises a left homology arm, a 2A cleavable peptide, a codon-optimized therapeutic protein and a right homology arm comprises the nucleic acid sequence of SEQ ID NO. 15. In various embodiments, the targeting construct is at least 75%, 80%, 85%, 90% or 95% identical to the nucleic acid sequence of SEQ ID NO. 15.

[0153] In various embodiments, the therapeutic protein is a chimeric receptor that expresses an extracellular domain of GPC3. In various embodiments, the therapeutic protein comprises the amino acid sequence of SEQ ID NO. 8. In various embodiments, the therapeutic protein is at least 75%, 80%, 85%, 90% or 95% identical to the amino acid sequence of SEQ ID NO. 8. In various embodiments, the targeting construct which comprises a left homology arm, a 2A cleavable peptide, a codon-optimized therapeutic protein and a right homology arm comprises the nucleic acid sequence of SEQ ID NO. 16. In various embodiments, the targeting construct is at least 75%, 80%, 85%, 90% or 95% identical to the nucleic acid sequence of SEQ ID NO. 16.

[0154] In various embodiments, the therapeutic protein is Interleukin 10 (IL-10). See for example FIG. **12**. In various embodiments, the therapeutic protein comprises the amino acid sequence of SEQ ID NO. 9. In various embodiments, the therapeutic protein is at least 75%, 80%, 85%, 90% or 95% identical to the amino acid sequence of SEQ ID NO. 9. In various embodiments, the targeting construct which comprises a left homology arm, a 2A cleavable peptide, a codon-optimized therapeutic protein and a right homology arm comprises the nucleic acid sequence of SEQ ID NO. 17. In various embodiments, the targeting construct is at least 75%, 80%, 85%, 90% or 95% identical to the nucleic acid sequence of SEQ ID NO. 17.

[0155] In various embodiments, the present disclosure relates to a method of expressing a therapeutic protein in a population of human B cells. In various embodiments, at least 20% of the human B cells express the therapeutic protein. In various embodiments at least 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90% or 95% of the engineered B cells express the therapeutic protein.

V. Methods of Treatment

[0156] In various aspects of the invention, the gene edited B cells will be delivered as a therapeutic to a patient in need thereof. In various embodiments, the gene edited B cells will be capable of treating or preventing various diseases or disorders.

[0157] In some embodiments, are methods for treating a rare disease or an orphan disease, where said therapy can be achieved by the replacement of a particular protein and/or enzyme. Such diseases include for example, but are not limited to Fabry disease, Pompe disease, Phenylketonuria (PKU) or Hemophilia A.

[0158] In some aspects, the invention comprises a pharmaceutical composition comprising a population of gene edited B cells comprising at least one therapeutic protein as described herein and a pharmaceutically acceptable excipient. In some embodiments, the pharmaceutical composition further comprises an additional active agent.

[0159] It will be appreciated that target doses for modified B cells can range from 1×10^6-2×10^{10} cells/kg, preferably 2×10^6 cells/kg, more preferably. It will be appreciated that doses above and below this range may be appropriate for certain subjects, and appropriate dose levels can be determined by the healthcare provider as needed. Additionally, multiple doses of cells can be provided in accordance with the invention.

[0160] In some embodiments, the expanded population of engineered B cells are autologous B cells. In some embodiments, the modified B cells are allogeneic B cells. In some embodiments, the modified B cells are heterologous B cells. In some embodiments, the modified B cells of the present application are transfected or transduced in vivo. In other embodiments, the engineered cells are transfected or transduced ex vivo.

[0161] As used herein, the term "subject" or "patient" means an individual. In some aspect, a subject is a mammal such as a human. In some aspect, a subject can be a non-human primate. Non-human primates include marmosets, monkeys, chimpanzees, gorillas, orangutans, and gibbons, to name a few. The term "subject" also includes domesticated animals, such as cats, dogs, etc., livestock (e.g., llama, horses, cows), wild animals (e.g., deer, elk, moose, etc.,), laboratory animals (e.g., mouse, rabbit, rat, gerbil, guinea pig, etc.) and avian species (e.g., chickens, turkeys, ducks, etc.). Preferably, the subject is a human subject. More preferably, the subject is a human patient.

[0162] In certain embodiments, compositions comprising gene edited B cells disclosed herein may be administered in conjunction with any number of chemotherapeutic agents. Examples of chemotherapeutic agents include alkylating agents such as thiotepa and cyclophosphamide (CYTOXAN™); alkyl sulfonates such as busulfan, improsulfan and piposulfan; aziridines such as benzodopa, carboquone, meturedopa, and uredopa; ethylenimines and methylamelamines including altretamine, triethylenemelamine, trietylenephosphoramide, triethylenethiophosphaoramide and trimethylolomelamine resume; nitrogen mustards such as chlorambucil, chlornaphazine, cholophosphamide, estramustine, ifosfamide, mechlorethamine, mechlorethamine oxide hydrochloride, melphalan, novembichin, phenesterine, prednimustine, trofosfamide, uracil mustard; nitrosureas such as carmustine, chlorozotocin, fotemustine, lomustine, nimustine, ranimustine; antibiotics such as aclacinomysins, actinomycin, authramycin, azaserine, bleomycins, cactinomycin, calicheamicin, carabicin, carminomycin, carzinophilin, chromomycins, dactinomycin, daunorubicin, detorubicin, 6-diazo-5-oxo-L-norleucine, doxorubicin, epirubicin, esorubicin, idarubicin, marcellomycin, mitomycins, mycophenolic acid, nogalamycin, olivomycins, peplomycin, potfiromycin, puromycin, que-lamycin, rodorubicin, streptonigrin, streptozocin, tubercidin, ubenimex, zinostatin, zorubicin; anti-metabolites such as methotrexate and 5-fluorouracil (5-FU); folic acid analogues such as denopterin, methotrexate, pteropterin, trimetrexate; purine analogs such as fludarabine, 6-mercaptopurine, thiamiprine, thioguanine; pyrimidine analogs such as ancitabine, azacitidine, 6-azauridine, carmofur, cytarabine, dideoxyuridine, doxifluridine, enocitabine, floxuridine, 5-FU; androgens such as calusterone, dromostanolone propionate, epitiostanol, mepitiostane, testolactone; anti-adrenals such as aminoglutethimide, mitotane, trilostane; folic acid replenisher such as frolinic acid; aceglatone; aldophosphamide glycoside; aminolevulinic acid; amsacrine; bestrabucil; bisantrene; edatraxate; defofamine; demecolcine; diaziquone; elformithine; elliptinium acetate; etoglucid; gallium nitrate; hydroxyurea; lentinan; lonidamine; mitoguazone; mitoxantrone; mopidamol; nitracrine; pentostatin; phenamet; pirarubicin; podophyllinic acid; 2-ethylhydrazide; procarbazine; PSK®; razoxane; sizofiran; spirogermanium; tenuazonic acid; triaziquone; 2, 2',2"-trichlorotriethylamine; urethan; vindesine; dacarbazine; mannomustine; mitobronitol; mitolactol; pipobroman; gacytosine; arabinoside ("Ara-C"); cyclophosphamide; thiotepa; taxoids, e.g. paclitaxel (TAXOL®, Bristol-Myers

Squibb) and doxetaxel (TAXOTERE®, Rhone-Poulenc Rorer); chlorambucil; gemcitabine; 6-thioguanine; mercaptopurine; methotrexate; platinum analogs such as cisplatin and carboplatin; vinblastine; platinum; etoposide (VP-16); ifosfamide; mitomycin C; mitoxantrone; vincristine; vinorelbine; navelbine; novantrone; teniposide; daunomycin; aminopterin; xeloda; ibandronate; CPT-11; topoisomerase inhibitor RFS2000; difluoromethylomithine (DMFO); retinoic acid derivatives such as TARGRETIN™ (bexarotene), PANRETIN™, (alitretinoin); ONTAK™ (denileukin diftitox); esperamicins; capecitabine; and pharmaceutically acceptable salts, acids or derivatives of any of the above. Also included in this definition are anti-hormonal agents that act to regulate or inhibit hormone action on tumors such as anti-estrogens including for example tamoxifen, raloxifene, aromatase inhibiting 4(5)-imidazoles, 4-hydroxytamoxifen, trioxifene, keoxifene, LY117018, onapristone, and toremifene (Fareston); and anti-androgens such as flutamide, nilutamide, bicalutamide, leuprolide, and goserelin; and pharmaceutically acceptable salts, acids or derivatives of any of the above. Combinations of chemotherapeutic agents are also administered where appropriate, including, but not limited to CHOP, i.e., Cyclophosphamide (CYTOXAN®) Doxorubicin (hydroxydoxorubicin), Fludarabine, Vincristine (ONCOVIN®), and Prednisone.

[0163] A variety of additional therapeutic agents may be used in conjunction with the compositions described herein. For example, potentially useful additional therapeutic agents include PD-1 (or PD-L1) inhibitors such as nivolumab (Opdivo®), pembrolizumab (Keytruda®), pembrolizumab, cemiplimaib (Libtayo®), and atezolizumab (Tecentriq®). Other additional therapeutics include anti-CTLA-4 antibodies (e.g., Ipilimumab®), anti-LAG-3 antibodies (e.g., Relatlimab, BMS), alone or in combination with PD-1 and/or PD-L1 inhibitors.

[0164] Additional therapeutic agents suitable for use in combination with the invention include, but are not limited to, ibrutinib (IMBRUVICA®), ofatumumab (ARZERRA®), rituximab (RITUXAN®), bevacizumab (AVASTIN®), trastuzumab (HERCEPTIN®), trastuzumab emtansine (KADCYLA®), imatinib (GLEEVEC®), cetuximab (ERBITUX®), panitumumab (VECTIBIX®), catumaxomab, ibritumomab, ofatumumab, tositumomab, brentuximab, alemtuzumab, gemtuzumab, erlotinib, gefitinib, vandetanib, afatinib, lapatinib, neratinib, axitinib, masitinib, pazopanib, sunitinib, sorafenib, toceranib, lestaurtinib, axitinib, cediranib, lenvatinib, nintedanib, pazopanib, regorafenib, semaxanib, sorafenib, sunitinib, tivozanib, toceranib, vandetanib, entrectinib, cabozantinib, imatinib, dasatinib, nilotinib, ponatinib, radotinib, bosutinib, lestaurtinib, ruxolitinib, pacritinib, cobimetinib, selumetinib, trametinib, binimetinib, alectinib, ceritinib, crizotinib, aflibercept, adipotide, denileukin diftitox, mTOR inhibitors such as Everolimus and Temsirolimus, hedgehog inhibitors such as sonidegib and vismodegib, CDK inhibitors such as CDK inhibitor (palbociclib).

[0165] In additional embodiments, the composition comprising gene edited B cells can be administered with an anti-inflammatory agent. Anti-inflammatory agents or drugs include, but are not limited to, steroids and glucocorticoids (including betamethasone, budesonide, dexamethasone, hydrocortisone acetate, hydrocortisone, hydrocortisone, methylprednisolone, prednisolone, prednisone, triamcinolone), nonsteroidal anti-inflammatory drugs (NSAIDS) including aspirin, ibuprofen, naproxen, methotrexate, sulfasalazine, leflunomide, anti-TNF medications, cyclophosphamide and mycophenolate. Exemplary NSAIDs include ibuprofen, naproxen, naproxen sodium, Cox-2 inhibitors, and sialylates. Exemplary analgesics include acetaminophen, oxycodone, tramadol of proporxyphene hydrochloride. Exemplary glucocorticoids include cortisone, dexamethasone, hydrocortisone, methylprednisolone, prednisolone, or prednisone. Exemplary biological response modifiers include molecules directed against cell surface markers (e.g., CD4, CD5, etc.), cytokine inhibitors, such as the TNF antagonists, (e.g., etanercept (ENBREL®), adalimumab (HUMIRA®) and infliximab (REMICADE®)), chemokine inhibitors and adhesion molecule inhibitors. The biological response modifiers include monoclonal antibodies as well as recombinant forms of molecules. Exemplary DMARDs include azathioprine, cyclophosphamide, cyclosporine, methotrexate, penicillamine, leflunomide, sulfasalazine, hydroxychloroquine, Gold (oral (auranofin) and intramuscular) and minocycline.

[0166] In certain embodiments, the compositions described herein are administered in conjunction with a cytokine. "Cytokine" as used herein is meant to refer to proteins released by one cell population that act on another cell as intercellular mediators. Examples of cytokines are lymphokines, monokines, and traditional polypeptide hormones. Included among the cytokines are growth hormones such as human growth hormone, N-methionyl human growth hormone, and bovine growth hormone; parathyroid hormone; thyroxine; insulin; proinsulin; relaxin; prorelaxin; glycoprotein hormones such as follicle stimulating hormone (FSH), thyroid stimulating hormone (TSH), and luteinizing hormone (LH); hepatic growth factor (HGF); fibroblast growth factor (FGF); prolactin; placental lactogen; mullerian-inhibiting substance; mouse gonadotropin-associated peptide; inhibin; activin; vascular endothelial growth factor; integrin; thrombopoietin (TPO); nerve growth factors (NGFs) such as NGF-beta; platelet-growth factor; transforming growth factors (TGFs) such as TGF-alpha and TGF-beta; insulin-like growth factor-I and —II; erythropoietin (EPO); osteoinductive factors; interferons such as interferon-alpha, beta, and -gamma; colony stimulating factors (CSFs) such as macrophage-CSF (M-CSF); granulocyte-macrophage-CSF (GM-CSF); and granulocyte-CSF (G-CSF); interleukins (ILs) such as IL-1, IL-1 alpha, IL-2, IL-3, IL-4, IL-5, IL-6, IL-7, IL-8, IL-9, IL-10, IL-11, IL-12; IL-15, a tumor necrosis factor such as TNF-alpha or TNF-beta; and other polypeptide factors including LIF and kit ligand (KL). As used herein, the term cytokine includes proteins from natural sources or from recombinant cell culture, and biologically active equivalents of the native sequence cytokines.

EXAMPLES

[0167] Although the foregoing invention has been described in some detail by way of illustration and example for purposes of clarity of understanding, it will be readily apparent to one of ordinary skill in the art in light of the teachings of this invention that certain changes and modifications may be made thereto without departing from the spirit or scope of the appended claims. The following examples are provided by way of illustration only and not by way of limitation. Those of skill in the art will readily recognize a variety of noncritical parameters that could be changed or modified to yield essentially similar results.

Example 1. Engineering Human B Cells—Optimal Nucleofection Conditions

[0168] The experimental design for the below described experiments is outlined in FIG. **1**. Briefly, PBMCs were isolated from healthy donors using magnetic beads and activated using CD40 ligand and IL4. B cells were engineered using nucleofection delivery of the CRISPR-Cas9 system and resulting engineered B cells were characterized by PCR and flow cytometry. FIG. **3** depicts the insertion site in the β2M gene and guide sequence used for editing.

[0169] Human B cell isolation, activation, expansion and electroporation. Buffy coats from healthy donors were obtained from Stanford Blood Center (Menlo Park, CA, USA). PBMCs were isolated from buffy coats using Ficoll-Paque (GE Healthcare, Chicago, IL). Primary human B cells were isolated using the EASYSEP™ Human B Cell Isolation Kit according to manufacturer's instruction (STEMCELL Technologies Inc., Cambridge, MA, USA). Isolated B cells were activated and expanded using the human B Cell Expansion Kit according to manufacturer's instruction (Miltenyi Biotec, Bergisch Gladbach, Germany).

[0170] Optimal Nucleofection Protocol Development. Nucleofection was performed using AMAXA™ 4D-NUCLEOFACTOR™ in P3 nucleofection solution (Lonza, Basel, Switzerland). 1 μg of pMAX-GFP plasmid DNA was used to electroporate 1 million activated human B cells in 20 μl volume for GFP expression. Various electroporation programs were examined both for efficiency of transfection (FIG. **2**A and FIG. **2**B ("GFP+%")) and for the percentage of viable cells achieved (FIG. **2**A and FIG. **2**B ("Viability %")).

[0171] Results. It was determined that program CM-137 achieved the most optimal combination of efficiency (79.017%) and viability (98.484%).

Example 2. Optimization of CRISPR Engineering Conditions

[0172] Next, parameters for delivery of the CRISPR-Cas9 complex were explored. PBMC-derived human B cells were isolated, activated and expanded as described in Example 1. Next, β2M targeting Cas9/sgRNA RNPs were prepared and electroporated into the B-cells. β2M knock-down and B-cell viability were evaluated.

[0173] β2M sgRNA and CRISPR engineering. A chemically modified sgRNA oligomer targeting β 2M was manufactured by IDT (Integrated DNA Technologies, Coralville, Iowa, USA). See, e.g., FIG. **3**. Recombinant *S. pyogenes* Cas9 enzyme was purchased from IDT (Integrated DNA Technologies, Coralville, Iowa, USA). Cas9 was incubated with sgRNA at a molar ratio of 1:1.2 at room temperature for 10 minutes prior to mixing with B cells. 100 pmol RNP was used for electroporation with 1 million activated human B cells in 20 μl volume (FIG. **4**A). Engineering of primary human B cells was carried out using an AMAXA™ 4D-NUCLEOFACTOR™ in P3 nucleofection solution with one of the 8 preset programs (Lonza, Basel, Switzerland). β2M knock-down and viability were assessed using flow cytometry.

[0174] Results. It was determined that program CM-137 achieved the most optimal combination of β2M knock-down (80.2% (donor 47) and 91.8% (donor 48)) and viability (85.6% (donor 47) and 92.8% (donor 48)). See, e.g., FIG. **4**C.

Example 3. P2M CRISPR Knockout INDELs Analysis

[0175] Next, the knockout of the β2M by CRISPR/cas9 was validated through INDEL analysis. PBMC-derived human B cells were isolated, activated and expanded as described in Example 1. Next, β2M targeting Cas9/sgRNA RNPs were prepared and electroporated into the B-cells.

[0176] Engineered cells were cultured for 2 days after electroporation. Genomic DNA was extracted using NucleoSpin Tissue, Mini kit for DNA from cells and tissue polymerase (Thermo Fisher Scientific, Waltham, MA, USA), according to the manufacturer's recommendations. To interrogate the sites of DNA cleavage after editing, PCR was performed using Q5 High-Fidelity polymerase (VWR International, LLC, Radnor, PA, USA) and primers flanking the region where double stranded breaks were generated. The PCR amplicons were purified using QIAquick PCR Purification Kit (Qiagen, Hilden, Germany) and sequenced by Sanger sequencing. The resulting sequences were used to calculate INDELs frequencies using ICE synthego (ice.synthego.com) web-based software. A list of the primer sequences is provided in Table 2.

TABLE 2

Primers for INDELs analysis of β2M specific sgRNA.

Primer ID	Sequence
InDels-B2M-FWD (SEQ ID NO: 27)	TGAGAGGGCATCAGAAGTCC
InDels-B2M-Rev (SEQ ID NO: 28)	AAGTCACATGGTTCACACGG

[0177] The results are depicted as FIG. **5**. FIG. **5**A shows the quantification of insertions and deletions generated at the cut site of the β2M sgRNA. Genomic DNA was extracted from B cells nucleofected with Cas9 and β2M sgRNA 2 days after editing and Sanger sequencing was performed to quantify INDELs at the cut site. FIG. **5**B shows an overview of the insertions and deletions generated at the cut site of the β2M sgRNA.

Example 4. Genome Targeting with rAAV6 and Assessment of HDR-Mediated Targeted Integration

[0178] Next, genome targeting with an rAAV6 expressing either green fluorescent protein (GFP) or GPC3-CAR HDR donor cassettes was assessed for HDR-mediated targeted integration.

[0179] β2M-targeting constructs. β2M-targeting constructs (either GFP (SEQ ID NO:25) or GPC3-CAR (SEQ ID NO. 16)) were synthesized by Genscript (Piscataway, NJ, USA), and cloned into pAAV6 vector (CellBiolabs, San Diego, CA, USA). rAAV6 viruses were produced by Vigene Biosciences (Rockville, MD, USA). See, e.g., FIGS. **6** and **7**.

[0180] Human B cell isolation, activation, expansion and electroporation. The experimental design for the below described experiments is outlined in FIG. **8**A. First, primary human B cells were isolated, activated and as described in Example 1 and 2 above. The growth curve of the cultured human B cells (FIG. **8**B) and the viability of cultured human B cells was evaluated over the course of expansion (FIG. **8**C). At day 9, gene editing was performed as described below.

[0181] Nucleofection Transduction. B cells were first nucleofected with the β2M-specific RNP using the protocol described in Examples 1 and 2, and then immediately transduced with AAV6 donor at a multiplicity of infection (MOI) of 10,000 viral genomes (vg)/μl or 100,000 vg/μl to maximize efficiency of transduction (Bak et al., 2018; Charlesworth et al., 2018). B cells were cultured as for an additional 3 or 6 days and efficiency of integration was assessed.

[0182] Efficiency and Integration. Rates of targeted integration of the GFP and GPC3 donors were measured by flow cytometry 3 or 6 days after electroporation and AAV6 transduction. Targeted integration of the GFP and GPC3 expression cassettes was measured by flow cytometry using Attune NxT Flow Cytometer (Invitrogen, Carlsbad, CA, USA). GPC3-CAR was detected using a biotinylated human Glypican 3 with His and Avi-tag (GP3-H82E5, Acro Biosystem, Newark, DE, USA), conjugated to a BV421-labeled streptavidin (Biolegend, San Diego, CA, USA). Additionally, cells were stained with LIVE/DEAD™ Fixable Near-IR (Invitrogen, Carlsbad, CA, USA) to discriminate live and dead cells according to manufacturer's instructions.

[0183] Results. Promoter-less GFP targeting constructs encoded in an AAV6 virus were efficiently integrated into the B2M locus in activated human B cells, leading to protein expression detected by flow cytometry. See, e.g., FIGS. **9**A and **9**B. AAV6 mediated promoter-less GPC3-CAR targeting into the β2M locus achieved similar efficiencies in human B cells. FIGS. **9**C and **9**D. At MOI of 100K, the targeting efficiency was greater than 40% for both the GFP and GPC3 CAR constructs. FIGS. **9**A and **9**C. Further, there did not appear to be any reduction in B cell viability. FIGS. **9**B and **9**D.

Example 5. Genome Targeting with dsRNA and Assessment of HDR-Mediated Targeted Integration

[0184] Next, genome targeting with a dsDNA HDR construct expressing green fluorescent protein (GFP) was assessed for HDR-mediated targeted integration. See, e.g., FIG. **10**.

[0185] Human B cell isolation, activation, expansion and electroporation. First, primary human B cells were isolated, activated and as described in Example 1 and 2 above. The growth curve of the cultured human B cells and the viability of cultured human B cells was evaluated over the course of expansion. At day 9, gene editing was performed as described below.

[0186] dsRNA. For viral-free engineering, β2M-targeting constructs were amplified by PCR using Q5 High-Fidelity polymerase (VWR International, LLC, Radnor, PA, USA) with forward primer 5'-GCTATGTCCCAGGCACTCTAC-3' (SEQ ID NO: 29) and reverse primer 5'-AGGATGCTAGGACAGCAGGA-3' (SEQ ID NO: 30). PCR products were purified using NUCLEOSPIN® Gel and PCR Clean-Up kits (TaKaRa Bio, Mountain View, CA, USA).

[0187] Nucleofection Transduction. B cells were first nucleofected with the β2M-specific RNP using the protocol described in Examples 1 and 2.

[0188] Efficiency and Integration. Rates of targeted integration of the GFP and GPC3 donors were measured by flow cytometry 3 or 6 days after electroporation and AAV6 transduction. Targeted integration of the GFP and GPC3 expression cassettes was measured by flow cytometry using Attune NxT Flow Cytometer (Invitrogen, Carlsbad, CA, USA). GPC3-CAR was detected using a biotinylated human Glypican 3 with His and Avi-tag (GP3-H82E5, Acro Biosystem, Newark, DE, USA), conjugated to a BV421-labeled streptavidin (Biolegend, San Diego, CA, USA). Additionally, cells were stained with LIVE/DEAD™ Fixable Near-IR (Invitrogen, Carlsbad, CA, USA) to discriminate live and dead cells according to manufacturer's instructions.

[0189] Results. Promoter-less GFP targeting constructs encoded in a dsDNA were integrated into the β2M locus in activated human B cells, leading to protein expression detected by flow cytometry. See, e.g., FIG. **10**. When electroporated with 3 μg of the dsDNA construct alone, no GFP expression was observed. FIG. **10**A. But when the 3 μg of the dsDNA construct was electroporated with the RNP complex, significant GFP expression (indicative of integration into the β2M locus was observed. FIG. **10**B.

Example 6. Genome Targeting with rAAV6 and Assessment of GLA Expression in Engineered B Cells

[0190] Next, genome targeting with an rAAV6 expressing the wild type GLA protein was assessed for in vitro GLA expression and secretion by the engineered B cells. See, e.g., FIGS. **13** and **14**.

[0191] Human B cell isolation, activation, expansion and electroporation. First, primary human B cells were isolated, activated and as described in Example 1 and 2 above. At day 7, gene editing was performed as described below.

[0192] β2M-targeting constructs. β2M-targeting constructs (SEQ ID NO. 31) were synthesized by Genscript (Piscataway, NJ, USA), and cloned into pAAV6 vector (CellBiolabs, San Diego, CA, USA). rAAV6 viruses were produced by Vigene Biosciences (Rockville, MD, USA).

[0193] Nucleofection Transduction. B cells were first nucleofected with the β2M-specific RNP using the protocol described in Examples 1 and 2, and then immediately transduced with AAV6 donor at a multiplicity of infection (MOI) of 10,000 viral genomes (vg)/μl to maximize efficiency of transduction (Bak et al., 2018; Charlesworth et al., 2018). B cells were cultured as for an additional 5 days. Next, efficiency of integration was assessed using qualitative PCR and expression of GLA in the supernatant and B cell lysates using ELISA.

[0194] Efficiency of Integration And Transgene Expression. Rates of targeted integration of the GLA donors were measured by qualitative PCR 5 days after electroporation and AAV6 transduction. Intracellular and secreted GLA was measured using an ELISA assay. Results. Promoter-less GLA constructs encoded in an AAV6 virus were integrated into the B2M locus in activated human B cells with an efficiency of about 20 to 30%. B cells engineered with the Cas9 RNP-GLA rAAV6 demonstrated a significant increase in GLA expression intracellularly, as well as an increase in extracellular secretion of GLA.

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Leu Asn Gly Thr Asn Gln Ser Phe His Pro Phe Ile His Asn Ser Lys
            260                 265                 270

Pro His Pro Gly Gln Leu Trp Ala Ala Asp Gln Met Ile Ser Leu Leu
        275                 280                 285

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Arg Asp His Glu Leu Ile Gln Asp Arg Tyr Ser Leu Arg Cys Leu Pro
305                 310                 315                 320

Gln Tyr Leu Gly Pro Ile Val Asp Gly Ile Ser Gln Ile Ala Lys Gln
                325                 330                 335

Ile Glu Ile Glu Ile Asn Ser Val Thr Asp Asn Pro Leu Ile Asp Val
            340                 345                 350

Asp Asn Gln Ala Ser Tyr His Gly Gly Asn Phe Leu Gly Gln Tyr Val
        355                 360                 365

Gly Met Gly Met Asp His Leu Arg Tyr Tyr Ile Gly Leu Leu Ala Lys
    370                 375                 380

His Leu Asp Val Gln Ile Ala Leu Leu Ala Ser Pro Glu Phe Ser Asn
385                 390                 395                 400

Gly Leu Pro Pro Ser Leu Leu Gly Asn Arg Glu Arg Lys Val Asn Met
                405                 410                 415

Gly Leu Lys Gly Leu Gln Ile Cys Gly Asn Ser Ile Met Pro Leu Leu
            420                 425                 430

Thr Phe Tyr Gly Asn Ser Ile Ala Asp Arg Phe Pro Thr His Ala Glu
        435                 440                 445
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Gln Phe Asn Gln Asn Ile Asn Ser Gln Gly Tyr Thr Ser Ala Thr Leu
    450                 455                 460

Ala Arg Arg Ser Val Asp Ile Phe Gln Asn Tyr Val Ala Ile Ala Leu
465                 470                 475                 480

Met Phe Gly Val Gln Ala Val Asp Leu Arg Thr Tyr Lys Lys Thr Gly
                485                 490                 495

His Tyr Asp Ala Arg Ala Ser Leu Ser Pro Ala Thr Glu Arg Leu Tyr
            500                 505                 510

Ser Ala Val Arg His Val Val Gly Gln Lys Pro Thr Ser Asp Arg Pro
        515                 520                 525

Tyr Ile Trp Asn Asp Asn Glu Gln Gly Leu Asp Glu His Ile Ala Arg
    530                 535                 540

Ile Ser Ala Asp Ile Ala Ala Gly Gly Val Ile Val Gln Ala Val Gln
545                 550                 555                 560

Asp Ile Leu Pro Ser Leu His
                565


<210> SEQ ID NO 5
<211> LENGTH: 952
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 5

Met Gly Val Arg His Pro Pro Cys Ser His Arg Leu Leu Ala Val Cys
1               5                   10                  15

Ala Leu Val Ser Leu Ala Thr Ala Ala Leu Leu Gly His Ile Leu Leu
            20                  25                  30

His Asp Phe Leu Leu Val Pro Arg Glu Leu Ser Gly Ser Ser Pro Val
        35                  40                  45

Leu Glu Glu Thr His Pro Ala His Gln Gln Gly Ala Ser Arg Pro Gly
    50                  55                  60

Pro Arg Asp Ala Gln Ala His Pro Gly Arg Pro Arg Ala Val Pro Thr
65                  70                  75                  80

Gln Cys Asp Val Pro Pro Asn Ser Arg Phe Asp Cys Ala Pro Asp Lys
                85                  90                  95

Ala Ile Thr Gln Glu Gln Cys Glu Ala Arg Gly Cys Cys Tyr Ile Pro
            100                 105                 110

Ala Lys Gln Gly Leu Gln Gly Ala Gln Met Gly Gln Pro Trp Cys Phe
        115                 120                 125

Phe Pro Pro Ser Tyr Pro Ser Tyr Lys Leu Glu Asn Leu Ser Ser Ser
    130                 135                 140

Glu Met Gly Tyr Thr Ala Thr Leu Thr Arg Thr Thr Pro Thr Phe Phe
145                 150                 155                 160

Pro Lys Asp Ile Leu Thr Leu Arg Leu Asp Val Met Met Glu Thr Glu
                165                 170                 175

Asn Arg Leu His Phe Thr Ile Lys Asp Pro Ala Asn Arg Arg Tyr Glu
            180                 185                 190

Val Pro Leu Glu Thr Pro His Val His Ser Arg Ala Pro Ser Pro Leu
        195                 200                 205

Tyr Ser Val Glu Phe Ser Glu Glu Pro Phe Gly Val Ile Val Arg Arg
    210                 215                 220

Gln Leu Asp Gly Arg Val Leu Leu Asn Thr Thr Val Ala Pro Leu Phe
225                 230                 235                 240
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Phe Ala Asp Gln Phe Leu Gln Leu Ser Thr Ser Leu Pro Ser Gln Tyr
                245                 250                 255

Ile Thr Gly Leu Ala Glu His Leu Ser Pro Leu Met Leu Ser Thr Ser
            260                 265                 270

Trp Thr Arg Ile Thr Leu Trp Asn Arg Asp Leu Ala Pro Thr Pro Gly
        275                 280                 285

Ala Asn Leu Tyr Gly Ser His Pro Phe Tyr Leu Ala Leu Glu Asp Gly
    290                 295                 300

Gly Ser Ala His Gly Val Phe Leu Leu Asn Ser Asn Ala Met Asp Val
305                 310                 315                 320

Val Leu Gln Pro Ser Pro Ala Leu Ser Trp Arg Ser Thr Gly Gly Ile
                325                 330                 335

Leu Asp Val Tyr Ile Phe Leu Gly Pro Glu Pro Lys Ser Val Val Gln
            340                 345                 350

Gln Tyr Leu Asp Val Val Gly Tyr Pro Phe Met Pro Pro Tyr Trp Gly
        355                 360                 365

Leu Gly Phe His Leu Cys Arg Trp Gly Tyr Ser Ser Thr Ala Ile Thr
    370                 375                 380

Arg Gln Val Val Glu Asn Met Thr Arg Ala His Phe Pro Leu Asp Val
385                 390                 395                 400

Gln Trp Asn Asp Leu Asp Tyr Met Asp Ser Arg Arg Asp Phe Thr Phe
                405                 410                 415

Asn Lys Asp Gly Phe Arg Asp Phe Pro Ala Met Val Gln Glu Leu His
            420                 425                 430

Gln Gly Gly Arg Arg Tyr Met Met Ile Val Asp Pro Ala Ile Ser Ser
        435                 440                 445

Ser Gly Pro Ala Gly Ser Tyr Arg Pro Tyr Asp Glu Gly Leu Arg Arg
    450                 455                 460

Gly Val Phe Ile Thr Asn Glu Thr Gly Gln Pro Leu Ile Gly Lys Val
465                 470                 475                 480

Trp Pro Gly Ser Thr Ala Phe Pro Asp Phe Thr Asn Pro Thr Ala Leu
                485                 490                 495

Ala Trp Trp Glu Asp Met Val Ala Glu Phe His Asp Gln Val Pro Phe
            500                 505                 510

Asp Gly Met Trp Ile Asp Met Asn Glu Pro Ser Asn Phe Ile Arg Gly
        515                 520                 525

Ser Glu Asp Gly Cys Pro Asn Asn Glu Leu Glu Asn Pro Pro Tyr Val
    530                 535                 540

Pro Gly Val Val Gly Gly Thr Leu Gln Ala Ala Thr Ile Cys Ala Ser
545                 550                 555                 560

Ser His Gln Phe Leu Ser Thr His Tyr Asn Leu His Asn Leu Tyr Gly
                565                 570                 575

Leu Thr Glu Ala Ile Ala Ser His Arg Ala Leu Val Lys Ala Arg Gly
            580                 585                 590

Thr Arg Pro Phe Val Ile Ser Arg Ser Thr Phe Ala Gly His Gly Arg
        595                 600                 605

Tyr Ala Gly His Trp Thr Gly Asp Val Trp Ser Ser Trp Glu Gln Leu
    610                 615                 620

Ala Ser Ser Val Pro Glu Ile Leu Gln Phe Asn Leu Leu Gly Val Pro
625                 630                 635                 640
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Leu Val Gly Ala Asp Val Cys Gly Phe Leu Gly Asn Thr Ser Glu Glu
                645                 650                 655

Leu Cys Val Arg Trp Thr Gln Leu Gly Ala Phe Tyr Pro Phe Met Arg
            660                 665                 670

Asn His Asn Ser Leu Leu Ser Leu Pro Gln Glu Pro Tyr Ser Phe Ser
        675                 680                 685

Glu Pro Ala Gln Gln Ala Met Arg Lys Ala Leu Thr Leu Arg Tyr Ala
    690                 695                 700

Leu Leu Pro His Leu Tyr Thr Leu Phe His Gln Ala His Val Ala Gly
705                 710                 715                 720

Glu Thr Val Ala Arg Pro Leu Phe Leu Glu Phe Pro Lys Asp Ser Ser
                725                 730                 735

Thr Trp Thr Val Asp His Gln Leu Leu Trp Gly Glu Ala Leu Leu Ile
            740                 745                 750

Thr Pro Val Leu Gln Ala Gly Lys Ala Glu Val Thr Gly Tyr Phe Pro
        755                 760                 765

Leu Gly Thr Trp Tyr Asp Leu Gln Thr Val Pro Val Glu Ala Leu Gly
    770                 775                 780

Ser Leu Pro Pro Pro Pro Ala Ala Pro Arg Glu Pro Ala Ile His Ser
785                 790                 795                 800

Glu Gly Gln Trp Val Thr Leu Pro Ala Pro Leu Asp Thr Ile Asn Val
                805                 810                 815

His Leu Arg Ala Gly Tyr Ile Ile Pro Leu Gln Gly Pro Gly Leu Thr
            820                 825                 830

Thr Thr Glu Ser Arg Gln Gln Pro Met Ala Leu Ala Val Ala Leu Thr
        835                 840                 845

Lys Gly Gly Glu Ala Arg Gly Glu Leu Phe Trp Asp Asp Gly Glu Ser
    850                 855                 860

Leu Glu Val Leu Glu Arg Gly Ala Tyr Thr Gln Val Ile Phe Leu Ala
865                 870                 875                 880

Arg Asn Asn Thr Ile Val Asn Glu Leu Val Arg Val Thr Ser Glu Gly
                885                 890                 895

Ala Gly Leu Gln Leu Gln Lys Val Thr Val Leu Gly Val Ala Thr Ala
            900                 905                 910

Pro Gln Gln Val Leu Ser Asn Gly Val Pro Val Ser Asn Phe Thr Tyr
        915                 920                 925

Ser Pro Asp Thr Lys Val Leu Asp Ile Cys Val Ser Leu Leu Met Gly
    930                 935                 940

Glu Gln Phe Leu Val Ser Trp Cys
945                 950
```

<210> SEQ ID NO 6
<211> LENGTH: 1457
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 6

```
Met Gln Ile Glu Leu Ser Thr Cys Phe Phe Leu Cys Leu Leu Arg Phe
1               5                   10                  15

Cys Phe Ser Ala Thr Arg Arg Tyr Tyr Leu Gly Ala Val Glu Leu Ser
            20                  25                  30

Trp Asp Tyr Met Gln Ser Asp Leu Gly Glu Leu Pro Val Asp Ala Arg
        35                  40                  45
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Phe Pro Pro Arg Val Pro Lys Ser Phe Pro Phe Asn Thr Ser Val Val
    50                  55                  60

Tyr Lys Lys Thr Leu Phe Val Glu Phe Thr Asp His Leu Phe Asn Ile
65                  70                  75                  80

Ala Lys Pro Arg Pro Pro Trp Met Gly Leu Leu Gly Pro Thr Ile Gln
                85                  90                  95

Ala Glu Val Tyr Asp Thr Val Val Ile Thr Leu Lys Asn Met Ala Ser
            100                 105                 110

His Pro Val Ser Leu His Ala Val Gly Val Ser Tyr Trp Lys Ala Ser
        115                 120                 125

Glu Gly Ala Glu Tyr Asp Asp Gln Thr Ser Gln Arg Glu Lys Glu Asp
    130                 135                 140

Asp Lys Val Phe Pro Gly Gly Ser His Thr Tyr Val Trp Gln Val Leu
145                 150                 155                 160

Lys Glu Asn Gly Pro Met Ala Ser Asp Pro Leu Cys Leu Thr Tyr Ser
                165                 170                 175

Tyr Leu Ser His Val Asp Leu Val Lys Asp Leu Asn Ser Gly Leu Ile
            180                 185                 190

Gly Ala Leu Leu Val Cys Arg Glu Gly Ser Leu Ala Lys Glu Lys Thr
        195                 200                 205

Gln Thr Leu His Lys Phe Ile Leu Leu Phe Ala Val Phe Asp Glu Gly
    210                 215                 220

Lys Ser Trp His Ser Glu Thr Lys Asn Ser Leu Met Gln Asp Arg Asp
225                 230                 235                 240

Ala Ala Ser Ala Arg Ala Trp Pro Lys Met His Thr Val Asn Gly Tyr
                245                 250                 255

Val Asn Arg Ser Leu Pro Gly Leu Ile Gly Cys His Arg Lys Ser Val
            260                 265                 270

Tyr Trp His Val Ile Gly Met Gly Thr Thr Pro Glu Val His Ser Ile
        275                 280                 285

Phe Leu Glu Gly His Thr Phe Leu Val Arg Asn His Arg Gln Ala Ser
    290                 295                 300

Leu Glu Ile Ser Pro Ile Thr Phe Leu Thr Ala Gln Thr Leu Leu Met
305                 310                 315                 320

Asp Leu Gly Gln Phe Leu Leu Phe Cys His Ile Ser Ser His Gln His
                325                 330                 335

Asp Gly Met Glu Ala Tyr Val Lys Val Asp Ser Cys Pro Glu Glu Pro
            340                 345                 350

Gln Leu Arg Met Lys Asn Asn Glu Glu Ala Glu Asp Tyr Asp Asp Asp
        355                 360                 365

Leu Thr Asp Ser Glu Met Asp Val Val Arg Phe Asp Asp Asp Asn Ser
    370                 375                 380

Pro Ser Phe Ile Gln Ile Arg Ser Val Ala Lys Lys His Pro Lys Thr
385                 390                 395                 400

Trp Val His Tyr Ile Ala Ala Glu Glu Glu Asp Trp Asp Tyr Ala Pro
                405                 410                 415

Leu Val Leu Ala Pro Asp Asp Arg Ser Tyr Lys Ser Gln Tyr Leu Asn
            420                 425                 430

Asn Gly Pro Gln Arg Ile Gly Arg Lys Tyr Lys Lys Val Arg Phe Met
        435                 440                 445
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Ala Tyr Thr Asp Glu Thr Phe Lys Thr Arg Glu Ala Ile Gln His Glu
    450                 455                 460

Ser Gly Ile Leu Gly Pro Leu Leu Tyr Gly Glu Val Gly Asp Thr Leu
465                 470                 475                 480

Leu Ile Ile Phe Lys Asn Gln Ala Ser Arg Pro Tyr Asn Ile Tyr Pro
                485                 490                 495

His Gly Ile Thr Asp Val Arg Pro Leu Tyr Ser Arg Arg Leu Pro Lys
            500                 505                 510

Gly Val Lys His Leu Lys Asp Phe Pro Ile Leu Pro Gly Glu Ile Phe
        515                 520                 525

Lys Tyr Lys Trp Thr Val Thr Val Glu Asp Gly Pro Thr Lys Ser Asp
    530                 535                 540

Pro Arg Cys Leu Thr Arg Tyr Tyr Ser Ser Phe Val Asn Met Glu Arg
545                 550                 555                 560

Asp Leu Ala Ser Gly Leu Ile Gly Pro Leu Leu Ile Cys Tyr Lys Glu
                565                 570                 575

Ser Val Asp Gln Arg Gly Asn Gln Ile Met Ser Asp Lys Arg Asn Val
            580                 585                 590

Ile Leu Phe Ser Val Phe Asp Glu Asn Arg Ser Trp Tyr Leu Thr Glu
        595                 600                 605

Asn Ile Gln Arg Phe Leu Pro Asn Pro Ala Gly Val Gln Leu Glu Asp
    610                 615                 620

Pro Glu Phe Gln Ala Ser Asn Ile Met His Ser Ile Asn Gly Tyr Val
625                 630                 635                 640

Phe Asp Ser Leu Gln Leu Ser Val Cys Leu His Glu Val Ala Tyr Trp
                645                 650                 655

Tyr Ile Leu Ser Ile Gly Ala Gln Thr Asp Phe Leu Ser Val Phe Phe
            660                 665                 670

Ser Gly Tyr Thr Phe Lys His Lys Met Val Tyr Glu Asp Thr Leu Thr
        675                 680                 685

Leu Phe Pro Phe Ser Gly Glu Thr Val Phe Met Ser Met Glu Asn Pro
    690                 695                 700

Gly Leu Trp Ile Leu Gly Cys His Asn Ser Asp Phe Arg Asn Arg Gly
705                 710                 715                 720

Met Thr Ala Leu Leu Lys Val Ser Ser Cys Asp Lys Asn Thr Gly Asp
                725                 730                 735

Tyr Tyr Glu Asp Ser Tyr Glu Asp Ile Ser Ala Tyr Leu Leu Ser Lys
            740                 745                 750

Asn Asn Ala Ile Glu Pro Arg Ser Phe Ser Gln Asn Pro Pro Val Leu
        755                 760                 765

Lys Arg His Gln Arg Glu Ile Thr Arg Thr Thr Leu Gln Ser Asp Gln
    770                 775                 780

Glu Glu Ile Asp Tyr Asp Asp Thr Ile Ser Val Glu Met Lys Lys Glu
785                 790                 795                 800

Asp Phe Asp Ile Tyr Asp Glu Asp Glu Asn Gln Ser Pro Arg Ser Phe
                805                 810                 815

Gln Lys Lys Thr Arg His Tyr Phe Ile Ala Ala Val Glu Arg Leu Trp
            820                 825                 830

Asp Tyr Gly Met Ser Ser Ser Pro His Val Leu Arg Asn Arg Ala Gln
        835                 840                 845

Ser Gly Ser Val Pro Gln Phe Lys Lys Val Val Phe Gln Glu Phe Thr
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    850                 855                 860

Asp Gly Ser Phe Thr Gln Pro Leu Tyr Arg Gly Glu Leu Asn Glu His
865                 870                 875                 880

Leu Gly Leu Leu Gly Pro Tyr Ile Arg Ala Glu Val Glu Asp Asn Ile
                885                 890                 895

Met Val Thr Phe Arg Asn Gln Ala Ser Arg Pro Tyr Ser Phe Tyr Ser
            900                 905                 910

Ser Leu Ile Ser Tyr Glu Glu Asp Gln Arg Gln Gly Ala Glu Pro Arg
        915                 920                 925

Lys Asn Phe Val Lys Pro Asn Glu Thr Lys Thr Tyr Phe Trp Lys Val
    930                 935                 940

Gln His His Met Ala Pro Thr Lys Asp Glu Phe Asp Cys Lys Ala Trp
945                 950                 955                 960

Ala Tyr Phe Ser Asp Val Asp Leu Glu Lys Asp Val His Ser Gly Leu
                965                 970                 975

Ile Gly Pro Leu Leu Val Cys His Thr Asn Thr Leu Asn Pro Ala His
            980                 985                 990

Gly Arg Gln Val Thr Val Gln Glu  Phe Ala Leu Phe Phe  Thr Ile Phe
        995                 1000                1005

Asp Glu  Thr Lys Ser Trp Tyr  Phe Thr Glu Asn Met  Glu Arg Asn
    1010                1015                1020

Cys Arg  Ala Pro Cys Asn Ile  Gln Met Glu Asp Pro  Thr Phe Lys
    1025                1030                1035

Glu Asn  Tyr Arg Phe His Ala  Ile Asn Gly Tyr Ile  Met Asp Thr
    1040                1045                1050

Leu Pro  Gly Leu Val Met Ala  Gln Asp Gln Arg Ile  Arg Trp Tyr
    1055                1060                1065

Leu Leu  Ser Met Gly Ser Asn  Glu Asn Ile His Ser  Ile His Phe
    1070                1075                1080

Ser Gly  His Val Phe Thr Val  Arg Lys Lys Glu Glu  Tyr Lys Met
    1085                1090                1095

Ala Leu  Tyr Asn Leu Tyr Pro  Gly Val Phe Glu Thr  Val Glu Met
    1100                1105                1110

Leu Pro  Ser Lys Ala Gly Ile  Trp Arg Val Glu Cys  Leu Ile Gly
    1115                1120                1125

Glu His  Leu His Ala Gly Met  Ser Thr Leu Phe Leu  Val Tyr Ser
    1130                1135                1140

Asn Lys  Cys Gln Thr Pro Leu  Gly Met Ala Ser Gly  His Ile Arg
    1145                1150                1155

Asp Phe  Gln Ile Thr Ala Ser  Gly Gln Tyr Gly Gln  Trp Ala Pro
    1160                1165                1170

Lys Leu  Ala Arg Leu His Tyr  Ser Gly Ser Ile Asn  Ala Trp Ser
    1175                1180                1185

Thr Lys  Glu Pro Phe Ser Trp  Ile Lys Val Asp Leu  Leu Ala Pro
    1190                1195                1200

Met Ile  Ile His Gly Ile Lys  Thr Gln Gly Ala Arg  Gln Lys Phe
    1205                1210                1215

Ser Ser  Leu Tyr Ile Ser Gln  Phe Ile Ile Met Tyr  Ser Leu Asp
    1220                1225                1230

Gly Lys  Lys Trp Gln Thr Tyr  Arg Gly Asn Ser Thr  Gly Thr Leu
    1235                1240                1245
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Met Val Phe Phe Gly Asn Val Asp Ser Ser Gly Ile Lys His Asn
1250 1255 1260

Ile Phe Asn Pro Pro Ile Ile Ala Arg Tyr Ile Arg Leu His Pro
1265 1270 1275

Thr His Tyr Ser Ile Arg Ser Thr Leu Arg Met Glu Leu Met Gly
1280 1285 1290

Cys Asp Leu Asn Ser Cys Ser Met Pro Leu Gly Met Glu Ser Lys
1295 1300 1305

Ala Ile Ser Asp Ala Gln Ile Thr Ala Ser Ser Tyr Phe Thr Asn
1310 1315 1320

Met Phe Ala Thr Trp Ser Pro Ser Lys Ala Arg Leu His Leu Gln
1325 1330 1335

Gly Arg Ser Asn Ala Trp Arg Pro Gln Val Asn Asn Pro Lys Glu
1340 1345 1350

Trp Leu Gln Val Asp Phe Gln Lys Thr Val Lys Val Thr Gly Val
1355 1360 1365

Thr Thr Gln Gly Val Lys Ser Leu Leu Thr Ser Met Tyr Val Lys
1370 1375 1380

Glu Phe Leu Ile Ser Ser Ser Gln Asp Gly His Gln Trp Thr Leu
1385 1390 1395

Phe Phe Gln Asn Gly Lys Val Lys Val Phe Gln Gly Asn Gln Asp
1400 1405 1410

Ser Phe Thr Pro Val Val Asn Ser Leu Asp Pro Pro Leu Leu Thr
1415 1420 1425

Arg Tyr Leu Arg Ile His Pro Gln Ser Trp Val His Gln Ile Ala
1430 1435 1440

Leu Arg Met Glu Val Leu Gly Cys Glu Ala Gln Asp Leu Tyr
1445 1450 1455

<210> SEQ ID NO 7
<211> LENGTH: 2351
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polypeptide

<400> SEQUENCE: 7

Met Gln Ile Glu Leu Ser Thr Cys Phe Phe Leu Cys Leu Leu Arg Phe
1 5 10 15

Cys Phe Ser Ala Thr Arg Arg Tyr Tyr Leu Gly Ala Val Glu Leu Ser
20 25 30

Trp Asp Tyr Met Gln Ser Asp Leu Gly Glu Leu Pro Val Asp Ala Arg
35 40 45

Phe Pro Pro Arg Val Pro Lys Ser Phe Pro Phe Asn Thr Ser Val Val
50 55 60

Tyr Lys Lys Thr Leu Phe Val Glu Phe Thr Asp His Leu Phe Asn Ile
65 70 75 80

Ala Lys Pro Arg Pro Pro Trp Met Gly Leu Leu Gly Pro Thr Ile Gln
85 90 95

Ala Glu Val Tyr Asp Thr Val Val Ile Thr Leu Lys Asn Met Ala Ser
100 105 110

His Pro Val Ser Leu His Ala Val Gly Val Ser Tyr Trp Lys Ala Ser

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        115                 120                 125

Glu Gly Ala Glu Tyr Asp Asp Gln Thr Ser Gln Arg Glu Lys Glu Asp
    130                 135                 140

Asp Lys Val Phe Pro Gly Gly Ser His Thr Tyr Val Trp Gln Val Leu
145                 150                 155                 160

Lys Glu Asn Gly Pro Met Ala Ser Asp Pro Leu Cys Leu Thr Tyr Ser
                165                 170                 175

Tyr Leu Ser His Val Asp Leu Val Lys Asp Leu Asn Ser Gly Leu Ile
            180                 185                 190

Gly Ala Leu Leu Val Cys Arg Glu Gly Ser Leu Ala Lys Glu Lys Thr
        195                 200                 205

Gln Thr Leu His Lys Phe Ile Leu Leu Phe Ala Val Phe Asp Glu Gly
    210                 215                 220

Lys Ser Trp His Ser Glu Thr Lys Asn Ser Leu Met Gln Asp Arg Asp
225                 230                 235                 240

Ala Ala Ser Ala Arg Ala Trp Pro Lys Met His Thr Val Asn Gly Tyr
                245                 250                 255

Val Asn Arg Ser Leu Pro Gly Leu Ile Gly Cys His Arg Lys Ser Val
            260                 265                 270

Tyr Trp His Val Ile Gly Met Gly Thr Thr Pro Glu Val His Ser Ile
        275                 280                 285

Phe Leu Glu Gly His Thr Phe Leu Val Arg Asn His Arg Gln Ala Ser
    290                 295                 300

Leu Glu Ile Ser Pro Ile Thr Phe Leu Thr Ala Gln Thr Leu Leu Met
305                 310                 315                 320

Asp Leu Gly Gln Phe Leu Leu Phe Cys His Ile Ser Ser His Gln His
                325                 330                 335

Asp Gly Met Glu Ala Tyr Val Lys Val Asp Ser Cys Pro Glu Glu Pro
            340                 345                 350

Gln Leu Arg Met Lys Asn Asn Glu Glu Ala Glu Asp Tyr Asp Asp Asp
        355                 360                 365

Leu Thr Asp Ser Glu Met Asp Val Val Arg Phe Asp Asp Asp Asn Ser
    370                 375                 380

Pro Ser Phe Ile Gln Ile Arg Ser Val Ala Lys Lys His Pro Lys Thr
385                 390                 395                 400

Trp Val His Tyr Ile Ala Ala Glu Glu Glu Asp Trp Asp Tyr Ala Pro
                405                 410                 415

Leu Val Leu Ala Pro Asp Asp Arg Ser Tyr Lys Ser Gln Tyr Leu Asn
            420                 425                 430

Asn Gly Pro Gln Arg Ile Gly Arg Lys Tyr Lys Lys Val Arg Phe Met
        435                 440                 445

Ala Tyr Thr Asp Glu Thr Phe Lys Thr Arg Glu Ala Ile Gln His Glu
    450                 455                 460

Ser Gly Ile Leu Gly Pro Leu Leu Tyr Gly Glu Val Gly Asp Thr Leu
465                 470                 475                 480

Leu Ile Ile Phe Lys Asn Gln Ala Ser Arg Pro Tyr Asn Ile Tyr Pro
                485                 490                 495

His Gly Ile Thr Asp Val Arg Pro Leu Tyr Ser Arg Arg Leu Pro Lys
            500                 505                 510

Gly Val Lys His Leu Lys Asp Phe Pro Ile Leu Pro Gly Glu Ile Phe
        515                 520                 525
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Lys Tyr Lys Trp Thr Val Thr Val Glu Asp Gly Pro Thr Lys Ser Asp
    530                 535                 540

Pro Arg Cys Leu Thr Arg Tyr Tyr Ser Ser Phe Val Asn Met Glu Arg
545                 550                 555                 560

Asp Leu Ala Ser Gly Leu Ile Gly Pro Leu Leu Ile Cys Tyr Lys Glu
                565                 570                 575

Ser Val Asp Gln Arg Gly Asn Gln Ile Met Ser Asp Lys Arg Asn Val
            580                 585                 590

Ile Leu Phe Ser Val Phe Asp Glu Asn Arg Ser Trp Tyr Leu Thr Glu
        595                 600                 605

Asn Ile Gln Arg Phe Leu Pro Asn Pro Ala Gly Val Gln Leu Glu Asp
    610                 615                 620

Pro Glu Phe Gln Ala Ser Asn Ile Met His Ser Ile Asn Gly Tyr Val
625                 630                 635                 640

Phe Asp Ser Leu Gln Leu Ser Val Cys Leu His Glu Val Ala Tyr Trp
                645                 650                 655

Tyr Ile Leu Ser Ile Gly Ala Gln Thr Asp Phe Leu Ser Val Phe Phe
            660                 665                 670

Ser Gly Tyr Thr Phe Lys His Lys Met Val Tyr Glu Asp Thr Leu Thr
        675                 680                 685

Leu Phe Pro Phe Ser Gly Glu Thr Val Phe Met Ser Met Glu Asn Pro
    690                 695                 700

Gly Leu Trp Ile Leu Gly Cys His Asn Ser Asp Phe Arg Asn Arg Gly
705                 710                 715                 720

Met Thr Ala Leu Leu Lys Val Ser Ser Cys Asp Lys Asn Thr Gly Asp
                725                 730                 735

Tyr Tyr Glu Asp Ser Tyr Glu Asp Ile Ser Ala Tyr Leu Leu Ser Lys
            740                 745                 750

Asn Asn Ala Ile Glu Pro Arg Ser Phe Ser Gln Asn Ser Arg His Pro
        755                 760                 765

Ser Thr Arg Gln Lys Gln Phe Asn Ala Thr Thr Ile Pro Glu Asn Asp
    770                 775                 780

Ile Glu Lys Thr Asp Pro Trp Phe Ala His Arg Thr Pro Met Pro Lys
785                 790                 795                 800

Ile Gln Asn Val Ser Ser Ser Asp Leu Leu Met Leu Leu Arg Gln Ser
                805                 810                 815

Pro Thr Pro His Gly Leu Ser Leu Ser Asp Leu Gln Glu Ala Lys Tyr
            820                 825                 830

Glu Thr Phe Ser Asp Asp Pro Ser Pro Gly Ala Ile Asp Ser Asn Asn
        835                 840                 845

Ser Leu Ser Glu Met Thr His Phe Arg Pro Gln Leu His His Ser Gly
    850                 855                 860

Asp Met Val Phe Thr Pro Glu Ser Gly Leu Gln Leu Arg Leu Asn Glu
865                 870                 875                 880

Lys Leu Gly Thr Thr Ala Ala Thr Glu Leu Lys Lys Leu Asp Phe Lys
                885                 890                 895

Val Ser Ser Thr Ser Asn Asn Leu Ile Ser Thr Ile Pro Ser Asp Asn
            900                 905                 910

Leu Ala Ala Gly Thr Asp Asn Thr Ser Ser Leu Gly Pro Pro Ser Met
        915                 920                 925
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Pro Val His Tyr Asp Ser Gln Leu Asp Thr Thr Leu Phe Gly Lys Lys
    930                 935                 940

Ser Ser Pro Leu Thr Glu Ser Gly Gly Pro Leu Ser Leu Ser Glu Glu
945                 950                 955                 960

Asn Asn Asp Ser Lys Leu Leu Glu Ser Gly Leu Met Asn Ser Gln Glu
                965                 970                 975

Ser Ser Trp Gly Lys Asn Val Ser Ser Thr Glu Ser Gly Arg Leu Phe
            980                 985                 990

Lys Gly Lys Arg Ala His Gly Pro  Ala Leu Leu Thr Lys  Asp Asn Ala
        995                 1000                 1005

Leu Phe  Lys Val Ser Ile Ser  Leu Leu Lys Thr Asn  Lys Thr Ser
    1010                 1015                 1020

Asn Asn  Ser Ala Thr Asn Arg  Lys Thr His Ile Asp  Gly Pro Ser
    1025                 1030                 1035

Leu Leu  Ile Glu Asn Ser Pro  Ser Val Trp Gln Asn  Ile Leu Glu
    1040                 1045                 1050

Ser Asp  Thr Glu Phe Lys Lys  Val Thr Pro Leu Ile  His Asp Arg
    1055                 1060                 1065

Met Leu  Met Asp Lys Asn Ala  Thr Ala Leu Arg Leu  Asn His Met
    1070                 1075                 1080

Ser Asn  Lys Thr Thr Ser Ser  Lys Asn Met Glu Met  Val Gln Gln
    1085                 1090                 1095

Lys Lys  Glu Gly Pro Ile Pro  Pro Asp Ala Gln Asn  Pro Asp Met
    1100                 1105                 1110

Ser Phe  Phe Lys Met Leu Phe  Leu Pro Glu Ser Ala  Arg Trp Ile
    1115                 1120                 1125

Gln Arg  Thr His Gly Lys Asn  Ser Leu Asn Ser Gly  Gln Gly Pro
    1130                 1135                 1140

Ser Pro  Lys Gln Leu Val Ser  Leu Gly Pro Glu Lys  Ser Val Glu
    1145                 1150                 1155

Gly Gln  Asn Phe Leu Ser Glu  Lys Asn Lys Val Val  Val Gly Lys
    1160                 1165                 1170

Gly Glu  Phe Thr Lys Asp Val  Gly Leu Lys Glu Met  Val Phe Pro
    1175                 1180                 1185

Ser Ser  Arg Asn Leu Phe Leu  Thr Asn Leu Asp Asn  Leu His Glu
    1190                 1195                 1200

Asn Asn  Thr His Asn Gln Glu  Lys Lys Ile Gln Glu  Glu Ile Glu
    1205                 1210                 1215

Lys Lys  Glu Thr Leu Ile Gln  Glu Asn Val Val Leu  Pro Gln Ile
    1220                 1225                 1230

His Thr  Val Thr Gly Thr Lys  Asn Phe Met Lys Asn  Leu Phe Leu
    1235                 1240                 1245

Leu Ser  Thr Arg Gln Asn Val  Glu Gly Ser Tyr Asp  Gly Ala Tyr
    1250                 1255                 1260

Ala Pro  Val Leu Gln Asp Phe  Arg Ser Leu Asn Asp  Ser Thr Asn
    1265                 1270                 1275

Arg Thr  Lys Lys His Thr Ala  His Phe Ser Lys Lys  Gly Glu Glu
    1280                 1285                 1290

Glu Asn  Leu Glu Gly Leu Gly  Asn Gln Thr Lys Gln  Ile Val Glu
    1295                 1300                 1305

Lys Tyr  Ala Cys Thr Thr Arg  Ile Ser Pro Asn Thr  Ser Gln Gln
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    1310                1315                1320

Asn Phe  Val Thr Gln Arg Ser  Lys Arg Ala Leu Lys  Gln Phe Arg
    1325                1330                1335

Leu Pro  Leu Glu Glu Thr Glu  Leu Glu Lys Arg Ile  Ile Val Asp
    1340                1345                1350

Asp Thr  Ser Thr Gln Trp Ser  Lys Asn Met Lys His  Leu Thr Pro
    1355                1360                1365

Ser Thr  Leu Thr Gln Ile Asp  Tyr Asn Glu Lys Glu  Lys Gly Ala
    1370                1375                1380

Ile Thr  Gln Ser Pro Leu Ser  Asp Cys Leu Thr Arg  Ser His Ser
    1385                1390                1395

Ile Pro  Gln Ala Asn Arg Ser  Pro Leu Pro Ile Ala  Lys Val Ser
    1400                1405                1410

Ser Phe  Pro Ser Ile Arg Pro  Ile Tyr Leu Thr Arg  Val Leu Phe
    1415                1420                1425

Gln Asp  Asn Ser Ser His Leu  Pro Ala Ala Ser Tyr  Arg Lys Lys
    1430                1435                1440

Asp Ser  Gly Val Gln Glu Ser  Ser His Phe Leu Gln  Gly Ala Lys
    1445                1450                1455

Lys Asn  Asn Leu Ser Leu Ala  Ile Leu Thr Leu Glu  Met Thr Gly
    1460                1465                1470

Asp Gln  Arg Glu Val Gly Ser  Leu Gly Thr Ser Ala  Thr Asn Ser
    1475                1480                1485

Val Thr  Tyr Lys Lys Val Glu  Asn Thr Val Leu Pro  Lys Pro Asp
    1490                1495                1500

Leu Pro  Lys Thr Ser Gly Lys  Val Glu Leu Leu Pro  Lys Val His
    1505                1510                1515

Ile Tyr  Gln Lys Asp Leu Phe  Pro Thr Glu Thr Ser  Asn Gly Ser
    1520                1525                1530

Pro Gly  His Leu Asp Leu Val  Glu Gly Ser Leu Leu  Gln Gly Thr
    1535                1540                1545

Glu Gly  Ala Ile Lys Trp Asn  Glu Ala Asn Arg Pro  Gly Lys Val
    1550                1555                1560

Pro Phe  Leu Arg Val Ala Thr  Glu Ser Ser Ala Lys  Thr Pro Ser
    1565                1570                1575

Lys Leu  Leu Asp Pro Leu Ala  Trp Asp Asn His Tyr  Gly Thr Gln
    1580                1585                1590

Ile Pro  Lys Glu Glu Trp Lys  Ser Gln Glu Lys Ser  Pro Glu Lys
    1595                1600                1605

Thr Ala  Phe Lys Lys Lys Asp  Thr Ile Leu Ser Leu  Asn Ala Cys
    1610                1615                1620

Glu Ser  Asn His Ala Ile Ala  Ala Ile Asn Glu Gly  Gln Asn Lys
    1625                1630                1635

Pro Glu  Ile Glu Val Thr Trp  Ala Lys Gln Gly Arg  Thr Glu Arg
    1640                1645                1650

Leu Cys  Ser Gln Asn Pro Pro  Val Leu Lys Arg His  Gln Arg Glu
    1655                1660                1665

Ile Thr  Arg Thr Thr Leu Gln  Ser Asp Gln Glu Glu  Ile Asp Tyr
    1670                1675                1680

Asp Asp  Thr Ile Ser Val Glu  Met Lys Lys Glu Asp  Phe Asp Ile
    1685                1690                1695
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Tyr Asp  Glu Asp Glu Asn Gln  Ser Pro Arg Ser Phe  Gln Lys Lys
    1700                 1705                 1710

Thr Arg  His Tyr Phe Ile Ala  Ala Val Glu Arg Leu  Trp Asp Tyr
    1715                 1720                 1725

Gly Met  Ser Ser Ser Pro His  Val Leu Arg Asn Arg  Ala Gln Ser
    1730                 1735                 1740

Gly Ser  Val Pro Gln Phe Lys  Lys Val Val Phe Gln  Glu Phe Thr
    1745                 1750                 1755

Asp Gly  Ser Phe Thr Gln Pro  Leu Tyr Arg Gly Glu  Leu Asn Glu
    1760                 1765                 1770

His Leu  Gly Leu Leu Gly Pro  Tyr Ile Arg Ala Glu  Val Glu Asp
    1775                 1780                 1785

Asn Ile  Met Val Thr Phe Arg  Asn Gln Ala Ser Arg  Pro Tyr Ser
    1790                 1795                 1800

Phe Tyr  Ser Ser Leu Ile Ser  Tyr Glu Glu Asp Gln  Arg Gln Gly
    1805                 1810                 1815

Ala Glu  Pro Arg Lys Asn Phe  Val Lys Pro Asn Glu  Thr Lys Thr
    1820                 1825                 1830

Tyr Phe  Trp Lys Val Gln His  His Met Ala Pro Thr  Lys Asp Glu
    1835                 1840                 1845

Phe Asp  Cys Lys Ala Trp Ala  Tyr Phe Ser Asp Val  Asp Leu Glu
    1850                 1855                 1860

Lys Asp  Val His Ser Gly Leu  Ile Gly Pro Leu Leu  Val Cys His
    1865                 1870                 1875

Thr Asn  Thr Leu Asn Pro Ala  His Gly Arg Gln Val  Thr Val Gln
    1880                 1885                 1890

Glu Phe  Ala Leu Phe Phe Thr  Ile Phe Asp Glu Thr  Lys Ser Trp
    1895                 1900                 1905

Tyr Phe  Thr Glu Asn Met Glu  Arg Asn Cys Arg Ala  Pro Cys Asn
    1910                 1915                 1920

Ile Gln  Met Glu Asp Pro Thr  Phe Lys Glu Asn Tyr  Arg Phe His
    1925                 1930                 1935

Ala Ile  Asn Gly Tyr Ile Met  Asp Thr Leu Pro Gly  Leu Val Met
    1940                 1945                 1950

Ala Gln  Asp Gln Arg Ile Arg  Trp Tyr Leu Leu Ser  Met Gly Ser
    1955                 1960                 1965

Asn Glu  Asn Ile His Ser Ile  His Phe Ser Gly His  Val Phe Thr
    1970                 1975                 1980

Val Arg  Lys Lys Glu Glu Tyr  Lys Met Ala Leu Tyr  Asn Leu Tyr
    1985                 1990                 1995

Pro Gly  Val Phe Glu Thr Val  Glu Met Leu Pro Ser  Lys Ala Gly
    2000                 2005                 2010

Ile Trp  Arg Val Glu Cys Leu  Ile Gly Glu His Leu  His Ala Gly
    2015                 2020                 2025

Met Ser  Thr Leu Phe Leu Val  Tyr Ser Asn Lys Cys  Gln Thr Pro
    2030                 2035                 2040

Leu Gly  Met Ala Ser Gly His  Ile Arg Asp Phe Gln  Ile Thr Ala
    2045                 2050                 2055

Ser Gly  Gln Tyr Gly Gln Trp  Ala Pro Lys Leu Ala  Arg Leu His
    2060                 2065                 2070
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Tyr Ser Gly Ser Ile Asn Ala Trp Ser Thr Lys Glu Pro Phe Ser
2075 2080 2085

Trp Ile Lys Val Asp Leu Leu Ala Pro Met Ile Ile His Gly Ile
2090 2095 2100

Lys Thr Gln Gly Ala Arg Gln Lys Phe Ser Ser Leu Tyr Ile Ser
2105 2110 2115

Gln Phe Ile Ile Met Tyr Ser Leu Asp Gly Lys Lys Trp Gln Thr
2120 2125 2130

Tyr Arg Gly Asn Ser Thr Gly Thr Leu Met Val Phe Phe Gly Asn
2135 2140 2145

Val Asp Ser Ser Gly Ile Lys His Asn Ile Phe Asn Pro Pro Ile
2150 2155 2160

Ile Ala Arg Tyr Ile Arg Leu His Pro Thr His Tyr Ser Ile Arg
2165 2170 2175

Ser Thr Leu Arg Met Glu Leu Met Gly Cys Asp Leu Asn Ser Cys
2180 2185 2190

Ser Met Pro Leu Gly Met Glu Ser Lys Ala Ile Ser Asp Ala Gln
2195 2200 2205

Ile Thr Ala Ser Ser Tyr Phe Thr Asn Met Phe Ala Thr Trp Ser
2210 2215 2220

Pro Ser Lys Ala Arg Leu His Leu Gln Gly Arg Ser Asn Ala Trp
2225 2230 2235

Arg Pro Gln Val Asn Asn Pro Lys Glu Trp Leu Gln Val Asp Phe
2240 2245 2250

Gln Lys Thr Met Lys Val Thr Gly Val Thr Thr Gln Gly Val Lys
2255 2260 2265

Ser Leu Leu Thr Ser Met Tyr Val Lys Glu Phe Leu Ile Ser Ser
2270 2275 2280

Ser Gln Asp Gly His Gln Trp Thr Leu Phe Phe Gln Asn Gly Lys
2285 2290 2295

Val Lys Val Phe Gln Gly Asn Gln Asp Ser Phe Thr Pro Val Val
2300 2305 2310

Asn Ser Leu Asp Pro Pro Leu Leu Thr Arg Tyr Leu Arg Ile His
2315 2320 2325

Pro Gln Ser Trp Val His Gln Ile Ala Leu Arg Met Glu Val Leu
2330 2335 2340

Gly Cys Glu Ala Gln Asp Leu Tyr
2345 2350

<210> SEQ ID NO 8
<211> LENGTH: 416
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polypeptide

<400> SEQUENCE: 8

Met Glu Thr Asp Thr Leu Leu Leu Trp Val Leu Leu Leu Trp Val Pro
1 5 10 15

Gly Ser Thr Gly Gln Ser Val Leu Thr Gln Pro Pro Ser Val Ser Ala
20 25 30

Ala Pro Gly Gln Arg Val Thr Ile Ser Cys Ser Gly Thr Arg Ser Asn
35 40 45

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Ile Gly Ser Asp Tyr Val Ser Trp Tyr Gln His Leu Pro Gly Thr Ala
    50                  55                  60

Pro Lys Leu Leu Val Tyr Gly Asp Asn Leu Arg Pro Ser Gly Ile Pro
65                  70                  75                  80

Asp Arg Phe Ser Ala Ser Lys Ser Gly Thr Ser Ala Thr Leu Gly Ile
                85                  90                  95

Thr Gly Leu Gln Thr Gly Asp Glu Ala Asp Tyr Tyr Cys Gly Thr Trp
            100                 105                 110

Asp Tyr Thr Leu Asn Gly Val Val Phe Gly Gly Gly Thr Lys Leu Thr
        115                 120                 125

Val Leu Gly Ser Arg Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly
    130                 135                 140

Gly Gly Gly Ser Leu Glu Met Ala Gln Val Gln Leu Val Glu Ser Gly
145                 150                 155                 160

Gly Gly Leu Val Gln Pro Gly Gly Ser Leu Arg Leu Ser Cys Ala Ala
                165                 170                 175

Ser Gly Phe Thr Phe Ser Ser Tyr Ala Met Ser Trp Val Arg Gln Ala
            180                 185                 190

Pro Gly Lys Gly Leu Glu Trp Val Ser Val Ile Tyr Ser Gly Gly Ser
        195                 200                 205

Ser Thr Tyr Tyr Ala Asp Ser Val Lys Gly Arg Phe Thr Ile Ser Arg
    210                 215                 220

Asp Asn Ser Lys Asn Thr Leu Tyr Leu Gln Met Asn Ser Leu Arg Ala
225                 230                 235                 240

Glu Asp Thr Ala Val Tyr Tyr Cys Ala Arg Thr Ser Tyr Leu Asn His
                245                 250                 255

Gly Asp Tyr Trp Gly Gln Gly Thr Leu Val Thr Val Ser Ser Ala Ala
            260                 265                 270

Ala Phe Val Pro Val Phe Leu Pro Ala Lys Pro Thr Thr Thr Pro Ala
        275                 280                 285

Pro Arg Pro Pro Thr Pro Ala Pro Thr Ile Ala Ser Gln Pro Leu Ser
    290                 295                 300

Leu Arg Pro Glu Ala Cys Arg Pro Ala Ala Gly Gly Ala Val His Thr
305                 310                 315                 320

Arg Gly Leu Asp Phe Ala Cys Asp Phe Trp Val Leu Val Val Val Gly
                325                 330                 335

Gly Val Leu Ala Cys Tyr Ser Leu Leu Val Thr Val Ala Phe Ile Ile
            340                 345                 350

Phe Trp Val Arg Lys Arg Trp Gln Asn Glu Lys Leu Gly Leu Asp Ala
        355                 360                 365

Gly Asp Glu Tyr Glu Asp Glu Asn Leu Tyr Glu Gly Leu Asn Leu Asp
    370                 375                 380

Asp Cys Ser Met Tyr Glu Asp Ile Ser Arg Gly Leu Gln Gly Thr Tyr
385                 390                 395                 400

Gln Asp Val Gly Ser Leu Asn Ile Gly Asp Val Gln Leu Glu Lys Pro
                405                 410                 415
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<210> SEQ ID NO 9
<211> LENGTH: 178
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

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<400> SEQUENCE: 9

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Met His Ser Ser Ala Leu Leu Cys Cys Leu Val Leu Leu Thr Gly Val
1               5                   10                  15

Arg Ala Ser Pro Gly Gln Gly Thr Gln Ser Glu Asn Ser Cys Thr His
            20                  25                  30

Phe Pro Gly Asn Leu Pro Asn Met Leu Arg Asp Leu Arg Asp Ala Phe
        35                  40                  45

Ser Arg Val Lys Thr Phe Phe Gln Met Lys Asp Gln Leu Asp Asn Leu
    50                  55                  60

Leu Leu Lys Glu Ser Leu Leu Glu Asp Phe Lys Gly Tyr Leu Gly Cys
65                  70                  75                  80

Gln Ala Leu Ser Glu Met Ile Gln Phe Tyr Leu Glu Glu Val Met Pro
                85                  90                  95

Gln Ala Glu Asn Gln Asp Pro Asp Ile Lys Ala His Val Asn Ser Leu
            100                 105                 110

Gly Glu Asn Leu Lys Thr Leu Arg Leu Arg Leu Arg Arg Cys His Arg
        115                 120                 125

Phe Leu Pro Cys Glu Asn Lys Ser Lys Ala Val Glu Gln Val Lys Asn
    130                 135                 140

Ala Phe Asn Lys Leu Gln Glu Lys Gly Ile Tyr Lys Ala Met Ser Glu
145                 150                 155                 160

Phe Asp Ile Phe Ile Asn Tyr Ile Glu Ala Tyr Met Thr Met Lys Ile
                165                 170                 175

Arg Asn
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<210> SEQ ID NO 10
<211> LENGTH: 2652
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polynucleotide

<400> SEQUENCE: 10

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gctatgtccc aggcactcta ctagacactt catacagttt agaaaatcag atgggtgtag     60

atcaaggcag gagcaggaac caaaaagaaa ggcataaaca taagaaaaaa aatggaaggg    120

gtggaaacag agtacaataa catgagtaat ttgatggggg ctattatgaa ctgagaaatg    180

aactttgaaa agtatcttgg ggccaaatca tgtagactct tgagtgatgt gttaaggaat    240

gctatgagtg ctgagagggc atcagaagtc cttgagagcc tccagagaaa ggctcttaaa    300

aatgcagcgc aatctccagt gacagaagat actgctagaa atctgctaga aaaaaaacaa    360

aaaaggcatg tatagaggaa ttatgaggga aagataccaa gtcacggttt attcttcaaa    420

atggaggtgg cttgttggga aggtggaagc tcatttggcc agagtggaaa tggaattggg    480

agaaatcgat gaccaaatgt aaacacttgg tgcctgatat agcttgacac caagttagcc    540

ccaagtgaaa taccctggca atattaatgt gtcttttccc gatattcctc aggtactccc    600

tgcagggagg gcagaggaag tcttctaaca tgcggtgacg tggaggagaa tcccggccct    660

actagtgcca ccatgcagct gagaaaccca gagttgcatc tgggatgcgc cctggcactt    720

cgatttctcg ccctcgtatc ttgggacatc cctggcgcgc gcgcactgga caacggactg    780

gctaggaccc ccaccatggg ttggttacac tgggaacggt ttatgtgtaa tctcgactgt    840
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caggaagaac cggactcttg tatctcagaa aaattgttta tggagatggc agaactgatg 900

gtgtcagagg ggtggaaaga tgctgggtac gaatacctct gcatagatga ctgttggatg 960

gcgccccagc gtgacagcga gggacgcctg caggctgatc ctcagagatt cccacatggg 1020

attcgacagc tcgctaatta cgtgcactca aaaggcctga aactggggat ctatgccgac 1080

gtcgggaaca aaacctgtgc aggattccct ggaagtttcg ggtactacga catcgacgcc 1140

caaacctttg cagactgggg cgttgatctg ctgaagtttg acggatgtta ttgtgacagc 1200

ctggagaacc ttgccgacgg ctataagcat atgagcttgg cattgaaccg caccggacga 1260

agcatagtgt actcttgcga gtggcctctc tatatgtggc cttttcagaa accaaactac 1320

acggaaatac ggcaatactg caatcattgg aggaacttcg ccgatattga cgattcttgg 1380

aaatccatca aaagcatatt agactggacc agtttcaatc aggaaagaat cgtggacgtt 1440

gctggtccag gaggttggaa cgatccagac atgctggtga ttgggaactt cggcctcagt 1500

tggaatcagc aggtcacaca aatggcgctg tgggctatca tggctgcacc gctgtttatg 1560

tcaaacgacc tgcgacacat cagccctcag gctaaggcat tgttgcaaga taaggatgta 1620

atagcaataa accaggaccc cctcggtaag cagggctatc agttacgcca gggcgataac 1680

tttgaagtgt gggaacgacc gttgtccggc ctggcctggg ctgtggccat gataaaccga 1740

caagagattg gagggcccag gtcttatact atcgctgtgg caagtcttgg aaaaggagtg 1800

gcttgcaacc cagcatgctt cattactcag ttgctcccag tcaaaagaaa gctcggcttt 1860

tacgagtgga ctagtcgact tcggagtcat attaatccaa ccggcaccgt cttacttcag 1920

ctggagaata caatgcagat gtccctgaag gacctcctct agaggctagg tggaggctca 1980

gtgaaactta ctcacgtcat ccagcagaga atggaaagtc aaatttcctg aattgctatg 2040

tgtctgggtt tcatccatcc gacattgaag ttgacttact gaagaatgga gagagaattg 2100

aaaaagtgga gcattcagac ttgtctttca gcaaggactg gtctttctat ctcttgtact 2160

acactgaatt caccccccact gaaaaagatg agtatgcctg ccgtgtgaac catgtgactt 2220

tgtcacagcc caagatagtt aagtggggta agtcttacat tcttttgtaa gctgctgaaa 2280

gttgtgtatg agtagtcata tcataaagct gctttgatat aaaaaaggtc tatggccata 2340

ctaccctgaa tgagtcccat cccatctgat ataaacaatc tgcatattgg gattgtcagg 2400

gaatgttctt aaagatcaga ttagtggcac ctgctgagat actgatgcac agcatggttt 2460

ctgaaccagt agtttccctg cagttgagca gggagcagca gcagcacttg cacaaataca 2520

tatacactct taacacttct tacctactgg cttcctctag cttttgtggc agcttcaggt 2580

atatttagca ctgaacgaac atctcaagaa ggtataggcc tttgtttgta agtcctgctg 2640

tcctagcatc ct 2652

<210> SEQ ID NO 11
<211> LENGTH: 2721
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polynucleotide

<400> SEQUENCE: 11

gctatgtccc aggcactcta ctagacactt catacagttt agaaaatcag atgggtgtag 60

atcaaggcag gagcaggaac caaaaagaaa ggcataaaca taagaaaaaa aatggaaggg 120

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gtggaaacag agtacaataa catgagtaat ttgatggggg ctattatgaa ctgagaaatg    180
aactttgaaa agtatcttgg ggccaaatca tgtagactct tgagtgatgt gttaaggaat    240
gctatgagtg ctgagagggc atcagaagtc cttgagagcc tccagagaaa ggctcttaaa    300
aatgcagcgc aatctccagt gacagaagat actgctagaa atctgctaga aaaaaaacaa    360
aaaaggcatg tatagaggaa ttatgaggga aagataccaa gtcacggttt attcttcaaa    420
atggaggtgg cttgttggga aggtggaagc tcatttggcc agagtggaaa tggaattggg    480
agaaatcgat gaccaaatgt aaacacttgg tgcctgatat agcttgacac caagttagcc    540
ccaagtgaaa taccctggca atattaatgt gtcttttccc gatattcctc aggtactccc    600
tgcagggagg gcagaggaag tcttctaaca tgcggtgacg tggaggagaa tcccggccct    660
actagtgcca ccatgtctac cgctgtgttg gagaacccag ggctcgggcg taaactgtca    720
gacttcggcc aggaaacttc ttatatagaa gacaactgta accagaatgg tgcaatctct    780
ctcattttca gtcttaagga ggaggtcggc gccttggcta aggttttacg tctgttcgag    840
gagaatgatg tgaacctgac acatatcgag tccaggccat cccgcttgaa gaaagatgag    900
tatgaattct ttacacatct cgacaaacgg agtctccctg ctctgacaaa tataattaaa    960
attctgaggc atgacatcgg agccacagtg catgaactga gccgtgataa aaagaaggat   1020
accgtcccat ggtttcctcg gacgattcag gaacttgacc gattcgctaa ccagatattg   1080
tcctacggtg ctgagctgga cgcggatcat cccggattca aggaccctgt atacagagct   1140
cgccggaagc agttcgcaga cattgcctat aattatagac acggtcagcc gattcctaga   1200
gtcgaatata tggaggagga aaagaaaacg tgggggacag tgtttaaaac attaaagagt   1260
ctctacaaga cacatgcttg ttatgagtat aatcacatct tcccccctgtt ggagaaatac   1320
tgcgggtttc acgaggataa cataccccag ctggaagacg ttagtcagtt ccttcagact   1380
tgcaccgggt ttaggctgcg cccagttgcc ggcctcctca gttcacggga ttttctcggc   1440
ggcctggctt tccgcgtgtt ccattgtaca cagtacattc gacatggatc taaaccaatg   1500
tatactcccg aaccagatat ttgccatgag ctccttgggc acgtgccact tttctctgac   1560
aggagtttcg ctcaatttag ccaggagatt ggactcgcca gcctgggtgc ccctgacgaa   1620
tacattgaaa aactcgcaac catttactgg ttcaccgtgg agtttggact ctgtaaacaa   1680
ggcgacagca taaaggctta cggagccggt ctcttgagct cattcgggga gctccagtac   1740
tgcctgtctg agaagcccaa gctgcttccc cttgaactcg aaaaaaacagc tatccaaaac   1800
tatacagtaa ctgagttcca gcccttatat tacgtggcag agagcttcaa cgacgccaag   1860
gagaaagtcc gcaactttgc cgcaaccatt cctcggcctt tttccgtgcg ttacgatcca   1920
tacactcaga ggatcgaagt gctcgataac actcaacagc tcaagatttt ggcagacagc   1980
attaactcag agattggcat actttgtagc gcccttcaga agattaaata gaggctaggt   2040
ggaggctcag tgaaacttac tcacgtcatc cagcagagaa tggaaagtca aatttcctga   2100
attgctatgt gtctgggttt catccatccg acattgaagt tgacttactg aagaatggag   2160
agagaattga aaaagtggag cattcagact tgtctttcag caaggactgg tctttctatc   2220
tcttgtacta cactgaattc acccccactg aaaaagatga gtatgcctgc cgtgtgaacc   2280
atgtgacttt gtcacagccc aagatagtta agtggggtaa gtcttacatt cttttgtaag   2340
ctgctgaaag ttgtgtatga gtagtcatat cataaagctg ctttgatata aaaaaggtct   2400
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atggccatac taccctgaat gagtcccatc ccatctgata taaacaatct gcatattggg   2460

attgtcaggg aatgttctta aagatcagat tagtggcacc tgctgagata ctgatgcaca   2520

gcatggtttc tgaaccagta gtttccctgc agttgagcag ggagcagcag cagcacttgc   2580

acaaatacat atacactctt aacacttctt acctactggc ttcctctagc ttttgtggca   2640

gcttcaggta tatttagcac tgaacgaaca tctcaagaag gtataggcct ttgtttgtaa   2700

gtcctgctgt cctagcatcc t                                             2721
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<210> SEQ ID NO 12
<211> LENGTH: 3066
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polynucleotide

<400> SEQUENCE: 12
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atcaaggcag gagcaggaac caaaaagaaa ggcataaaca taagaaaaaa aatggaaggg    120

gtggaaacag agtacaataa catgagtaat ttgatggggg ctattatgaa ctgagaaatg    180

aactttgaaa agtatcttgg ggccaaatca tgtagactct tgagtgatgt gttaaggaat    240

gctatgagtg ctgagagggc atcagaagtc cttgagagcc tccagagaaa ggctcttaaa    300

aatgcagcgc aatctccagt gacagaagat actgctagaa atctgctaga aaaaaaacaa    360

aaaaggcatg tatagaggaa ttatgaggga aagataccaa gtcacggttt attcttcaaa    420

atggaggtgg cttgttggga aggtggaagc tcatttggcc agagtggaaa tggaattggg    480

agaaatcgat gaccaaatgt aaacacttgg tgcctgatat agcttgacac caagttagcc    540

ccaagtgaaa taccctggca atattaatgt gtcttttccc gatattcctc aggtactccc    600

tgcagggagg gcagaggaag tcttctaaca tgcggtgacg tggaggagaa tcccggccct    660

actagtgcca ccatgaaaac gctgagccag gctcagagta aaacctcttc tcaacagttc    720

agtttcacag gcaactccag cgctaacgtc ataatcggta accagaaact gactattaat    780

gacgtggccc gagtcgctcg gaacgggaca ctcgtcagcc tgacaaataa tacagatatc    840

ctgcaaggca ttcaggccag ttgcgactac ataaacaacg cggttgagtc aggcgagcct    900

atttacgggg ttacgtccgg tttcggagga atggccaacg tggccatcag cagggagcag    960

gcttctgaat tgcagactaa tctcgtctgg ttcttgaaaa ccggggctgg gaataagctc   1020

ccactcgcag atgtcagggc cgctatgctg ttgcgggcta attcacacat gcgtggggca   1080

agcggcatcc ggcttgaact gattaagcga atggagattt ttttgaatgc cggggtgaca   1140

ccctacgtgt acgagttcgg cagcattggc gctagcgggg atcttgtccc cttatcatat   1200

attacaggca gcctgattgg actggacccc tcttttaagg ttgattttaa tggcaaggaa   1260

atggacgccc ccacagcctt gcgacagctg aacctgtctc ctcttactct gcttcctaaa   1320

gaaggcttgg ctatgatgaa tgggacatcc gtgatgactg gcattgcagc taactgtgtt   1380

tatgatactc aaattctgac agcaatagct atgggcgtcc atgccctcga cattcaggcc   1440

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polynucleotide

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polynucleotide

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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
polynucleotide

<400> SEQUENCE: 16

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aactttgaaa agtatcttgg ggccaaatca tgtagactct tgagtgatgt gttaaggaat 240

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<211> LENGTH: 1899
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
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<400> SEQUENCE: 17

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aatgcagcgc aatctccagt gacagaagat actgctagaa atctgctaga aaaaaaacaa    360

aaaaggcatg tatagaggaa ttatgaggga aagataccaa gtcacggttt attcttcaaa    420

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<210> SEQ ID NO 18
<211> LENGTH: 1368
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

<400> SEQUENCE: 18

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Lys Val Leu Gly Asn Thr Asp Arg His Ser Ile Lys Lys Asn Leu Ile
        35                  40                  45

Gly Ala Leu Leu Phe Asp Ser Gly Glu Thr Ala Glu Ala Thr Arg Leu
    50                  55                  60

Lys Arg Thr Ala Arg Arg Arg Tyr Thr Arg Arg Lys Asn Arg Ile Cys
65                  70                  75                  80

Tyr Leu Gln Glu Ile Phe Ser Asn Glu Met Ala Lys Val Asp Asp Ser
                85                  90                  95

Phe Phe His Arg Leu Glu Glu Ser Phe Leu Val Glu Glu Asp Lys Lys
            100                 105                 110

His Glu Arg His Pro Ile Phe Gly Asn Ile Val Asp Glu Val Ala Tyr
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His Glu Lys Tyr Pro Thr Ile Tyr His Leu Arg Lys Lys Leu Val Asp
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Ser Thr Asp Lys Ala Asp Leu Arg Leu Ile Tyr Leu Ala Leu Ala His
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Met Ile Lys Phe Arg Gly His Phe Leu Ile Glu Gly Asp Leu Asn Pro
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Asp Asn Ser Asp Val Asp Lys Leu Phe Ile Gln Leu Val Gln Thr Tyr
            180                 185                 190

Asn Gln Leu Phe Glu Glu Asn Pro Ile Asn Ala Ser Gly Val Asp Ala
        195                 200                 205

Lys Ala Ile Leu Ser Ala Arg Leu Ser Lys Ser Arg Arg Leu Glu Asn
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Leu Ile Ala Gln Leu Pro Gly Glu Lys Lys Asn Gly Leu Phe Gly Asn
225                 230                 235                 240

Leu Ile Ala Leu Ser Leu Gly Leu Thr Pro Asn Phe Lys Ser Asn Phe
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Asp Leu Ala Glu Asp Ala Lys Leu Gln Leu Ser Lys Asp Thr Tyr Asp
            260                 265                 270

Asp Asp Leu Asp Asn Leu Leu Ala Gln Ile Gly Asp Gln Tyr Ala Asp
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Leu Phe Leu Ala Ala Lys Asn Leu Ser Asp Ala Ile Leu Leu Ser Asp
    290                 295                 300

Ile Leu Arg Val Asn Thr Glu Ile Thr Lys Ala Pro Leu Ser Ala Ser
305                 310                 315                 320

Met Ile Lys Arg Tyr Asp Glu His His Gln Asp Leu Thr Leu Leu Lys
                325                 330                 335

Ala Leu Val Arg Gln Gln Leu Pro Glu Lys Tyr Lys Glu Ile Phe Phe
            340                 345                 350

Asp Gln Ser Lys Asn Gly Tyr Ala Gly Tyr Ile Asp Gly Gly Ala Ser
        355                 360                 365

Gln Glu Glu Phe Tyr Lys Phe Ile Lys Pro Ile Leu Glu Lys Met Asp
    370                 375                 380

Gly Thr Glu Glu Leu Leu Val Lys Leu Asn Arg Glu Asp Leu Leu Arg
385                 390                 395                 400

Lys Gln Arg Thr Phe Asp Asn Gly Ser Ile Pro His Gln Ile His Leu
                405                 410                 415

Gly Glu Leu His Ala Ile Leu Arg Arg Gln Glu Asp Phe Tyr Pro Phe
            420                 425                 430

Leu Lys Asp Asn Arg Glu Lys Ile Glu Lys Ile Leu Thr Phe Arg Ile
        435                 440                 445

Pro Tyr Tyr Val Gly Pro Leu Ala Arg Gly Asn Ser Arg Phe Ala Trp
    450                 455                 460

Met Thr Arg Lys Ser Glu Glu Thr Ile Thr Pro Trp Asn Phe Glu Glu
465                 470                 475                 480

Val Val Asp Lys Gly Ala Ser Ala Gln Ser Phe Ile Glu Arg Met Thr
                485                 490                 495

Asn Phe Asp Lys Asn Leu Pro Asn Glu Lys Val Leu Pro Lys His Ser
            500                 505                 510

Leu Leu Tyr Glu Tyr Phe Thr Val Tyr Asn Glu Leu Thr Lys Val Lys
        515                 520                 525

Tyr Val Thr Glu Gly Met Arg Lys Pro Ala Phe Leu Ser Gly Glu Gln
    530                 535                 540
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Lys Lys Ala Ile Val Asp Leu Leu Phe Lys Thr Asn Arg Lys Val Thr
545                 550                 555                 560

Val Lys Gln Leu Lys Glu Asp Tyr Phe Lys Lys Ile Glu Cys Phe Asp
                565                 570                 575

Ser Val Glu Ile Ser Gly Val Glu Asp Arg Phe Asn Ala Ser Leu Gly
            580                 585                 590

Thr Tyr His Asp Leu Leu Lys Ile Ile Lys Asp Lys Asp Phe Leu Asp
        595                 600                 605

Asn Glu Glu Asn Glu Asp Ile Leu Glu Asp Ile Val Leu Thr Leu Thr
    610                 615                 620

Leu Phe Glu Asp Arg Glu Met Ile Glu Glu Arg Leu Lys Thr Tyr Ala
625                 630                 635                 640

His Leu Phe Asp Asp Lys Val Met Lys Gln Leu Lys Arg Arg Arg Tyr
                645                 650                 655

Thr Gly Trp Gly Arg Leu Ser Arg Lys Leu Ile Asn Gly Ile Arg Asp
            660                 665                 670

Lys Gln Ser Gly Lys Thr Ile Leu Asp Phe Leu Lys Ser Asp Gly Phe
        675                 680                 685

Ala Asn Arg Asn Phe Met Gln Leu Ile His Asp Asp Ser Leu Thr Phe
    690                 695                 700

Lys Glu Asp Ile Gln Lys Ala Gln Val Ser Gly Gln Gly Asp Ser Leu
705                 710                 715                 720

His Glu His Ile Ala Asn Leu Ala Gly Ser Pro Ala Ile Lys Lys Gly
                725                 730                 735

Ile Leu Gln Thr Val Lys Val Val Asp Glu Leu Val Lys Val Met Gly
            740                 745                 750

Arg His Lys Pro Glu Asn Ile Val Ile Glu Met Ala Arg Glu Asn Gln
        755                 760                 765

Thr Thr Gln Lys Gly Gln Lys Asn Ser Arg Glu Arg Met Lys Arg Ile
    770                 775                 780

Glu Glu Gly Ile Lys Glu Leu Gly Ser Gln Ile Leu Lys Glu His Pro
785                 790                 795                 800

Val Glu Asn Thr Gln Leu Gln Asn Glu Lys Leu Tyr Leu Tyr Tyr Leu
                805                 810                 815

Gln Asn Gly Arg Asp Met Tyr Val Asp Gln Glu Leu Asp Ile Asn Arg
            820                 825                 830

Leu Ser Asp Tyr Asp Val Asp His Ile Val Pro Gln Ser Phe Leu Lys
        835                 840                 845

Asp Asp Ser Ile Asp Asn Lys Val Leu Thr Arg Ser Asp Lys Asn Arg
    850                 855                 860

Gly Lys Ser Asp Asn Val Pro Ser Glu Glu Val Val Lys Lys Met Lys
865                 870                 875                 880

Asn Tyr Trp Arg Gln Leu Leu Asn Ala Lys Leu Ile Thr Gln Arg Lys
                885                 890                 895

Phe Asp Asn Leu Thr Lys Ala Glu Arg Gly Gly Leu Ser Glu Leu Asp
            900                 905                 910

Lys Ala Gly Phe Ile Lys Arg Gln Leu Val Glu Thr Arg Gln Ile Thr
        915                 920                 925

Lys His Val Ala Gln Ile Leu Asp Ser Arg Met Asn Thr Lys Tyr Asp
    930                 935                 940

Glu Asn Asp Lys Leu Ile Arg Glu Val Lys Val Ile Thr Leu Lys Ser
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945                 950                 955                 960

Lys Leu Val Ser Asp Phe Arg Lys Asp Phe Gln Phe Tyr Lys Val Arg
                965                 970                 975

Glu Ile Asn Asn Tyr His His Ala His Asp Ala Tyr Leu Asn Ala Val
            980                 985                 990

Val Gly Thr Ala Leu Ile Lys Lys  Tyr Pro Lys Leu Glu  Ser Glu Phe
        995                 1000                1005

Val Tyr  Gly Asp Tyr Lys Val  Tyr Asp Val Arg Lys  Met Ile Ala
    1010                 1015                 1020

Lys Ser  Glu Gln Glu Ile Gly  Lys Ala Thr Ala Lys  Tyr Phe Phe
    1025                 1030                 1035

Tyr Ser  Asn Ile Met Asn Phe  Phe Lys Thr Glu Ile  Thr Leu Ala
    1040                 1045                 1050

Asn Gly  Glu Ile Arg Lys Arg  Pro Leu Ile Glu Thr  Asn Gly Glu
    1055                 1060                 1065

Thr Gly  Glu Ile Val Trp Asp  Lys Gly Arg Asp Phe  Ala Thr Val
    1070                 1075                 1080

Arg Lys  Val Leu Ser Met Pro  Gln Val Asn Ile Val  Lys Lys Thr
    1085                 1090                 1095

Glu Val  Gln Thr Gly Gly Phe  Ser Lys Glu Ser Ile  Leu Pro Lys
    1100                 1105                 1110

Arg Asn  Ser Asp Lys Leu Ile  Ala Arg Lys Lys Asp  Trp Asp Pro
    1115                 1120                 1125

Lys Lys  Tyr Gly Gly Phe Asp  Ser Pro Thr Val Ala  Tyr Ser Val
    1130                 1135                 1140

Leu Val  Val Ala Lys Val Glu  Lys Gly Lys Ser Lys  Lys Leu Lys
    1145                 1150                 1155

Ser Val  Lys Glu Leu Leu Gly  Ile Thr Ile Met Glu  Arg Ser Ser
    1160                 1165                 1170

Phe Glu  Lys Asn Pro Ile Asp  Phe Leu Glu Ala Lys  Gly Tyr Lys
    1175                 1180                 1185

Glu Val  Lys Lys Asp Leu Ile  Ile Lys Leu Pro Lys  Tyr Ser Leu
    1190                 1195                 1200

Phe Glu  Leu Glu Asn Gly Arg  Lys Arg Met Leu Ala  Ser Ala Gly
    1205                 1210                 1215

Glu Leu  Gln Lys Gly Asn Glu  Leu Ala Leu Pro Ser  Lys Tyr Val
    1220                 1225                 1230

Asn Phe  Leu Tyr Leu Ala Ser  His Tyr Glu Lys Leu  Lys Gly Ser
    1235                 1240                 1245

Pro Glu  Asp Asn Glu Gln Lys  Gln Leu Phe Val Glu  Gln His Lys
    1250                 1255                 1260

His Tyr  Leu Asp Glu Ile Ile  Glu Gln Ile Ser Glu  Phe Ser Lys
    1265                 1270                 1275

Arg Val  Ile Leu Ala Asp Ala  Asn Leu Asp Lys Val  Leu Ser Ala
    1280                 1285                 1290

Tyr Asn  Lys His Arg Asp Lys  Pro Ile Arg Glu Gln  Ala Glu Asn
    1295                 1300                 1305

Ile Ile  His Leu Phe Thr Leu  Thr Asn Leu Gly Ala  Pro Ala Ala
    1310                 1315                 1320

Phe Lys  Tyr Phe Asp Thr Thr  Ile Asp Arg Lys Arg  Tyr Thr Ser
    1325                 1330                 1335
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Thr Lys  Glu Val Leu Asp Ala  Thr Leu Ile His Gln  Ser Ile Thr
    1340                 1345                 1350

Gly Leu  Tyr Glu Thr Arg Ile  Asp Leu Ser Gln Leu  Gly Gly Asp
    1355                 1360                 1365


<210> SEQ ID NO 19
<211> LENGTH: 100
<212> TYPE: RNA
<213> ORGANISM: Homo sapiens
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (1)..(4)
<223> OTHER INFORMATION: modified with 2'-O-methyl phosphorothioate
<220> FEATURE:
<221> NAME/KEY: misc_feature
<222> LOCATION: (97)..(100)
<223> OTHER INFORMATION: modified with 2'-O-methyl phosphorothioate

<400> SEQUENCE: 19

cgugaguaaa ccugaaucuu guuuuagagc uagaaauagc aaguuaaaau aaggcuaguc     60

cguuaucaac uugaaaaagu ggcaccgagu cggugcuuuu                          100


<210> SEQ ID NO 20
<211> LENGTH: 598
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polynucleotide

<400> SEQUENCE: 20

gctatgtccc aggcactcta ctagacactt catacagttt agaaaatcag atgggtgtag     60

atcaaggcag gagcaggaac caaaaagaaa ggcataaaca taagaaaaaa aatggaaggg    120

gtggaaacag agtacaataa catgagtaat ttgatggggg ctattatgaa ctgagaaatg    180

aactttgaaa agtatcttgg ggccaaatca tgtagactct tgagtgatgt gttaaggaat    240

gctatgagtg ctgagagggc atcagaagtc cttgagagcc tccagagaaa ggctcttaaa    300

aatgcagcgc aatctccagt gacagaagat actgctagaa atctgctaga aaaaaaacaa    360

aaaaggcatg tatagaggaa ttatgaggga aagataccaa gtcacggttt attcttcaaa    420

atggaggtgg cttgttggga aggtggaagc tcatttggcc agagtggaaa tggaattggg    480

agaaatcgat gaccaaatgt aaacacttgg tgcctgatat agcttgacac caagttagcc    540

ccaagtgaaa taccctggca atattaatgt gtcttttccc gatattcctc aggtactc      598


<210> SEQ ID NO 21
<211> LENGTH: 665
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polynucleotide

<400> SEQUENCE: 21

ttactcacgt catccagcag agaatggaaa gtcaaatttc ctgaattgct atgtgtctgg     60

gtttcatcca tccgacattg aagttgactt actgaagaat ggagagagaa ttgaaaaagt    120

ggagcattca gacttgtctt tcagcaagga ctggtctttc tatctcttgt actacactga    180

attcaccccc actgaaaaag atgagtatgc ctgccgtgtg aaccatgtga ctttgtcaca    240
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gcccaagata gttaagtggg gtaagtctta cattcttttg taagctgctg aaagttgtgt 300

atgagtagtc atatcataaa gctgctttga tataaaaaag gtctatggcc atactaccct 360

gaatgagtcc catcccatct gatataaaca atctgcatat tgggattgtc agggaatgtt 420

cttaaagatc agattagtgg cacctgctga gatactgatg cacagcatgg tttctgaacc 480

agtagtttcc ctgcagttga gcagggagca gcagcagcac ttgcacaaat acatatacac 540

tcttaacact tcttacctac tggcttcctc tagcttttgt ggcagcttca ggtatattta 600

gcactgaacg aacatctcaa gaaggtatag gcctttgttt gtaagtcctg ctgtcctagc 660

atcct 665

<210> SEQ ID NO 22
<211> LENGTH: 360
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 22

atgtctcgct ccgtggcctt agctgtgctc gcgctactct ctctttctgg cctggaggct 60

atccagcgta ctccaaagat tcaggtttac tcacgtcatc cagcagagaa tggaaagtca 120

aatttcctga attgctatgt gtctgggttt catccatccg acattgaagt tgacttactg 180

aagaatggag agagaattga aaaagtggag cattcagact tgtctttcag caaggactgg 240

tctttctatc tcttgtacta cactgaattc acccccactg aaaaagatga gtatgcctgc 300

cgtgtgaacc atgtgacttt gtcacagccc aagatagtta agtgggatcg agacatgtaa 360

<210> SEQ ID NO 23
<211> LENGTH: 119
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 23

Met Ser Arg Ser Val Ala Leu Ala Val Leu Ala Leu Leu Ser Leu Ser
1   5   10   15

Gly Leu Glu Ala Ile Gln Arg Thr Pro Lys Ile Gln Val Tyr Ser Arg
            20   25   30

His Pro Ala Glu Asn Gly Lys Ser Asn Phe Leu Asn Cys Tyr Val Ser
        35   40   45

Gly Phe His Pro Ser Asp Ile Glu Val Asp Leu Leu Lys Asn Gly Glu
    50   55   60

Arg Ile Glu Lys Val Glu His Ser Asp Leu Ser Phe Ser Lys Asp Trp
65   70   75   80

Ser Phe Tyr Leu Leu Tyr Tyr Thr Glu Phe Thr Pro Thr Glu Lys Asp
                85   90   95

Glu Tyr Ala Cys Arg Val Asn His Val Thr Leu Ser Gln Pro Lys Ile
            100   105   110

Val Lys Trp Asp Arg Asp Met
        115

<210> SEQ ID NO 24
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic

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oligonucleotide

<400> SEQUENCE: 24

cgtgagtaaa cctgaatctt tgg 23

<210> SEQ ID NO 25
<211> LENGTH: 2079
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic polynucleotide

<400> SEQUENCE: 25

gctatgtccc aggcactcta ctagacactt catacagttt agaaaatcag atgggtgtag 60

atcaaggcag gagcaggaac caaaaagaaa ggcataaaca taagaaaaaa aatggaaggg 120

gtggaaacag agtacaataa catgagtaat ttgatggggg ctattatgaa ctgagaaatg 180

aactttgaaa agtatcttgg ggccaaatca tgtagactct tgagtgatgt gttaaggaat 240

gctatgagtg ctgagagggc atcagaagtc cttgagagcc tccagagaaa ggctcttaaa 300

aatgcagcgc aatctccagt gacagaagat actgctagaa atctgctaga aaaaaaacaa 360

aaaaggcatg tatagaggaa ttatgaggga aagataccaa gtcacggttt attcttcaaa 420

atggaggtgg cttgttggga aggtggaagc tcatttggcc agagtggaaa tggaattggg 480

agaaatcgat gaccaaatgt aaacacttgg tgcctgatat agcttgacac caagttagcc 540

ccaagtgaaa taccctggca atattaatgt gtcttttccc gatattcctc aggtactccc 600

tgcagggagg gcagaggaag tcttctaaca tgcggtgacg tggaggagaa tcccggccct 660

actagtgcca ccatggtgag caagggcgag gagctgttca ccggggtggt gcccatcctg 720

gtcgagctgg acggcgacgt aaacggccac aagttcagcg tgtccggcga gggcgagggc 780

gatgccacct acggcaagct gaccctgaag ttcatctgca ccaccggcaa gctgcccgtg 840

ccctggccca ccctcgtgac caccctgacc tacggcgtgc agtgcttcag ccgctacccc 900

gaccacatga agcagcacga cttcttcaag tccgccatgc ccgaaggcta cgtccaggag 960

cgcaccatct tcttcaagga cgacggcaac tacaagaccc gcgccgaggt gaagttcgag 1020

ggcgacaccc tggtgaaccg catcgagctg aagggcatcg acttcaagga ggacggcaac 1080

atcctggggc acaagctgga gtacaactac aacagccaca acgtctatat catggccgac 1140

aagcagaaga acggcatcaa ggtgaacttc aagatccgcc acaacatcga ggacggcagc 1200

gtgcagctcg ccgaccacta ccagcagaac acccccatcg gcgacggccc cgtgctgctg 1260

cccgacaacc actacctgag cacccagtcc gccctgagca aagaccccaa cgagaagcgc 1320

gatcacatgg tcctgctgga gttcgtgacc gccgccggga tcactctcgg catggacgag 1380

ctgtacaagt aactcgagaa gcttaagttt aaacttactc acgtcatcca gcagagaatg 1440

gaaagtcaaa tttcctgaat tgctatgtgt ctgggtttca tccatccgac attgaagttg 1500

acttactgaa gaatggagag agaattgaaa aagtggagca ttcagacttg tctttcagca 1560

aggactggtc tttctatctc ttgtactaca ctgaattcac ccccactgaa aaagatgagt 1620

atgcctgccg tgtgaaccat gtgactttgt cacagcccaa gatagttaag tggggtaagt 1680

cttacattct tttgtaagct gctgaaagtt gtgtatgagt agtcatatca taaagctgct 1740

ttgatataaa aaaggtctat ggccatacta ccctgaatga gtcccatccc atctgatata 1800

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aacaatctgc atattgggat tgtcagggaa tgttcttaaa gatcagatta gtggcacctg   1860

ctgagatact gatgcacagc atggtttctg aaccagtagt ttccctgcag ttgagcaggg   1920

agcagcagca gcacttgcac aaatacatat acactcttaa cacttcttac ctactggctt   1980

cctctagctt ttgtggcagc ttcaggtata tttagcactg aacgaacatc tcaagaaggt   2040

ataggccttt gtttgtaagt cctgctgtcc tagcatcct                           2079


&lt;210&gt; SEQ ID NO 26
&lt;211&gt; LENGTH: 288
&lt;212&gt; TYPE: PRT
&lt;213&gt; ORGANISM: Artificial Sequence
&lt;220&gt; FEATURE:
&lt;223&gt; OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polypeptide

&lt;400&gt; SEQUENCE: 26

Met Ser Arg Ser Val Ala Leu Ala Val Leu Ala Leu Leu Ser Leu Ser
1               5                   10                  15

Gly Leu Glu Ala Ile Gln Arg Thr Pro Cys Arg Glu Gly Arg Gly Ser
            20                  25                  30

Leu Leu Thr Cys Gly Asp Val Glu Glu Asn Pro Gly Pro Thr Ser Ala
        35                  40                  45

Thr Met Val Ser Lys Gly Glu Glu Leu Phe Thr Gly Val Val Pro Ile
    50                  55                  60

Leu Val Glu Leu Asp Gly Asp Val Asn Gly His Lys Phe Ser Val Ser
65                  70                  75                  80

Gly Glu Gly Glu Gly Asp Ala Thr Tyr Gly Lys Leu Thr Leu Lys Phe
                85                  90                  95

Ile Cys Thr Thr Gly Lys Leu Pro Val Pro Trp Pro Thr Leu Val Thr
            100                 105                 110

Thr Leu Thr Tyr Gly Val Gln Cys Phe Ser Arg Tyr Pro Asp His Met
        115                 120                 125

Lys Gln His Asp Phe Phe Lys Ser Ala Met Pro Glu Gly Tyr Val Gln
    130                 135                 140

Glu Arg Thr Ile Phe Phe Lys Asp Asp Gly Asn Tyr Lys Thr Arg Ala
145                 150                 155                 160

Glu Val Lys Phe Glu Gly Asp Thr Leu Val Asn Arg Ile Glu Leu Lys
                165                 170                 175

Gly Ile Asp Phe Lys Glu Asp Gly Asn Ile Leu Gly His Lys Leu Glu
            180                 185                 190

Tyr Asn Tyr Asn Ser His Asn Val Tyr Ile Met Ala Asp Lys Gln Lys
        195                 200                 205

Asn Gly Ile Lys Val Asn Phe Lys Ile Arg His Asn Ile Glu Asp Gly
    210                 215                 220

Ser Val Gln Leu Ala Asp His Tyr Gln Gln Asn Thr Pro Ile Gly Asp
225                 230                 235                 240

Gly Pro Val Leu Leu Pro Asp Asn His Tyr Leu Ser Thr Gln Ser Ala
                245                 250                 255

Leu Ser Lys Asp Pro Asn Glu Lys Arg Asp His Met Val Leu Leu Glu
            260                 265                 270

Phe Val Thr Ala Ala Gly Ile Thr Leu Gly Met Asp Glu Leu Tyr Lys
        275                 280                 285
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<210> SEQ ID NO 27
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      oligonucleotide

<400> SEQUENCE: 27

tgagagggca tcagaagtcc 20

<210> SEQ ID NO 28
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      oligonucleotide

<400> SEQUENCE: 28

aagtcacatg gttcacacgg 20

<210> SEQ ID NO 29
<211> LENGTH: 21
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      oligonucleotide

<400> SEQUENCE: 29

gctatgtccc aggcactcta c 21

<210> SEQ ID NO 30
<211> LENGTH: 20
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      oligonucleotide

<400> SEQUENCE: 30

aggatgctag gacagcagga 20

<210> SEQ ID NO 31
<211> LENGTH: 3677
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of Artificial Sequence: Synthetic
      polynucleotide

<400> SEQUENCE: 31

tgctttctct gacctgcatt ctctcccctg ggcctgtgcc gctttctgtc tgcagcttgt 60

ggcctgggtc acctctacgg ctggcccaga tccttccctg ccgcctcctt caggttccgt 120

cttcctccac tccctcttcc ccttgctctc tgctgtgttg ctgcccaagg atgctctttc 180

cggagcactt ccttctcggc gctgcaccac gtgatgtcct ctgagcggat cctccccgtg 240

tctgggtcct ctccgggcat ctctcctccc tcacccaacc ccatgccgtc ttcactcgct 300

gggttccctt ttccttctcc ttctggggcc tgtgccatct ctcgtttctt aggatggcct 360

tctccgacgg atgtctccct tgcgtcccgc ctccccttct tgtaggcctg catcatcacc 420

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gtttttctgg acaaccccaa agtaccccgt ctccctggct ttagccacct ctccatcctc    480
ttgctttctt tgcctggaca ccccgttctc ctgtggattc gggtcacctc tcactccttt    540
catttgggca gctcccctac ccccttacc tctctagtct gtgctagctc ttccagcccc    600
ctgtcatggc atcttccagg ggtccgagag ctcagctagt cttcttcctc caacccgggc    660
ccctatgtcc acttcaggac agcatgtttg ctgcctccag ggatcctgtg tccccgagct    720
gggaccacct tatattccca gggccggtta atgtggctct ggttctgggt acttttatct    780
gtcccctcca ccccacagtg gggccctgca ggtgaaagac cccacctgta ggtttggcaa    840
gatagctgca gtaacgccat tttgcaaggc atggaaaaat accaaaccaa gaatagagaa    900
gttcagatca agggcgggta catgaaaata gctaacgttg ggccaaacag gatatctgcg    960
gtgagcagtt tcggccccgg cccggggcca agaacagatg gtcaccgcag tttcggcccc   1020
ggcccgaggc caagaacaga tggtccccag atatggccca accctcagca gtttcttaag   1080
acccatcaga tgtttccagg ctcccccaag gacctgaaat gaccctgcgc cttatttgaa   1140
ttaaccaatc agcctgcttc tcgcttctgt tcgcgcgctt ctgcttcccg agctctataa   1200
aagagctcac aacccctcac tcggcgcgcc agtcctccga ttgactgagt cgccctgatc   1260
agtactacta gtgccaccat gcagctgaga aacccagagt tgcatctggg atgcgccctg   1320
gcacttcgat ttctcgccct cgtatcttgg gacatccctg gcgcgcgcgc actggacaac   1380
ggactggcta ggacccccac catgggttgg ttacactggg aacggtttat gtgtaatctc   1440
gactgtcagg aagaaccgga ctcttgtatc tcagaaaaat tgtttatgga gatggcagaa   1500
ctgatggtgt cagaggggtg gaaagatgct gggtacgaat acctctgcat agatgactgt   1560
tggatggcgc cccagcgtga cagcgaggga cgcctgcagg ctgatcctca gagattccca   1620
catgggattc gacagctcgc taattacgtg cactcaaaag gcctgaaact ggggatctat   1680
gccgacgtcg ggaacaaaac ctgtgcagga ttccctggaa gtttcgggta ctacgacatc   1740
gacgcccaaa cctttgcaga ctggggcgtt gatctgctga agtttgacgg atgttattgt   1800
gacagcctgg agaaccttgc cgacggctat aagcatatga gcttggcatt gaaccgcacc   1860
ggacgaagca tagtgtactc ttgcgagtgg cctctctata tgtggccttt tcagaaacca   1920
aactacacgg aaatacggca atactgcaat cattggagga acttcgccga tattgacgat   1980
tcttggaaat ccatcaaaag catattagac tggaccagtt tcaatcagga aagaatcgtg   2040
gacgttgctg gtccaggagg ttggaacgat ccagacatgc tggtgattgg gaacttcggc   2100
ctcagttgga atcagcaggt cacacaaatg gcgctgtggg ctatcatggc tgcaccgctg   2160
tttatgtcaa acgacctgcg acacatcagc cctcaggcta aggcattgtt gcaagataag   2220
gatgtaatag caataaacca ggacccccctc ggtaagcagg gctatcagtt acgccagggc   2280
gataactttg aagtgtggga acgaccgttg tccggcctgg cctgggctgt ggccatgata   2340
aaccgacaag agattggagg gcccaggtct tatactatcg ctgtggcaag tcttggaaaa   2400
ggagtggctt gcaacccagc atgcttcatt actcagttgc tcccagtcaa aagaaagctc   2460
ggcttttacg agtggactag tcgacttcgg agtcatatta atccaaccgg caccgtctta   2520
cttcagctgg agaatacaat gcagatgtcc ctgaaggacc tcctctagag gctaggtgga   2580
ggctcagtgg tttaaaccgc tgatcactgt gccttctagt tgccagccat ctgttgtttg   2640
cccctccccc gtgccttcct tgaccctgga aggtgccact cccactgtcc tttcctaata   2700
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aaatgaggaa attgcatcgc attgtctgag taggtgtcat tctattctgg ggggtggggt 2760

ggggcaggac agcaaggggg aggattggga agacaatagc aggcatgctg gggatgcggt 2820

gggctctatg ggcccgggca caggattggt gacagaaaag ccccatcctt aggcctcctc 2880

cttcctagtc tcctgatatt gggtctaacc cccacctcct gttaggcaga ttccttatct 2940

ggtgacacac ccccatttcc tggagccatc tctctccttg ccagaacctc taaggtttgc 3000

ttacgatgga gccagagagg atcctgggag ggagagcttg gcagggggtg ggagggaagg 3060

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```

1-78. (canceled)

79. An engineered human B cell comprising a therapeutic protein encoded by a nucleic acid sequence inserted into a locus of a β2M gene.

80. The engineered human B cell of claim **79**, wherein expression of the β2M gene has been disrupted.

81. The engineered human B cell of claim **80**, wherein the nucleic acid sequence encoding the therapeutic protein has been inserted into exon 2 of the locus of the β2M gene.

82. The engineered human B cell of claim **79**, wherein the nucleic acid sequence encoding the therapeutic protein has been inserted into an intron of the locus of the β2M gene, and wherein expression of the β2M gene is maintained at a percentage greater than 50% when compared with a wild type human B cell.

83. The engineered human B cell of claim **79**, wherein the therapeutic protein is alpha-galactosidase A (GLA), acid alpha-glucosidase (GAA), phenylalanine hydroxylase (PAH), phenylalanine ammonia-lyase (PAL), full length or B domain-deleted (BDD) FVIII, a GPC3 chimeric receptor, a cytokine or a chemokine.

84. The engineered human B cell of claim **82**, wherein the therapeutic protein is selected from the amino acid sequences consisting of SEQ ID NOs. 2-9.

85. A method of producing an engineered B cell expressing a therapeutic protein, the method comprising delivering to a human B cell:

a. an RNA-guided nuclease;

b. a gRNA targeting a β2M gene; and

c. a construct comprising a nucleic acid sequence encoding the therapeutic protein.

86. The method of claim **85**, wherein the RNA-guided nuclease comprises the amino acid sequence of SEQ ID NO. 18.

87. The method of claim **85**, wherein the gRNA comprises the nucleic acid sequence of SEQ ID NO. 19.

88. The method of claim **85**, wherein the gRNA specifically targets exon 2 of a locus of the β2M gene.

89. The method of claim **85**, wherein the gRNA specifically targets an intron of a locus of the β2M gene.

90. The method of claim **85**, wherein the β2M gene expresses at a percentage greater than 50% when compared to a wild-type human B cell.

91. The method of claim **85**, wherein the construct comprises a codon-optimized nucleic acid sequence selected from the group consisting of SEQ ID NOs. 10-17 and 31.

92. The method of claim **85**, wherein the construct comprises a left homology arm of SEQ ID NO. 20 and a right homology arm of SEQ ID NO. 21.

93. A method of treating a patient in need thereof comprising administering to said patient a therapeutically effective amount of an engineered human B cell, wherein the engineered human B cell comprises a therapeutic protein, wherein a gene expressing said therapeutic protein has been inserted into a locus of a β2M gene.

94. The method of claim **93**, wherein the therapeutic protein is for the treatment of Fabry disease, Pompe disease, Phenylketonuria (PKU) or Hemophilia A.

95. The method of claim **94**, wherein expression of the β2M gene has been disrupted.

96. The method of claim **95**, wherein a nucleic acid sequence encoding the therapeutic protein has been inserted into exon 2 of the locus of the β2M gene.

97. The method of claim **94**, wherein the therapeutic protein is alpha-galactosidase A (GLA), acid alpha-glucosidase (GAA), phenylalanine hydroxylase (PAH), phenylalanine ammonia-lyase (PAL) or B domain-deleted (BDD) FVIII.

98. The method of claim **97**, wherein the therapeutic protein is selected from the amino acid sequences consisting of SEQ ID NOs. 2-7.

* * * * *